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GENERAL HOMOLOGY AND COHOMOLOGY THEORIES. CURRENT STATE AND TYPICAL APPLICATIONS

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A contemporary approach is given to the construction of homology and cohomology theories and from this point of view, the most common concrete theories. Typical applications of the theory as developed at the present time are cited.

Even if originally one deals only with such "good" objects as manifolds and polyhedra, in studying many concrete problems concerning such objects one encouters more general formations such as subsets and quotient-spaces of maps, orbits of points, and spaces of orbits, sets of fixed and periodic points, points of coincidence of maps, etc. It is clear that this presupposes the development of homology and cohomology theories more general than the classical ones, which are adaptable for application to any situations which arise in the course of study. However the reason indicated for developing such theories is not the only one and maybe not even the main one (although now and then it is assumed to be). The fact is that both in topology itself and in some contiguous areas (differential geometry, algebraic geometry, functional and complex analysis, etc.) objects arise including triangulable ones which are not defined with the help of triangulations (which are usually difficult to give) but by their subdivision into separate comparatively simple constituent parts, i.e., actually with the help of coverings (manifolds, analytic spaces, algebraic varieties, etc. are defined in this way). Here it occurs not infrequently that the domain of coefficients for homology or cohomology defined separately on each such part does not happen to be identical as an object in the large which was one of the reasons for the development of the theory with nonconstant coefficients (cf. Chap. 1, and point 6.1 of Chap. 4 for other examples at least as important).

It is well known that although the singular theory is defined for any topological spaces without restrictions, for applications to general situations it turns out to be unsuitable. Thus, for a compactum S homeomorphic to a solenoid (and similar compacta occur as attractors in the theory of dynamical systems), since S is a union of a continuum of its path connected components with trivial singular homology and cohomology, the zero-dimensional homology and cohomology coincide, respectively, with the continual direct sum and direct product of the coefficient group (despite the connectedness of S!), while at the same time the one-dimensional ones are equal to zero (despite the fact that S admits nontrivial maps onto a circle!).

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Conversely, in three-dimensional Euclidean space there exist compacta with nontrivial singular groups of arbitrarily high dimension (cf. [77, 78]), although from the commonsense point of view such compacta should not support cycles of dimension $\geqslant 3$. Some other deficiencies of singular theory are also known (cf. Sec. 1 of Chap. 3). All such "anomalies" indicate that outside certain limits singular theory generally stops being a homology and cohomology theory.

At the present time a large set of different approaches to the construction of homology and cohomology of general spaces has accumulated. One also encounters some which lead to different results (in identical categories). It would seem, since there are many logically possible paths of generalization this is completely natural. This concerns homology theory especially in contrast with cohomology theory exhibits its own kind of "stability" in relation to the many possible initial definitions (different definitions as a rule lead to the same theory), homology usually reacts sensitively to logical nuances in the original definitions and it may seem that almost any new definition leads to a new theory. The reason for the difficulties observed in constructing homology is often assumed to be the inexactness of the inverse limit functor due to which the Čech homology $\check{H}_n(X)$ for example, differs from the groups $H_n(\check{C}_*)$, where $\check{C}_*$ is the inverse limit (if it exists) of the chains of nerves of coverings of X from a refinement of the system. In the case of noncompact X however one should in addition consider the difference of the groups $\check{H}_n(X)$, $H_n(\check{C}_*)$, respectively, from $\check{H}_n^c(X)$ and $H_n(\check{C}_*^c)$, where $\check{H}_n^c$ and $\check{C}_*^c$ are the direct limits of the groups $\check{H}_n$ and $\check{C}_*$ corresponding to all compact subspaces $C \subset X$ (Čech homology and Čech type chains with compact supports). In defining homology by chains of the type $\check{C}_*$ the result may also change in considering the topology of the inverse limit which arises in such chain complexes. Cf. also the remarks in Sec. 4 of Chap. 3 and Sec. 7 of Chap. 3.

But despite the large variety of approaches corresponding to the goals set and the various categories of topological spaces, the idea held up to the sixties that beyond the limits of the categories of polyhedra there exists a set of mutually inequivalent homology and cohomology theories is erroneous. One of the most essential results of the development in this area was the isolation (for a set of methods of description) of actually one natural homology theory and one single cohomology theory. In the case of cohomology this is the theory of sheaf cohomology (including cohomology with respect to different families of supports), and its uniqueness is one of the congruences of the apparatus of sheaf theory used there. In the case of homology this is the theory with compact (and also closed locally compact) supports where there are also many arguments in favor of its uniqueness. It is precisely these theories which occur in any applications and the practice is the test of their truth.

As to other theories, for all variants without exception, their development in the best case consists of verifying the Eilenberg–Steenrod axioms or modifications of them (usually this ends it). A tendency not infrequently observed on the basis of the new definitions given (isolated from the others) is to verify essentially the same properties included in the axioms, slightly adjusted; the theories which arise in this way are either equivalent to the accepted one or differing from it do not have the prospect of independent development, turning out to be homology (or cohomology) theories only in name. One should consider moreover that the properties included in the Eilenberg–Steenrod axioms determine the theory uniquely only on the category of finite polyhedra, and beyond its limits (even in categories of polyhedra), along with homology and cohomology quite random functors rather far from homology and cohomology may satisfy them (cf. [112]). Due to this experiment, rather than ending the new theory with the verification of the axioms discussed slightly adjusted, it is better to start with this since only the minimum of the requirements imposed on the theory are contained in the axioms (cf. point 5.3 of Chap. 3). Without regard to this the homology (rare cohomology) devised in general categories of topological spaces may turn out to differ from the usual ones in subcategories of polyhedra, but this contradicts common sense since from the very initial definition of homology and cohomology of polyhedra, these groups are defined by the boundary formula for a simplex.

The apparatus of sheaf theory which arose initially as the natural language for describing cohomology, opens a broad possibility of applying methods of homological algebra, completely changed the face of classical cohomology theory and immeasurably enriched its contents. Actually a new theory arose in relation to which the former one is only one of the concrete manifestations of the new one which united all such manifestations in a single scheme, which (in its turn!) turned out to be one of the most typical examples of a still more general construction from homological algebra, one of four possible versions of the construction of product functors. Since then however as a glance at the way cohomology was finally assembled in the fifties, on the one hand, in scientific journals there was found a set of simplifications, and many ideas which at first glance seemed rather complicated have been clarified, and on the other, due to insufficiently broad clarification of all the ideas and constructions used in the literature of educational character, some of them have become forgotten and sometimes even lost.* The primary value of the present survey is the clarification of all the basic ideas and an account of the simplified approach to the construction of sheaf cohomology.

In recent decades the values of the language of sheaf theory for homology also and not just for the description of chains has been discovered. As to the chains, the fact that they can be combined into flabby sheaves plays a large role. This in particular lets us exhibit the duality between the parallel basic constructions used in describing homology and cohomology which is clearly compatible with all the Poincaré duality relations. And although in its development homology theory yields somewhat to cohomology theory, homology entered firmly into the treatment, occupying the corresponding place in many applications. Both theories together, elucidating many constructions previously known from different points of view not infrequently leads to results and connections which turn out to be new, even in the classical categories of polyhedra.

CHAPTER 1

COHOMOLOGY WITH COEFFICIENTS IN A SHEAF

Reflecting only one aspect of the subject, namely the possibility of using cohomology with systems of coefficients of quite general type, this title may not be entirely accurate. For example, it gave rise to at most unsuccessful attempts to define homology with coefficients in sheaves also. It is helpful however in representing the far more important other side, the necessity and naturality of the language of sheaf theory both in the description of cochains and for the construction, while also developing numerous applications in general of cohomology theory itself. This language revealed unlimited possibilities for the application to cohomology theory of all the tools of modern homological algebra. Among the most important achievements reached by the methods of homological algebra one should indubitably cite the interpretation of the cohomology of a space as the derived functors of the zero-dimensional cohomology functor (in the category of sheaves on this space). This interpretation which determines the nature of the cohomology uniquely lets us in particular confidently separate from it constructions externally similar to cohomology and clearly outline the limits within which the constructions used nevertheless lead to cohomology groups.

1. Sheaves and Presheaves

1.1 A presheaf A on a topological space X with values in the category of Abelian groups or modules is a contravariant functor from the category of open subsets $U \subset X$ and their inclusion maps into the category of Abelian groups (or modules). By a homomorphism $A \to B$ of presheaves on X is meant an ordinary transformation of functors.

If the values A(U) of the presheaf A are any type of function, then A obviously satisfies the following conditions:

(S1) if the restrictions of $\xi \in A(U)$ to some $U_\lambda \subset U$ for which $U = \bigcup_\lambda U_\lambda$ are equal to zero, then $\xi = 0$ also.

(S2) If $\xi_\lambda \in A(U_\lambda)$ are such that the restrictions of any pair ξ_λ, ξ_μ to $U_\lambda \cap U_\mu$ coincide, then on $U = \bigcup_\lambda U_\lambda$ there exists an element $\xi \in A(U)$ such that its restrictions to all U_λ coincide with ξ_λ.

*This is testified to, for example, by the recently published book [110] which is devoted to sheaf cohomology. In it, sheaves are, rather than a given mode, the natural necessary language of cohomology theory. The author in partiuclar even "knew" to manage without the basic construction of a sheaf as a covering space, restricting himself to satisfying the standard formal requirements of being a contravariant functor on the category of open subsets of a topological space. True to "succeed" in this is to make some valuable improvements. In particular, the definition of the cohomology of a subspace (not even suitable for all subspaces) turned out to be laborious. The cohomology of a pair turned out to be a not quite finished concept (cf. point 9 in Chap. II and point 8 in Chap. IV of [110]).

Clearly in general neither of these conditions holds [for example, for the presheaves whose values are the cohomology $H^n(U)$ or the homology of the pair $H_n(X, X \setminus U)$].

Typical examples of presheaves are defined by cochains of any concrete type. Since cochains are usually functions with values in the coefficient groups defined not at points x but on certain objects located in the topological space X (singular simplices, collections of points, intersections of elements of coverings, collections of tangent vectors, etc.), presheaves of cochains (to open sets $U \subset X$ one associates functions on objects in U) always satisfy (S2) obviously. Of course in general (S1) fails since a concrete cochain may be nonzero only on "large" objects (ones not contained in elements U_λ of a sufficiently fine covering).

1.2. A <u>sheaf</u> of Abelian groups (or modules) on X is a topological space $\mathscr{A}$ (generally non-Hausdorff) together with a locally homeomorphic projection $\pi:\mathscr{A}\to X$ whose stalks $\mathscr{A}_x=\pi^{-1}(x)$, $x\in X$ are Abelian groups (or modules) with algebraic operations which are continuous (in the topology of all of $\mathscr{A}$). By a homomorphism of sheaves $\mathscr{A}\to\mathscr{B}$ is meant any continuous map which is compatible with the projections to X (which automatically turns out to be a local homeomorphism) whose restriction to a stalk is a homomorphism $\mathscr{A}_x\to\mathscr{B}_x$.

A continuous map $s:N\to\mathscr{A}$ of an arbitrary subset $N \subset X$ for which $s(x)\in\mathscr{A}_x$ is called a section of the sheaf $\mathscr{A}$ on N. Over any set the zero section formed by the zeros of the stalks $\mathscr{A}_x$ is always defined. The existence of nontrivial sections over sufficiently small open sets is guaranteed by the fact that π is a local homeomorphism. The groups (or modules) $\mathscr{A}(U)$ consisting of all sections of $\mathscr{A}$ on U obviously form a presheaf satisfying (S1) and (S2), the presheaf of sections of the sheaf $\mathscr{A}$. Conversely, for any presheaf A it is easy to construct a sheaf $\mathscr{A}=r(A)$ corresponding to it. Its stalks at points $x\in X$ are the direct limits $\mathscr{A}_x=\varinjlim_{x\in U} A(U)$ and the topology is defined by all sets U_a consisting of images of $a\in A(U)$ upon passing to the direct limits in those fibers $\mathscr{A}_x$ for which $x\in U$. One says that the sheaf $\mathscr{A}$ is generated by the presheaf A. The functor r which associates with the presheaf A the sheaf $\mathscr{A}=r(A)$ is sometimes called the reflector. Like the direct limit which participates in its definition, this functor is exact (short exact sequences remain exact).

Obviously there is a natural transformation of functors $r:A(U)\to\mathscr{A}(U)$. It is an equivalence of functors if and only if the presheaf A satisfies (S1) and (S2), cf. Sec. 1.2 of Chap. 2 of [107]. Since (S1) does not always hold for presheaves of cochains, it is useful to note that the one condition (S2) guarantees that $A(U)\to\mathscr{A}(U)$ is an epimorphism for any paracompact $U \subset X$, Sec. 3.9 of Chap. 2 of [107]. Analogously, if X is paracompact, it follows from (S2) that any section s defined on a closed set $Y \subset X$ (and if X is hereditarily paracompact, then also on any Y), is the restriction to Y of a section of the form $r(a)$, $a\in A(U)$, U being a neighborhood of Y.

<u>1.3. Remark.</u> The tendency some times observed (as in [110]) to identify sheaves with presheaves satisfying (S1) and (S2) is not justified. Under this approach in particular difficulties even arise in defining a quotient sheaf (consisting of the lostness property in the requirements of (S1) and (S2): an exact sequence of sheaves in the usual sense stops being exact upon passage to the presheaf of sections). In a sheaf of cochains the natural clarity of any constructions in which one must consider supports of cochains (considered as sections of sheaves) is lost. The loss of the possibility of considering restrictions of sheaves to subspaces also leads to unreasonable difficulties in describing the cohomology of subsets, to artificial methods of definition of the cohomology of a pair (using unnecessary restrictions in addition).

2. What is Sheaf Cohomology?

<u>2.1. Sheaves of Cochains.</u> Sheaves $\mathscr{A}^n$ generated by presheaves of any concrete cochains by means of the coboundary operator d for these cochains are combined in a differential graded sheaf $\mathscr{A}^*$, i.e., into a sequence of the form $\ldots\to\mathscr{A}^{n-1}\xrightarrow[d]{}\mathscr{A}^n\xrightarrow[d]{}\mathscr{A}^{n+1}\to\ldots$ in which clearly $d^2 = dd = 0$. Under typical restrictions on X usually depending on the form of the cochains used (such a restriction often turns out to be the requirement of X being paracompact), the map $r:A^*(X)\to\mathscr{A}^*(X)$ of the complex of cochains into the complex of sections is either an isomorphism or induces an isomorphism in cohomology. Thus, in typical cases the cohomology of X coincides with the cohomology $H^n(\mathscr{A}^*(X))$ of the complex of sections of $\mathscr{A}^*$.

2.2. Soft and Flabby Sheaves. The properties of the constituents $\mathscr{A}^n$ of the differential sheaf of cochains $\mathscr{A}^*$ are determined by the following simple observations. The concrete cochains used in defining cohomology usually have the following obvious property: any cochain defined on an arbitrary subset of the space X extends to a cochain of X. In correspondence with point 1.2 this means that if X is a paracompact space, the sheaves of cochains satisfy the following condition: any section defined on a closed subset $Y \subset X$ extends to a section on all of X. Sheaves having this property are called soft. When X is hereditarily paracompact, the sections of sheaves of cochains also extend to the whole space from any subsets. Sheaves (on some space X) having this property for open $U \subset X$ are called flabby.

Thus, in typical cases arising in describing the cohomology of a topological space, sheaves of cochains turn out to be soft or flabby. In accord with point 1.2, on a paracompact space any flabby sheaf is soft. Analogously, in a hereditarily paracompact space the restrictions of flabby sheaves to any subspaces are flabby sheaves (Corollary 2 of Sec. 3.3 of Chap. 2 of [107]). It is clear that the restriction of a flabby sheaf to any open set is flabby and the restriction of a soft sheaf to any closed set is a soft sheaf.

2.3. Resolutions. Since the sections of sheaves $\mathscr{A}^*$ defining the cohomology of a space X play the role of cochains it is natural to expect that their restrictions to subspaces (at least sufficiently "good" ones) should give the cohomology of these subspaces. According to the properties of soft and flabby sheaves discussed in the case of a point $x \in X$ (we assume that points are closed sets) the restrictions of sections to x is simply the stalk $\mathscr{A}_x^*$ of the differential sheaf $\mathscr{A}^*$. Since the cohomology of a point is trivial, this means that the following sequence of sheaves is exact:

$$0 \to \mathscr{G} \to \mathscr{A}^0 \to \mathscr{A}^1 \to \ldots \to \mathscr{A}^n \to \ldots . \tag{1}$$

Here $\mathscr{G}$ is simply the kernel of $d: \mathscr{A}^0 \to \mathscr{A}^1$ for now. In the case of cohomology with coefficients in G this is the constant sheaf $X \times G$ (cf. below) usually denoted simply by G.

Any exact sequence of the form (1) is called a resolution of the sheaf $\mathscr{G}$.

The exactness of the sequence $\mathscr{A}^*$ at terms of positive dimension means that each cocycle is locally cohomologous to zero: if $d\xi = 0$, $\xi \in A^p(U)$ for the element $a_x^p \in \mathscr{A}_x^p$ defined by ξ at the point $x \in U$ we have $da_x^p = 0$ so $a_x^p = da_x^{p-1}$ for some $a_x^{p-1} \in \mathscr{A}_x^{p-1}$ and in a sufficiently small neighborhood $V \subset U$ of this point $\xi|_V = d\eta$, where $\eta \in A^{p-1}(V)$ defines a_x^{p-1}. This condition is very natural and confirms the intuitively obvious idea that any cohomology class h of positive dimension must occupy some "volume" in the space X in the sense that for each point one can find a small enough neighborhood U_x such that the restriction of h to U_x is equal to zero. The condition is equivalent to the requirement that the sequence (1) be exact. It means that at each point $x \in X$ for $p > 0$ the relations $\varinjlim_{x \in U} H^p(U) = 0$ hold and since $\mathscr{A}_x^* = \varinjlim A^*(U)$ and the exact functor $\varinjlim$ commutes with passage to homology, this means that $\mathscr{A}_x^*$ is acyclic is positive dimensions, i.e., that the sequence $\mathscr{A}^*$ is exact.

2.4. The zero-dimensional cohomology group $H^0(X; G)$ with coefficients in G was realized above as the group of sections of the kernel $\mathscr{K}$ of the homomorphism $d: \mathscr{A}^0 \to \mathscr{A}^1$ having stalks isomorphic to G, and since each point $x \in X$ is a retract of X, it can be represented as a direct sum $H^0(X; G) = G \oplus \Gamma_x'$, where Γ_x' is the kernel of the restriction of $H^0(X; G)$ to x [and the first direct summand is the image of $G = H^0(x; G)$ under the retraction $X \to x$ and this image is independent of x]. Elements $g \in G \subset H^0(X; G)$ considered as sections of $\mathscr{K}$ obviously give, under restriction to $\mathscr{K}$ the same element g of the stalk G of the sheaf $x \in X$, i.e., are constant sections of it. But this means that $\mathscr{K} = X \times G$.

One can show analogously that for cohomology with coefficients in any locally constant system $\mathscr{G}$ the kernel $d: \mathscr{A}^0 \to \mathscr{A}^1$ turns out to be $\mathscr{G}$ itself. Hence in the more general situations when resolutions of the form (1) (with soft or flabby sheaves $\mathscr{A}^p$) arise, the sheaf is called the sheaf of coefficients. The treatment given of zero-dimensional groups can also be followed well on any concrete construction of cohomology, cf. Secs. 2, 3, 4, and 7 of Chap. 3.

2.5. Basic Idea. Thus, in natural situations the differential graded sheaf $\mathscr{A}^*$ whose sections define the cohomology of X is a resolution of the coefficient sheaf $\mathscr{G}$ consisting of soft or flabby sheaves. The following assertion is a basic fact: for all resolutions of the sheaf $\mathscr{G}$ independently of their nature and of how they arise, the cohomology $\mathscr{A}^*(X)$ of the sections are canonically isomorphic to one another if the constituents of the resolution of the sheaf $\mathscr{A}^p$ are soft or flabby (more generally, acyclic, cf. below). A consequence of this is that the cohomology $H^p(X;\mathscr{G})$ of the topological space X with coefficients in the sheaf $\mathscr{G}$ is well-defined as the cohomology $H^p(\mathscr{A}^*(X))$ of the complex of sections on X of any resolution $\mathscr{A}^*$ of the sheaf $\mathscr{G}$ consisting of sheaves of the type indicated.

The set of assertions given in the entire paper below is the fundamental value of this interpretation of cohomology. But even from what has been said it is clear that first of all defines a theory uniquely (the simple axiomatic description of cohomology as a functor of the argument $\mathscr{G}$ also speaks of this, cf. point 3.3 of Chap. 3 in [69] and below point 4.6). In fact, although concrete cochains (such as singular) are occasionally defined for any topological spaces without any restrictions, the cohomology they describe is usually used under some conditions or other. In all exceptional cases such conditions (usually describing a category of topological spaces most natural for applications of the given theory) are exhibited upon comparing the cohomology considered with the sheaf cohomology. In order to formulate the conditions discussed it is necessary to answer two concrete questions: a) when do the sheaves of cochains which arise form a resolution? (local conditions); b) are the sheaves which arise soft, flabby, or simply acyclic? (global conditions). These determine the limits within which the concrete cohomology theory given coincides with the sheaf theory. We note that in a number of typical cases (for example, for Cech type cochains, in Alexander–Spanier cohomology, for de Rham cohomology, etc.) the answer to a) is positive without any restrictions. Not infrequently however this leads to very essential restrictions (for example, in the case of singular theory). Conditions guaranteeing b) are usually paracompactness requirements. Cf. also the end of point 4.6 below.

In any examples of "anomalies" in the behavior of any cohomology such phenomena are explained by the deviation of this cohomology such phenomena are explained by the deviation of this cohomology from the sheaf cohomology. Conversely, upon the coincidence of concrete cohomology and sheaf cohomology no claims are imposed on their behavior.

2.6. Examples. It happens that resolutions composed of acyclic sheaves are defined by objects outwardly far from any cochains. Similar situations arise in problems of differential calculus, in functional and complex analysis, in algebraic geometry (and in other areas). Since the sections of these resolutions defining cohomology coincide as a rule with the objects considered themselves, in all such cases the influence on the solvability of such problems appears through the cohomology, outwardly not connected in any way with the topology of the topological structure of the ambient space.

A typical example is the differential sheaf Ω^* of germs of differential forms on a smooth manifold with the operation d of exterior differentiation being (by virtue of the famous Poincaré lemma) a resolution of the constant sheaf R (R being the field of real numbers). This resolution is φ -acyclic for any paracompactifying family of supports φ (cf. Sec. 3 below) and hence defines cohomology (including that with supports in the φ indicated) of the manifold with coefficients in R (de Rham's theorem). Thus the question of the solvability for a closed k-form $\omega\in\Omega^k$ (i.e., for $d\omega = 0$) of the equation $\omega = d\tilde{\omega}$ depends only on the topological structure of the manifold (or domain in which the form ω is defined). We note that in terms of differential forms one can express concrete tensors in geometry, mechanics, physics, and in terms of the operation d, some differential equations.

Analogs of de Rham's theorem hold for complex-valued differential forms on a complex manifold X. Each form $\omega\in\mathscr{E}^r(X)$ of degree r of the full algebra $\mathscr{E}^*(X)=\bigoplus_r\mathscr{E}^r(X)$ of exterior complex-valued differential forms can be written locally in the form of linear combinations of forms of the form $dz^{i_1}\wedge\ldots\wedge dz^{i_p}\wedge d\bar{z}^{j_1}\wedge\ldots\wedge d\bar{z}^{j_q}$; $p+q=r$. so $\mathscr{E}^r(X)$ is the direct sum $\bigoplus_{p+q=r}\mathscr{E}^{p,q}(X)$ of differential forms of bidegree (p, q) and the exterior derivative $d:\mathscr{E}^r(X)\to\mathscr{E}^{r+1}(X)$ can be represented in homogeneous terms as the direct sum $d=\partial\oplus\bar{\partial}$ of differentiations $\partial:\mathscr{E}^{p,q}(X)\to\mathscr{E}^{p+1,q}(X)$, $\bar{\partial}:\mathscr{E}^{p,q}(X)\to\mathscr{E}^{p,q+1}(X)$. For fixed $p\geqslant 0$ (on an n-dimensional complex manifold X) there arises a differential sheaf $0\to\Omega^p\to\mathscr{E}^{p,0}\to\mathscr{E}^{p,1}\to\ldots\to\mathscr{E}^{p,n}\to 0$, where Ω^p is the sheaf of holomorphic forms of type (p, 0) called holomorphic forms of degree p), while $\mathscr{E}^{p,q}(X)$ are the sheaves $\mathscr{E}^{p,q}$. By virtue of Grothendieck's analog for $\bar{\partial}$ of Poincaré's lemma the differential sheaf $\mathscr{E}^{p,*}$ is a resolution (of Dolbeault–Grothendieck) of the sheaf Ω^p and since (as in de Rham's theorem) it consists of fine sheaves, for any paracompactifying family φ of supports $H^q(\Gamma_\varphi(\mathscr{E}^{p,*}))$† is the cohomology of X with coefficients in the sheaf Ω^p and with

†$\Gamma_\varphi(\mathscr{A})$ is the sections of the sheaf $\mathscr{A}$ with support in φ.

supports in Φ (Dolbeault's theorem [147]).

There are analogs of de Rham's and Dolbeault's theorems for differential forms with values in vector bundles and, respectively, in holomorphic vector bundles over manifolds. Resolutions of Dolbault–Grothendieck type for sheaves Ω^p are also constructed in the case of forms of bidegree (p, q), in whose decompositions analytic functions on $R^{2n}(=C^n)$ distributions in the sense of Schwartz [109], and also hyperfunctions in the sense of Sato (cf. [139]) can figure as coefficients. The latter lies at the base of the proof of Sato's theorem giving one of the basic interpretations of hyperfunctions: for an open set $U\subset R^n\subset C^n$ the cohomology $H^p_U(C^n;\mathcal{O})$ (with supports situated in U and with coefficients in the sheaf $\mathcal{O}$ of germs of holomorphic functions on C^n) is equal to zero for p ≠ n and the presheaf U → $H^n_U(C^n;\mathcal{O})$ coincides with the (flabby) sheaf of hyperfunctions on R^n (cf. [139]).

2.7. Some properties of cohomology following directly from the definition. The obvious property of Π-additivity of sheaf cohomology: $H^n(X;\mathcal{G})$ is the direct product of $H^n(X_\lambda;\mathcal{G}_\lambda)$ if $X=\bigcup\limits_\lambda X_\lambda$, where X_λ are pairwise disjoint open–closed subspaces of X and $\mathcal{G}_\lambda$ is the restriction of $\mathcal{G}$ to X_λ. The analogous property of Σ-additivity of cohomology $H^p_c(X;\mathcal{G})$ with compact supports: in the situation indicated $H^p_c(X;\mathcal{G})$ is the direct sum of $H^p_c(X_\lambda;\mathcal{G}_\lambda)$. Despite all their obviousness these properties play a very important role in questions of axiomatics (cf. point 5.3 of Chap. 3).

The possibility of extending a section from closed subsets Y of a paracompact space X to neighborhoods of them, and in the case of a hereditarily paracompact X also from any subsets Y ⊂ X lets us (by virtue of the acyclicity of the restrictions to such Y of flabby or soft sheaves, cf. point 2.2) get the cohomology of Y starting from the restrictions to Y of sections of $\mathcal{A}^*$, and the cohomology of the pair (X, Y) starting from the kernel of the restriction of $\mathcal{A}^*(X)$ to Y, interpreting them as the cohomology of X defined by all cochains whose supports do not touch Y (i.e., lie in X \ Y). Moreover, for the indicated Y, $H^p(Y;\mathcal{G})=\varinjlim H^p(U;\mathcal{G})$, where U are all neighborhoods of Y in X (rigidity property of cohomology). Cf. also Sec. 5.

3. Cohomology as Derived Functor. Comparison Homomorphism. Supports

3.1. Derived Functors. For cohomology with constant or locally constant coefficients the existence of the resolutions indicated in Sec. 2 is a corollary of the description of cohomology by means of some concrete cochains. The existence of similar resolutions for an arbitrary sheaf $\mathcal{G}$ is a fact established in the theory of sheaves, first of all in the category of sheaves of Abelian groups (or modules). In this category (or an arbitrary topological space) there are sufficiently many injective objects: each sheaf $\mathcal{G}$ can be imbedded in an injective $\mathcal{J}$ (cf. [87, 107, 108]). Setting $\mathcal{J}^0=\mathcal{J}$, taking an injective sheaf containing the quotient $\mathcal{J}^1$ as $\mathcal{J}^0/\mathcal{G}$ and continuing this process in the obvious way, we get an injective resolution $\mathcal{J}^*$ of the sheaf $\mathcal{G}$. Since injective sheaves are always flabby (Sec. 5 of Chap. 2 of [87] or Sec. 7.1 of Chap. 2 of [107]), the sections of $\mathcal{J}^*$ define the cohomology $H^n(X;\mathcal{G})$. This is the interpretation of cohomology as the derived functors of the zero-dimensional cohomology functor.

We recall that a covariant functor F is called left exact if for anv short exact sequence of objects of the base category $0\to\mathcal{G}'\to\mathcal{G}\to\mathcal{G}''\to 0$ the sequence $0\to F(\mathcal{G}')\to F(\mathcal{G})\to F(\mathcal{G}'')$ is exact. The functor F is called additive if t the difference of any morphisms $f,g:\mathcal{A}\to\mathcal{B}$, it associates the difference $f_* - g_* = (f-g)$ of the corresponding homomorphisms $f_* = F(f)$, $g_* = F(g)$. For such F (under the condition that in the base category there is a sufficient supply of injective objects) there are defined right derived functors $F^n: F^n(\mathcal{G})=H^n(F(\mathcal{J}^*))$, where $\mathcal{J}^*$ is an arbitrary injective resolution of the object $\mathcal{G}$.

In our case, F is the functor of sections, usually denoted by $\Gamma:\Gamma(\mathcal{G})=\mathcal{G}(X)$ is the group (or module) of sections of the sheaf $\mathcal{G}$ on X. The left exactness and additivity of Γ are obvious. In accord with Sec. 2, $\Gamma(\mathcal{G})=H^0(X;\mathcal{G})$. Thus, in the category of sheaves on X the cohomology groups $H^n(X;\mathcal{G})$ are the right derived functors of the left exact covariant additive functor of sections Γ, the zero-dimensional cohomology functor. Cohomology groups with coefficients in sheaves are one of the most typical examples of derived functors, all the basic ideas of the theory of derived functors can be traced most completely, simply, and accurately on them.

3.2. Comparison Homomorphism. If $\mathcal{L}^*$ is an arbitrary resolution of the object $\mathcal{B}$ then any morphism $\mathcal{B}\to\mathcal{G}$ as is known extends to a morphism of resolutions $\mathcal{L}^*\to\mathcal{J}^*$ defined up to homotopy (morphisms $f, g: \mathscr{L}^* \to \mathscr{I}^*$ are called homotopic if $f - g = Dd + dD$, where $D: \mathscr{L}^n \to \mathscr{I}^{n-1}$ are morphisms defined for all $n > 0$ they are called homotopies between f and g). For $\mathscr{B} = \mathscr{G}$ in particular there is always a natural transformation $\gamma: H^n(F(\mathscr{L}^*)) \to F^n(\mathscr{G})$ called the comparison homomorphism (one compares with the derived functors on the object $\mathscr{G}$ the cohomology groups defined by a random resolution $\mathscr{L}^*$ of the object $\mathscr{G}$ when $\mathscr{L}^*$ also consists of injective objects, γ is a natural isomorphism). This homomorphism also arises in a number of other typical situations (cf. below).

In particular, for any resolution $\mathscr{L}^*$ of the sheaf $\mathscr{G}$ there is defined a natural comparison homomorphism $\gamma: H^n(\Gamma(\mathscr{L}^*)) \to H^n(X; \mathscr{G})$.

3.3. Cohomology with Supports. By the support of the cochain ξ is meant the set $|\xi| \subset X$ closed by the definition itself, and such that its complement consists of all points $x \in X$ with the following property: the restrictions of ξ to sufficiently small neighborhoods of x are equal to zero. When the cochain is represented as a section of the sheaf $\mathscr{A}$, $|\xi|$ is the set of all $x \in X$ at which this section is different from zero, the support of the section. Since the projection $\pi: \mathscr{A} \to X$ is a local homeomorphism, and all the zeros in $\mathscr{A}$ form a section on X, if the section ξ vanishes at a point, then it is also equal to zero in a neighborhood of this point.

There is a set of natural situations when along with ordinary cohomology one must consider cohomology with some special supports. Thus, despite the fact that they reduce to ordinary ones under passage to one point compactifications, in the category of locally compact spaces and their proper maps the cohomology groups with compact supports are of independent interest. As noted in point 2.7, in suitable cases the cohomology groups of the pair (X, Y) coincide with the cohomology groups of X with supports located in $X \setminus Y$.

By a family of supports φ in the space X is meant any system of closed sets satisfying the following conditions: a) if $F_1, F_2 \in \varphi$ then $F_1 \cup F_2 \in \varphi$; b) $F' \in \varphi$ if $F' \subset D$ and $F \in \varphi$. A family of supports is called paracompactifying if all $F \in \varphi$ are paracompact and for any $F \in \varphi$ one can find an $F' \in \varphi$ such that $F \subset \operatorname{Int} F'$. As examples of paracompactifying families one has the family of compact subspaces of a locally compact space, the family of closed subsets contained in $X \setminus Y$ in the case of a closed subset Y of the paracompact space X.

Just as usual one defines the cohomology $H^n_\varphi(X; \mathscr{G})$ of the space X with supports in the family φ (and with coefficients in the sheaf $\mathscr{G}$). Here one actually uses the subfunctor Γ_φ of the functor $\Gamma(\mathscr{G})$ of the argument $\mathscr{G}$ consisting of all sections of sheaves $\mathscr{G}$ with supports in φ which, like Γ, is left exact and additive. Hence, like $H^n(X; \mathscr{G})$ the cohomology groups $H^n_\varphi(X; \mathscr{G})$ coincide in the category of sheaves on X, with the right derived functors of the functor Γ_φ. Just as above, $\Gamma_\varphi(\mathscr{G}) = H^0_\varphi(X; \mathscr{G})$ and the comparison homomorphism $\gamma: H^n(\Gamma_\varphi(\mathscr{L}^*)) \to H^n_\varphi(X; \mathscr{G})$ is defined.

4. Other Typical Methods of Comparison

4.1. Iterated Connecting Homomorphism. Let $\mathscr{Z}^k$ be the kernel of the morphism $d: \mathscr{L}^k \to \mathscr{L}^{k+1}$ in the resolution $\mathscr{L}^*$ of an object $\mathscr{G}$. Short exact sequences $0 \to \mathscr{Z}^k \to \mathscr{L}^k \to \mathscr{Z}^{k+1} \to 0$ define connecting homomorphisms $\delta: F^{n-k-1}(\mathscr{Z}^{k+1}) \to F^{n-k}(\mathscr{Z}^k)$ of derived functors whose composition together with the analogous homomorphism $H^n(F(\mathscr{L}^*)) \to F^1(\mathscr{Z}^{n-1})$ corresponding to the case $k = n - 1$, defines the iterated connecting homomorphism $\Delta: H^n(F(\mathscr{L}^*)) \to F^n(\mathscr{G})$ (we assume that $\mathscr{Z}^0 = \mathscr{G}$). Using a scheme of arguments analogous to the construction from Sec. 7 of Chap. 5 in [92] (adapted there to a different version of derived functors), by induction on n one can establish that the comparison theorem γ coincides with $(-1)^{\frac{n(n+1)}{2}} \Delta$ (these homomorphisms are erroneously identified by some authors, cf., e.g., Sec. 4.7 of Chap. 2 of [107], Sec. 4 of Chap. 2 of [87]).

4.2. Acyclic Objects (Sheaves). Sheaves of cochains which arise in practice do not happen to be injective. In connection with this it is natural to know which resolutions one can use for defining cohomology, analogously for calculating derived functors.

In relation to the functor F, an object $\mathscr{L}$ is said to be F-acyclic if $F^n(\mathscr{L}) = 0$ for $n > 0$. Analogously, the sheaf $\mathscr{L}$ is called acyclic if $H^n(X; \mathscr{L}) = 0$ for all $n > 0$ and, respectively, φ-acyclic if $H^n_\varphi(X; \mathscr{G})$ is zero. Examples of φ-acyclic sheaves (for any family φ) are all flabby sheaves (Sec. 3.1 of Chap. 2 of [107]). One also establishes the acyclicity of soft sheaves on a paracompact Hausdorff space straightforwardly (Sec. 3.5 of Chap. 2 of [107]).

The construction of the preceding point discloses the role of acyclic objects: for any resolution $\mathscr{L}^*$ consisting of F-acyclic objects $\mathscr{L}^p$ all δ and hence also their composition Δ are obviously isomorphisms. Consequently (since γ is an isomorphism for them) any such resolutions of the object $\mathscr{G}$ define $F^n(\mathscr{G})$. This is the precise meaning of the basic assertion formulated in point 2.5 (for $F = \Delta$ or Γ_φ). It is clear from the construction of $\Delta = \pm\gamma$ that the requirement of F-acyclicity is far from accidental (it is necessary, for example, in order that the connecting homomorphisms δ which are the constituents of Δ be isomorphisms).

4.3. Godement's Canonical Resolution. Godement's canonical resolution occupies a special place in the theory of sheaf cohomology. In some situations it is practically irreplaceable (cf., e.g., Sec. 5).

For an arbitrary sheaf $\mathscr{G}$ let $\mathscr{C}^0(\mathscr{G})$ be the sheaf of germs of arbitrary sections (including any discontinuous ones) of the sheaf $\mathscr{G}$, $\mathscr{C}^1(\mathscr{G})$ be the sheaf $\mathscr{C}^0(\mathscr{C}^0(\mathscr{G})/\mathscr{G})$, etc. There arises a flabby resolution $\mathscr{C}^*(\mathscr{G})$ which turns out to be along with the complexes $\Gamma_\varphi(\mathscr{C}^*(\mathscr{G}))$ (for any families φ) an exact functor of the argument $\mathscr{G}$. This makes it possible almost automatically to get all the usual cohomology properties (functoriality with respect to $\mathscr{G}$, various spectral sequences, etc.). In particular, to any short exact sequence of sheaves of coefficients $0\to\mathscr{G}'\to\mathscr{G}\to\mathscr{G}''\to 0$ obviously corresponds a cohomology exact sequence

$$\ldots\to H^n_\varphi(X;\mathscr{G}')\to H^n_\varphi(X;\mathscr{G})\to H^n_\varphi(X;\mathscr{G}'')\to H^{n+1}_\varphi(X;\mathscr{G}')\to\ldots \tag{2}$$

(φ being any family of supports). Thanks to its qualities the resolution can be used easily for independent description of the foundations of sheaf cohomology theory (cf. [107]).

4.4. Double Complexes. Using the functorial properties of the Godement resolution, $\mathscr{L}^*$ for any graded differential sheaf $\mathscr{C}^*(\mathscr{L}^*)=\{\mathscr{C}^p(\mathscr{L}^q)\}$ with total differential $d = d' + (-1)^p d''$ (here $d^2 = 0$), where d' and d'' are the homomorphisms induced, respectively, by the differential in $\mathscr{C}^*$ and the coboundary operator in $\mathscr{L}^*$. With the help of a simple diagram chase (of the type of the arguments in Lemma B. 32 in [139], for example) it is easy to see that when $\mathscr{L}^*$ is a resolution of $\mathscr{G}$ the inclusion of $\mathscr{C}^*(\mathscr{G})$ in $\mathscr{C}^*(\mathscr{L}^0)\subset\mathscr{C}^*(\mathscr{L}^*)$ lets one identify $H_\varphi^*(X;\mathscr{G})$ with the cohomology of the double complex $\Gamma_\varphi(\mathscr{C}^*(\mathscr{L}^*))$ considered in the total grading (the n-dimensional cochains are $\bigoplus_{p+q=n}\Gamma_\varphi(\mathscr{C}^p(\mathscr{L}^q))$). Since the map $\mathscr{L}^*\to\mathscr{J}^*$ defining γ extends to a map of bigraded differential sheaves $\mathscr{C}^*(\mathscr{L}^*)\to\mathscr{C}^*(\mathscr{J}^*)$ which is the identity on the differential graded sheaf $\mathscr{C}^*(\mathscr{G})$ the map of $H^n(\Gamma_\varphi(\mathscr{L}^*))$ which arises from the inclusion of $\mathscr{L}^*$ in $\mathscr{C}^0(\mathscr{L}^*)\subset\mathscr{C}^*(\mathscr{L}^*)$ (which is also an isomorphism for acyclic $\mathscr{L}^q$) coincides with the comparison homomorphism γ.

One can make analogous inferences based on the injective bigraded sheaf $\mathscr{J}^*(\mathscr{L}^*)$ [which however is constructed in a more complicated way than $\mathscr{C}^*(\mathscr{L}^*)$]. Bigraded differential sheaves of type $\mathscr{C}^*(\mathscr{L}^*)$ or $\mathscr{J}^*(\mathscr{L}^*)$ and the double complexes of sections corresponding to them are the source for all typical spectral sequences which arise in the homology and cohomology theories of topological spaces (cf. Sec. 3 of Chap. 3 of [69]; cf. also below the end of Chap. 2 and Sec. 5 of Chap. 4).

4.5. Defining Properties of the Homomorphism γ. The comparison homomorphism can also arise in many other concrete situations (for example, in spectral sequences), while it is not always simple to establish its identity with the original definition. Hence an axiomatic description of it is of interest. We give it in terms of abstract derived functors F^n.

We note that the transformation γ is the identity in dimension 0: If the morphism $f: \mathscr{G}\to\mathscr{F}$ extends to a morphism of resolutions $\tilde{f}: \mathscr{L}^*\to\mathscr{M}^*$ then $\gamma\tilde{f}_* = f_*\gamma$. Finally, if a short exact sequence of objects is completed to a short exact sequence of resolutions which also remain exact after application of the functor F, then $\gamma\tilde{\delta} = \delta\gamma$ ($\tilde{\delta}$ being the connecting homomorphism for the cohomology of the complexes obtained by application of the functor F to the resolutions). It turns out that there exists only one transformation $\gamma: H^n(F(\mathscr{L}^*))\to F^n(\mathscr{G})$ with these properties, cf. point 3.2 of Chap. 3 in [69] (here $\mathscr{L}^*$ is a resolution of the object $\mathscr{G}$].

4.6. Comparison with Cohomology Functors. One can compare with the derived functors $F^n(\mathscr{G})$ not only the cohomologies defined by different resolutions, but also any functors of cohomology type. Let $\{\Phi^n\}$, $n\geqslant 0$ be any connected sequence of cohomology functors (this means that just as for F^n any short exact sequence of arguments $0\to\mathscr{G}'\to\mathscr{G}\to\mathscr{G}''\to 0$ corresponds to an exact sequence

$$\ldots\to\Phi^n(\mathscr{G}')\to\Phi^n(\mathscr{G})\to\Phi^n(\mathscr{G}'')\xrightarrow[\delta_\Phi]{}\Phi^{n+1}(\mathscr{G}')\to\ldots,$$

and for a map f of the short exact sequence indicated into any other one, there is a corresponding map of the sequences of the cohomology functors satisfying the identity $f_*^{\Phi}\delta_{\Phi} = \delta_{\Phi} f_*^{\Phi}$). Under the assumption that $\Phi^0 = F$ there is a unique natural transformation $\mu: F^n(\mathcal{G}) \to \Phi^n(\mathcal{G})$ (universality property of the derived functors F^n) which is the identity in dimension 0 and is such that for any morphism $f: \mathcal{G} \to \mathcal{F}$ one has $f_*^{\Phi}\mu = \mu f_*$ and for any short exact sequence of arguments $\delta_{\Phi}\mu = \mu\delta$. The transformation μ is an isomorphism if and only if for $n > 0$ all the Φ^n vanish on injective objects (hence also on any F-acyclic ones). This is an axiomatic characterization of derived functors (in particular, of sheaf cohomology, cf., e.g., Sec. 4.8 of Chap. 2 of [107]) and point 3.3 of Chap. 3 of [69]).

Thus (cf. point 2.5 above), sheaf cohomology differs further by the remarkable property that any other (sufficiently "decent") cohomology theory can be compared with it (by means of μ). This is precisely what lets one get the simplest proof (under natural restrictions) of the coincidence of the Cech cohomology with $H_{\varphi}^*(X;\mathcal{G})$ (cf. Chap. 4 of [69]).

5. Cohomology of Spaces and Pairs. Rigidity and Excitation Properties

5.1. Subspaces. As noted above in point 2.7, under certain typical conditions, the restrictions $\mathcal{A}|_Y$ of acyclic sheaves $\mathcal{A}$ (in particular, the terms $\mathcal{A}^q$ of acyclic resolutions $\mathcal{A}^*$) to subspaces $Y \subset X$ are acyclic. One has this for flabby $\mathcal{A}$ and open Y, for soft $\mathcal{A}$ and closed Y, when X is paracompact, for flabby $\mathcal{A}$ and any Y when X is hereditarily paracompact (in particular metrizable), for soft $\mathcal{A}$ and locally closed (in particular for open) Y when X is hereditarily paracompact (cf. point 2.2 above and Sec. 3.4 of Chap. 2 in [107]). It is clear that for such $\mathcal{A}^*$ and Y the cohomology of Y with coefficients in $\mathcal{G}|_Y$ usually denoted simply by $H^*(Y;\mathcal{G})$ can be obtained starting from the resolution $\mathcal{A}^*|_Y$ of the sheaves $\mathcal{G}|_Y$. Moreover, the restrictions of sections of $\mathcal{A}^*(X)$ to Y define natural homomorphisms $H^n(X;\mathcal{G}) \to H^n(Y;\mathcal{G})$. For cohomology $H_{\varphi}^n(X;\mathcal{G})$ with supports in the family φ it is natural to compare it with $H^n_{\varphi\cap Y}(Y;\mathcal{G})$, where $\varphi\cap Y$ is the intersection of the family φ with Y (consisting of all $F\cap Y$, $F\in\varphi$). Analogous conclusions are valid when the restriction to Y of some φ-acyclic resolution $\mathcal{A}^*$ of the sheaf $\mathcal{G}$ on X is $(\varphi\cap Y)$-acyclic. For flabby $\mathcal{A}^*$ this is so in each of the following ases (cf. Sec. 10 of Chap. 2 of [87]): a) for open Y and any φ; b) for paracompactifying families φ if each $F\cap Y$, $F\in\varphi$ has a fundamental system of paracompact neighborhoods in X; c) for any φ if X is hereditarily paracompact (since the sheaves $\mathcal{A}^*|_Y$ are flabby); d) for paracompactifying φ if Y is closed in X; e) for the family of all closed subsets of X if Y is a compact subspace of an arbitrary Hausdorff space X.

In general the restriction of an acyclic sheaf $\mathcal{A}$ to a subspace may not be acyclic. For example, the constant sheaf G is acyclic on any space X which has trivial cohomology (with coefficients in G) but X may contain a subspace with nontrivial cohomology in some positive dimensions. However the restriction homomorphisms $i^*: H_{\varphi}^n(X;\mathcal{G}) \to H^n_{\varphi\cap Y}(Y;\mathcal{G})$ induced by the inclusion $i: Y \subset X$ can be defined with the help of any φ-acyclic resolution $\mathcal{A}^*$ of the sheaf $\mathcal{G}$ on X. Indeed the comparison homomorphisms γ for $\mathcal{A}^*$ and $\mathcal{A}^*|_Y$ (expressed for example in terms of the iterated connecting homomorphism Δ) are compatible with the restriction of the sheaves and their sections to Y. Hence i^* is defined as the composition of the cohomology map induced by the restriction to Y of sections of $\mathcal{A}^*$ with supports in φ and the comparison homomorphism γ corresponding to $\mathcal{A}^*|_Y$ on Y. The independence of i^* from the choice of $\mathcal{A}^*$ follows, for example, from consideration of homomorphisms of $\mathcal{A}^*$ into any injective resolutions $\mathcal{J}^*$ for $\mathcal{G}$ for which it is not difficult to verify independence (of the choice of $\mathcal{J}^*$).

The homomorphism i^* is described from other points of view in Sec. 10 of Chap. 2 of [87].

Below (point 5.3) we shall use another construction of the restriction homomorphism i^* entirely depending on special qualities of the Godement resolution. Let $\tilde{\mathcal{F}}$ temporarily denote the restriction of the sheaf $\mathcal{F}$ to Y. Since any section of the sheaf $\mathcal{C}(\tilde{\mathcal{G}}) = \mathcal{C}^0(\tilde{\mathcal{G}})$ extends to a section of $\mathcal{C}^0(\mathcal{G})$ (in addition it is easy to arrange that the support of the extended section is the closure of the support of the original one), there arises an epimorphism of sheaves $\widetilde{\mathcal{C}^0(\mathcal{G})} \xrightarrow[\rho]{} \mathcal{C}^0(\tilde{\mathcal{G}})$ guaranteeing the existence of an epimorphism $\Gamma_{\varphi}(\mathcal{C}^0(\mathcal{G})) \to \Gamma_{\varphi\cap Y}(\mathcal{C}^0(\tilde{\mathcal{G}}))$. There arises an epimorphism of sheaves $\widetilde{\mathcal{Z}^1(\mathcal{G})} = (\mathcal{C}^0(\mathcal{G})/\mathcal{G})|_Y = \widetilde{\mathcal{C}^0(\mathcal{G})}/\tilde{\mathcal{G}} \xrightarrow[\rho']{} \mathcal{C}^0(\tilde{\mathcal{G}})/\tilde{\mathcal{G}} = \mathcal{Z}^1(\tilde{\mathcal{G}})$.

Repeating the arguments, we get epimorphisms $\widetilde{\mathcal{C}^1(\mathcal{G})} = \widetilde{\mathcal{C}^0(\mathcal{Z}^1(\mathcal{G}))} \xrightarrow[\rho]{} \mathcal{C}^0(\widetilde{\mathcal{Z}^1(\mathcal{G})}) \to \mathcal{C}^0(\mathcal{Z}^1(\tilde{\mathcal{G}})) = \mathcal{C}^1(\tilde{\mathcal{G}})$.

As noted above, the map ρ guarantees that $\Gamma_{\varphi}(\mathcal{C}^1(\mathcal{G})) = \Gamma_{\varphi}(\mathcal{C}^0(\mathcal{Z}^1(\mathcal{G}))) \to \Gamma_{\varphi\cap Y}(\mathcal{C}^0(\widetilde{\mathcal{Z}^1(\mathcal{G})}))$ is an epimorphism and together with the second map (induced by ρ') that $\Gamma_{\varphi}(\mathcal{C}^1(\mathcal{G})) \to \Gamma_{\varphi\cap Y}(\mathcal{C}^1(\tilde{\mathcal{G}}))$ is an epimorphism. Again we have an epimorphism $\widetilde{\mathscr{Z}^2(\mathscr{G})} \to \mathscr{Z}^2(\tilde{\mathscr{G}})$. Continuation of the arguments shows the existence of differential sheaves $\widetilde{\mathscr{C}^*(\mathscr{G})} \xrightarrow[\alpha]{} \mathscr{C}^*(\tilde{\mathscr{G}})$ which provide epimorphisms $\Gamma_\varphi(\alpha)\colon \Gamma_\varphi(\mathscr{C}^* \times (\mathscr{G})) \to \Gamma_{\varphi\cap Y}(\mathscr{C}^*(\tilde{\mathscr{G}}))$ for any $Y \subset X$ and arbitrary families of supports φ which also induce i^* [since the composition of α with a map of $\mathscr{C}^*(\mathscr{G})$ preserving the cohomology of Y into any injective resolution $\tilde{\mathscr{G}}$ defines, for the resolution $\widetilde{\mathscr{C}^*(\mathscr{G})}$, a comparison homomorphism γ and its composition with the restriction homomorphism to Y for $\mathscr{C}^*(\mathscr{G})$ is again i^*].

5.2. Rigidity. If in addition to the fact that the restriction to Y of a φ-acyclic resolution $\mathscr{A}^*$ of the sheaf $\mathscr{G}$ on X is $(\varphi\cap Y)$-acyclic, any section of $\mathscr{A}^*|_Y$ on Y with support in $\varphi\cap Y$ extends to a section on a neighborhood $N \supset Y$ (with support in $\varphi\cap N$), then as is easy to note, $H^n_{\varphi\cap Y}(Y;\mathscr{G}) = \varinjlim_{N} H^n_{\varphi\cap N}(N;\mathscr{G})$ is the φ-rigidity property of cohomology. This will be so in all cases of acyclicity or $(\varphi\cap Y)$-acyclicity of the restriction of $\mathscr{A}^*$ to Y listed in the preceding point except for case c) which fits into the general rule under the additional requirement that φ is paracompactifying (Sec. 10 of Chap. 2 of [87]). Without this condition the relation is lost: when φ and Y are one point $y \in X$ for all its neighborhoods $H^n_{\varphi\cap N}(N;\mathscr{G}) = H^n(X, X\setminus y;\mathscr{G})$ (cf. the next point) are the local cohomology groups which can clearly be different from zero at the same time that $H^n_\varphi(y;\mathscr{G}) = 0$ for $n > 0$ as the cohomology of a point.

5.3. Pairs. The existence of the epimorphisms $\Gamma_\varphi(\alpha)$ lets us define the cohomology of a pair $H_\varphi^n(X, Y;\mathscr{G})$ (the relative cohomology of X mod Y) as the cohomology of the kernel of $\Gamma_\varphi(\alpha)$. For any $Y \subset X$ and any family of supports φ there arises the cohomology exact sequence of the pair

$$\ldots \to H^n_\varphi(X, Y;\mathscr{G}) \to H^n_\varphi(X;\mathscr{G}) \xrightarrow[i^*]{} H^n_{\varphi\cap Y}(Y;\mathscr{G}) \to H^{n+1}_\varphi(X, Y;\mathscr{G}) \to \ldots \tag{3}$$

Actually this construction is given completely in Sec. 4.9 of Chap. 2 of [107] (the author did not note however in the general case that there exists an epimorphism $\Gamma_\varphi(\alpha)$ so the exact sequence (3) is only defined for closed Y in [107]). The sequence (3) is given in complete form in Sec. 12 of Chap. 2 of [87] (although the derivation is somewhat different from ours). The cohomology groups of pairs are defined in [110] only for the special cases when Y is an open or a closed subset of a locally compact space X (the definition is given separately for open Y and the cohomology groups which arise are for some reason called local).

In dimension zero the exact sequence (3) gives the identity $H^0_\varphi(X, Y;\mathscr{G}) = \Gamma_{\varphi|(X\setminus Y)}(\mathscr{G}) = H^0_{\varphi|(X\setminus Y)}(X;\mathscr{G})$, where $\varphi|(X\setminus Y)$ is the restriction of the family φ to X \ Y (composed of all $F \in \varphi$, $F \subset X\setminus Y$). It is natural to expect that for $n > 0$ too the relative cohomology groups should coincide with the derivatives $H^n_{\varphi|(X\setminus Y)}(X;\mathscr{G})$ of the functor $\Gamma_{\varphi|(X\setminus Y)}(\mathscr{G})$. However this will not always be so, for this it is necessary that the inclusion $\Gamma_{\varphi|(X\setminus Y)}(C^*(\mathscr{G})) \subset \operatorname{Ker}\Gamma_\varphi(\alpha)$ induce a cohomology isomorphism, in particular, itself be an isomorphism. The divergences in general situations can be explained by the fact that for general methods of definition of the homomorphism i^*, besides the restrictions to Y of acyclic resolutions $\mathscr{A}^*$ of the sheaf $\mathscr{G}$ on X (for which the kernel again consists of sections with supports in $\varphi|(X\setminus Y)$) one should also consider the comparison homomorphisms corresponding to $\mathscr{A}^*|_Y$ and when the restriction $\Gamma_\varphi(\mathscr{A}^*) \to \Gamma_{\varphi\cap Y}(\mathscr{A}^*|_Y)$ is not an epimorphism also the cohomology homomorphism corresponding to the inclusion $\operatorname{Im}\Gamma_\varphi(\mathscr{A}^*) \subset \Gamma_{\varphi\cap Y}(\mathscr{A}^*|_Y)$. Hence $H^*_\varphi(X, Y;\mathscr{G}) = H^*_{\varphi|(X\setminus Y)}(X;\mathscr{G})$ at least in those cases when it turns out that when the restriction of a φ-acyclic resolution turns out to be $(\varphi\cap Y)$-acyclic and the restriction of the corresponding sections is an epimorphism (cf. with points 5.1 and 5.2). In such cases the independence of the definition of $H^*_\varphi(X, Y;\mathscr{G})$ from the choice of $\mathscr{A}^*$ is a consequence of the independence of i^* and arguments of the type of the five-lemma.

The essential character of the condition of surjectivity of the restriction to Y of sections of $\mathscr{A}^*$ is discovered in the attempt to use soft resolutions of $Y \subset X$ (for hereditarily paracompact X the sheaves $\mathscr{A}^*|_Y$ remain soft, cf. point 5.1). For $\dim X = n < \infty$ there exist, as is known, soft resolutions $\mathscr{A}^*$ of the constant sheaf Z of length n (cf. point 4.2 of Chap. 4). For such $\mathscr{A}^*$ the groups $H^q(X, Y; Z)$ do not coincide with the cohomology groups of the complex $\Gamma_{X\setminus Y}(\mathscr{A}^*)$: it can happen, for example, that $H^{n+1}(X, Y; Z) \neq 0$ (cf. point 3.3 of Chap. 4) although for $q > n$ all the $\mathscr{A}^q$ are zero. Nevertheless, for open $Y \subset X$ and paracompactifying φ (as is established with the help of flabby resolutions $\mathscr{A}^*$) $H^*_\varphi(X, Y;\mathscr{G}) = H^*_{\varphi|(X\setminus Y)}(X;\mathscr{G})$.

This isomorphism holds in all cases listed at the beginning of point 5.1 except case c) for noncompactifying φ (cf. with point 5.2): if φ and Y are one point $y \in X$ then for n > 1 [by virtue of the exact sequence (3)] $H^n_\varphi(X, y; \mathcal{G}) = H^n_\varphi(X; \mathcal{G}) = H^n(X, X \setminus y; \mathcal{G})$ while at the same time $H^n_{\varphi|(X \setminus y)}(X; \mathcal{G}) = 0$ since the family $\varphi|(X \setminus y)$ is empty.

When the subspace Y is closed, for any family of supports φ all terms of the exact sequence (3) admit interpretation as the cohomology groups with suitable coefficients of the space X itself (with the same supports). Let $\mathcal{G}_{X \setminus Y}$ be the subsheaf of $\mathcal{G}$ which coincides with $\mathcal{G}$ on X\Y and is equal to zero on Y, $\mathcal{G}_Y = \mathcal{G}/\mathcal{G}_{X \setminus Y}$ be the quotient sheaf (coinciding on Y with the restriction of $\mathcal{G}$ and with zero on X \ Y). The sections $\mathcal{C}^*(\mathcal{G}|_Y)$ with supports in $\varphi \cap Y$ obviously coincide with the sections (with supports in φ) of the sheaves $\mathcal{C}^*(\mathcal{G}_Y)$ and the map $\Gamma_\varphi(\alpha)$ is induced by the homomorphism of differential sheaves $\mathcal{C}^*(\mathcal{G}) \to \mathcal{C}^*(\mathcal{G}_Y)$ so the cohomology sequence (3) coincides with the sequence (2) of point 4.3

$$\ldots \to H^n_\varphi(X; \mathcal{G}_{X \setminus Y}) \to H^n_\varphi(X; \mathcal{G}) \to H^n_\varphi(X; \mathcal{G}_Y) \to H^{n+1}_\varphi(X; \mathcal{G}_{X \setminus Y}) \to \ldots, \tag{4}$$

corresponding to the short exact sequence of coefficient sheaves

$$0 \to \mathcal{G}_{X \setminus Y} \to \mathcal{G} \to \mathcal{G}_Y \to 0.$$

For paracompactifying families φ the cohomology of any locally closed subspace Y ⊂ X can also be reduced to cohomology of the space X. One has the isomorphisms $H^n_{\varphi|Y}(Y; \mathcal{G}) = H^n_\varphi(X; \mathcal{G}^X)|$ (Sec. 10 of Chap. 2 of [87]). Here $\mathcal{G}$ is a sheaf on Y and $\mathcal{G}^X$ is the unique sheaf on X which induces $\mathcal{G}$ on Y and zero on X \ Y.

Under suitable restrictions on the space X (conditions of the type of paracompactness) and its subsets in terms of sections of sheaves of cochains on X (sections of one and the same acyclic, for example flabby, resolution $\mathcal{G}$) of the coefficient sheaf $\mathcal{A}^*$ the cohomology of pairs (Y, A) of subsets of X can also be defined. If the restrictions of $\mathcal{A}^*$ to Y and A are acyclic (in relation to the situation corresponding to a paracompactifying family of supports φ), the groups $H^n_{\varphi \cap Y}(Y, A; \mathcal{G})$ are defined by the restriction to Y of all sections of $\mathcal{A}^*$ with supports in φ vanishing on A, i.e., sections on Y of sheaves $\mathcal{A}^*$ with supports in the family $(\varphi \cap Y)|_{(Y \setminus A)} = \varphi|_{(X \setminus A)} \cap Y$. Exact cohomology sequences of such pairs, triples, various triads can be described in the standard way (cf. also Sec. 2 of Chap. 4).

5.4. Excision. From the fact that $H^n_\varphi(X, Y; \mathcal{G}) = H^n_\varphi(X; \mathcal{G}_{X \setminus Y})$ for closed Y ⊂ X the excision property for such pairs follows, i.e., the existence of isomorphisms $H^n_\varphi(X, Y; \mathcal{G}) = H^n_{\varphi|(X \setminus U)}(X \setminus U, Y \setminus U; \mathcal{G})$ for any sets U ⊂ Y open in X (in particular, for U = Int Y). In general, when Y is not closed, one can analogously excise arbitrary (not necessarily open) sets contained with their closures in Int Y. Indeed the map α constructed in point 5.1 of sheaves on Y onto Int Y, is obviously an isomorphism so the sections of sheaves $\mathcal{C}^*(\mathcal{G})$ defining the cohomology groups of the pairs (X, Y) and (X \ U, Y \ U) coincide.

In each case when $H^*_\varphi(X, Y; \mathcal{G}) = H^*_{\varphi|(X \setminus Y)}(X; \mathcal{G})$ (cf. the preceding point) and the family $\varphi|_{(X \setminus Y)}$ remains paracompactifying (and analogously for any φ if Y is closed), the union of all the $F \in \varphi|_{(X \setminus Y)}$ is contained in $\overline{\mathrm{Int}}(X \setminus Y)$ so the inclusion $X \setminus Y \subset X$ induces an excision isomorphism in the form of $H^*_\varphi(X, Y; \mathcal{G}) = H^*_{\varphi|(X \setminus Y)}(X \setminus Y; \mathcal{G})$. There is no such isomorphism for arbitrary φ: $H^n(X, X \setminus y; \mathcal{G}) = H^n_y(X; \mathcal{G}) \neq H^n_y(y; \mathcal{G})$ (example from point 5.2).

In these cases when the groups of (X, Y) reduce to cohomology groups of the space X \ Y with suitable supports, these groups of course differ from the usual cohomology groups of X \ Y (excision of Y in such cases is rather conditional) and depend both on the topology of X \ Y and on the inclusion X \ Y ⊂ X (with which the suitable family of supports in X \ Y is just defined). Excition in the pure form holds however for the cohomology of locally compact spaces with compact supports: if Y is closed, $H^n_c(X, Y; \mathcal{G}) = H^{nc}(X \setminus Y; \mathcal{G})$. In particular, for constant coefficients, the cohomology groups of a locally compact X coincide with those of the one-point compactification of X.

For $\varphi = c$ the symbol $\varphi|_{(X \setminus Y)}$ above can always be replaced by c, and for closed Y analogously one can also replace $\varphi \cap Y$. If φ is the family of all closed sets, the symbols φ and $\varphi \cap Y$ are usually omitted in the exact sequence (3).

CHAPTER 2

HOMOLOGY

The homology groups of polyhedra are geometrically considerably more intuitive than the cohomology groups so they appeared considerably earlier in topology. For reasons discussed in the introduction, their development however turned out to be considerably more difficult. In particular, despite the set of rather general constructions of homology with variable coefficients proposed up to now, they have not been successfully interpreted as derived functors of the zero dimensional homology functor H_0, to the end of a proper homology theory (as a set of indicators dual to cohomology) at the present time constructed only for locally constant coefficients.

Many arguments were given above in favor of the uniqueness of cohomology theory, the deviations from certain natural and easily formulated conditions for concrete methods of construction there lead to "pathologies." It is clear that arguments corroborating the uniqueness of homology theory must be different. A rather important role in finding a "proper" homology theory among the set of logically possible versions belongs to its connection with cohomology. The existence of a large number of problems and observations which serve as a distinctive test for its recognition in which "... even if one is mainly interested in assertions containing only cohomology, in the proofs one must use groups playing the role of homology groups and hence this presumes the existence of a homology theory" (p. 23 of [85]) promoted this.

1. Compactness Factor

1.1. Compact Supports. First of all it became clear that "ordinary" homology (which occurs in applications) is homology H_*^c with compact supports: the homology groups of a space X coincide with the direct limit $\varinjlim H_n(C)$ taken over all compact subsets $C \subset X$ (analogously for any pairs $(X, Y): H_n^c(X, Y) = \varinjlim H_n(C, C')$, where C and C' are all compact subsets of X and Y). Namely this is how it turned out to be for the groups in the classical theories, simplicial and cellular homology and singular: although their definitions do not seem to be in any way connected with compactness, the chains used in their construction are finite and hence have compact supports.

Many facts speak in favor of homology with compact supports. For example, for sets X and $Y = S^n \setminus X$ which are complementary in the n-dimensional sphere S^n, the complements $U = S^n \setminus C$ of all compact subsets $C \subset X$ are all neighborhoods of Y. Each such neighborhood U is a polyhedron and the groups $H^p(U; G)$ are uniquely defined, and by the rigidity property $H^p(Y; G) = \varinjlim_U H^p(U; G)$. But $H^p(Y; G) = H_q(X; G)$, where $q = n - p - 1$ (Sitnikov duality), and $H^p(U; G) = H_q(C; G)$ (Steenrod duality), cf. point 5.6 of Chap. 8 in [69], so $H_q(X; G) = H_q^c(X; G)$ is homology with compact supports. Analogously, if X is an (orientable and piecewise-linear) subspace of a compact manifold M^n the groups $H_p(X; G)$ should coincide with $H^{n-p} \times (M^n, Y; G)$, where $Y = M^n \setminus X$ (Poincaré–Lefschetz duality, cf. point 5.6 of Chap. 8 of [69]), and $H_p(C; G)$ for compact $C \subset X$ with $H^{n-p}(M^n, U; G)$, where $U = M^n \setminus C$ are polyhedral neighborhoods of Y, and since $H^{n-p}(M^n, Y; G) = \varinjlim_U H^{n-p}(M^n, U; G)$ (due to the rigidity property for Y, the exact cohomology sequences of the pairs (M^n, Y), (M^n, U), and the five lemma), again $H_p(X; G) = H_p^c(X; G)$. Thus, at least for subspaces of Euclidean spaces, homology theory with compact supports is the only acceptable one.

All these observations speak of the fact that the compactness factor, inherent in the nature of homology itself, is inseparable from it (cf. also point 1.3 below). The groups H_*^c not infrequently (starting with Milnor [128]) called the Steenrod–Sitnikov homology groups have been widely used in topology starting with the beginning of the sixties (exercises on Chap. 9 of [106, 1, 86, 128, 137], etc.), and the applications of these groups have not exposed any defects in them up to now. Nevertheless, up to the present time papers appear whose authors for some reason feel as if before extending homology theory beyond the limits of the compact category there are some difficulties in connection with which they offer rather complicated new definitions, independent of the indicated factor. Not infrequently the new theories proposed are called Steenrod–Sitnikov or Steenrod homology by the authors, cf., e.g., [94, 116, 121, 122, 130] (here, as is not strange, Milnor [128] is usually cited). The construction of new theories is usually motivated to the unsuitability for the categories of shapes and general topological spaces, of singular theory, its known inadequacies are listed, however it is unclear whether the new homology proposed is free from these inadequacies (judging over all, hardly, in addition to some inadequacies of the singular groups new ones are added (cf. [124]): difference from zero in negative dimensions, loss of additivity properties for infinite discrete unions, impossibility of expressing the homology of such unions in terms of the homology of the summands). Here the homology H_*^c with compact supports is for some reason ignored,† although as a rule connections with it are easily discovered, also exposing the essence of the new theory itself (which in the best case can be considered as an auxiliary construction for the study of the behavior of the ordinary ones, cf. in this connection point 6.3 of Chap. 8 of [69]). In the category of compact spaces their nature (including coincidence with the ordinary ones) is determined in [43] (the results of [123] in particular follows from it).

Clearly a few satisfactory constructions of homology in which the influence of compactness could not have been explicitly traced up to now have been found. Besides, no reasons have yet been discovered for the rejection of the theory H_*^c in favor of some more perfect one (which apparently simply does not exist). Cf. also point 5.3 of Chap. 3 below.

1.2. Compact Spaces. Thanks to the property of compact supports, clearing up the question of uniqueness of homology theory depends entirely on the category $\mathscr{B}_0$ of compact (Hausdorff) spaces. Striking arguments in favor of such uniqueness in the category $\mathscr{B}_0$ are the following. For a compact C which lies in the sphere S^n the groups $H_p(C; G)$ should coincide with the uniquely defined $H^{n-p-1}(S^n\setminus C; G)$ (Steenrod duality, cf. the preceding point). Clearly $H^p(C; G)=H^c_{n-p-1}(S^n\setminus C; G)$ also (Pontryagin duality).

An arbitrary metrizable compactum C can be imbedded, by gluing together the mapping cylinders of piecewise linear maps of nerves of coverings from some refining sequence of coverings of C, in a contractible compactum $\tilde{C}$ such that $\tilde{C} \setminus C$ is a locally finite polyhedron (cf. [62, 63]), so $H_p(C; G)=H_{p+1}(\tilde{C}, C; G)=H_{p+1}(\tilde{C}\setminus C; G)$ is the homology of the second kind of the polyhedron, cf. the following point (in particular, the manifold of groups $H_p(C)$, $C\in\mathscr{B}_0$ is not larger than the analogous manifold of homology of the second kind of polyhedra). The "validity" of the last excision is corroborated by the existence of the analogous identity $H_p(K, L) = H_p(K \setminus L)$ for any finite polyhedra. Analogous arguments are clearly applicable to cohomology also and they give $H^p(C; G)=H_c^{p+1}(\tilde{C}\setminus C; G)$.

Thus, independent of the method of description of chains, at least in the subcategory of metrizable compact the homology (analogously cohomology) groups are uniquely defined. However, any chains and cochains defining these groups can be constructed without difficulty for compacta without the metrizability condition also which corroborates the uniqueness of the theories defined by them in this more general case also. Incidentally, each concrete construction of such chains and cochains is only applicable in the category $\mathscr{B}_0$ and beyond its limits must be changed essentially (cf. Secs. 3 and 4 of Chap. 3). In view of the connection discovered with simplicial theories of the second kind, the existence in the category $\mathscr{B}_0$ of free cochains defining the integral cohomology and the natural coincidence of complexes of homomorphisms of such cochains into Abelian groups G with the complexes of chains with coefficients in G is not unexpected. (The construction of $C \subset \tilde{C}$ also sheds light on the axiomatics of homology and cohomology in the category of metric compacta, cf. Sec. 2 of Chap. 7 of [69].)

1.3. Theories of the Second Kind. Just as naturally as surfaces of the type of the sphere or torus, one also considers surfaces which go to infinity of the type of the paraboloid or hyperboloid of one sheet. The homology H_* of locally compact spaces based on the consideration of chains with arbitrary closed supports (not just compact ones) is usually called homology of the second kind (cf. Chap. 3). In the category $\mathscr{B}$ of locally compact (Hausdorff) spaces and their proper maps, the groups H_* are of just as much natural interest as H_*^c. The dual theory of cohomology of the second kind is the cohomology H_c^* with compact supports which we met above.

The theory H_* seems exceptional at first glance, i.e., not included in the framework outlined above. Doubts are dissipated however upon passing in $\mathscr{B}$ to compactifications $\tilde{X}$ of locally compact X by one point * (and adding to the compact spaces from the subcategory $\mathscr{B}_0\subset\mathscr{B}$ the isolated distinguished point *) there arises the category $\tilde{\mathscr{B}}$ of compact spaces with distinguished point and with morphisms carrying the complements of the distinguished point into the same kind of complements. Actually $\mathscr{B}$ can be replaced by the isomorphic category $\tilde{\mathscr{B}}$ of compact pairs of the form $(\tilde{X}, *)$ (and the maps indicated above). Since on the (full) subcategory $\mathscr{B}_0\subset\tilde{\mathscr{B}}$ there exists only one natural theory (of homology or cohomology

†It is a strange style not to note the path proposed by K. A. Sitnikov and at the same time to attribute to N. Steenrod and K. A. Sitnikov something these authors had not the slightest relation to.

and for each $\tilde{X}\in\tilde{\mathscr{B}}$ one can find a homeomorphic (possibly up to one isolated point) capactum from $\mathscr{B}_0$ only one natural theory is possible in the category $\mathscr{B}$ also. Here the homology (or cohomology) groups of a locally compact $X\in\mathscr{B}$ which coincide with the groups of the pair $(\tilde{X}, *)$ are simply the reduced homology (or cohomology) groups of the one-point compactification of X. Thus, compactness informs about itself in the case of homology of the second kind also.

We note that the isomorphism of cohomology of the second kind H_* of a locally compact space X with the (reduced) cohomology of the one-point compactification $\tilde{X}$ was established in point 5.4 of Chap. 1 above. We also note that by virtue of the observations made of the connection between theories of the second kind and the ordinary one, one can write the (homology and cohomology) exact sequences of the pair $(\tilde{X}, X)$ (cf. also point 2.4 of Chap. 2 of [69]).

The uniqueness of the extension to the category $\mathscr{B}$ of theories of the second kind from the subcategory of polyhedra can also be established directly without appeal to one-point compactifications as was done for the category $\mathscr{B}_0$ in point 1.2 above (cf. [63]).

Until recently it was considered that the theories of the second kind H_* and H_c^* are only defined in the category $\mathscr{B}$. As to the homology H_* this is not so. In particular, even for a polyhedron (or cell complex) K along with the ordinary (finite) chains $C_*^c(K; G)$ one can also consider arbitrary locally ones $C_*(K; G)$ containing them which also define groups $H_n(K; G)$. In general (cf. below) the groups $H_n(X; G)$ are the direct limits $\varinjlim_B H_n(B; G)$ defined over all closed locally compact subspaces B of the space X. Induced homomorphisms f_* of the homology groups H_* are defined for all perfect (i.e., closed and with compact preimages of points) continuous maps $f:X \to Y$ (coinciding with proper ones in the subcategory $\mathscr{B}$). Homology gropus of the second kind of general spaces were first considered by Ivanadze [27].

2. Sheaves of Chains

2.1. Formulation of the Problem. All kinds of homology groups, starting with the classical ones, are dual to cohomology groups, so it is natural to expect that homology theory in its final form (which should contain the treatment of the homology groups as derived functors) the concept of cosheaf should figure just as widely as the concept of sheaf in cohomology. As noted at the beginning of the chapter, unfortunately there is no such theory yet. However it was noted recently that sheaves also play a not unimportant role in homology. They arise, in particular, in considering chains.

In contrast with cohomology, for which (in typical cases) the cochains of a subspace $\mathscr{B}$) $Y \subset X$ are obtained by restriction to Y of cochains of X (in particular, as sections of sheaves of cochains $\mathscr{A}^*$) and the cochains of a pair (X, Y) are cochain-sections on X with supports in $X \setminus Y$ the chains of $Y \subset X$ are the parts of chains of X whose supports happen to be in Y, and at the same time the homology groups of the pair (X, Y) are obtained by factoring all chains by the chains of Y (a distinctive "restriction" of chains of X to $X \setminus Y$). Thus, in the use of any chains C_* defining homology groups on X, there arise graded differential presheaves $U \to C_*(X, X \setminus U)$. In most cases however no remarkable connection is observed (other than the natural homomorphism into sections of the sheaves which arise) between the original chains and the differential sheaves $\mathscr{C}_*$ which arise. In the case of singular chains, for example, one can only assert (cf. point 2.9 of Chap. 4 of [87]) that the sheaves $\mathscr{C}_*$ are homotopically fine and the natural transformations $C_*(X, X\setminus U)\to\mathscr{C}_*(U)$ are monomorphisms, which turn out to be isomorphisms in general only for U = X. (There is also a description of these sheaves at the end of Sec. 2 of Chap. 2 in [143], and of the homotopy fineness of the sheaves in Proposition 7 and its corollary in Chap. 6 of this book.)

2.2. Chains Gathered into Sheaves. As will be clear from the remaining content of the present paper, there is a set of examples corroborating the naturality and effectiveness, and in some cases the irreplaceability of the language and methods of sheaf theory in problems connected with the application of homology. In particular, the existence of flabby sheaves of chains is of great value. However such sheaves do not arise in each approach to the construction of homology groups (for example, the constructions based on the consideration of "regular" or "true" cycles of the ancestors of the theory of N. Steenrod and K. A. Sitnikov are unsuitable for this purpose), and far from all chains giving necessary homology groups can be gathered into sheaves.

Sheaves of chains first appeared in [85, 86]. The basic construction of the authors, intended for arbitrary sheaves of coefficients and hence saturated with homological algebra however happened not only to the complicated, but also to a considerable extent formal and artificial. The chains for the homology of a locally compact space X were defined only for coefficients in the ground ring R. They turned out to be sections of the flabby sheaves defined by them $\mathscr{C}_{\cdot}(R):\mathscr{C}_{\cdot}(R)(U)=C_{\cdot}(X, X\setminus U; R)$ for any open sets $U \subset X$. Now as to any other coefficients $\mathscr{G}$ there are no chains as such for them, just differential graded sheaves for the homology $\mathscr{C}_{\cdot}(\mathscr{G})$ are constructed and as chains of any closed pair (X, Y) one uses the sections $\mathscr{C}_{\cdot}(\mathscr{G})(U)$, $U=X\setminus Y$, which do not have a concrete interpretation. The sheaves $\mathscr{C}_{\cdot}(\mathscr{G})$ turn out to be fine (consequently soft).

As was later discovered, the constructions discussed are proper all the way only for locally constant coefficients $\mathscr{G}$ with finitely generated stalks. This is corroborated, in particular, by the fact that all the basic applications are restricted to precisely such coefficients (cf. [87]). The artificiality of the construction is indicated by the fact tht the constituents $\mathscr{C}_n$ of the differential sheaves $\mathscr{C}_{\cdot}$ vanish only for $n\leqslant -2$. The inadequacies of the theory are discussed at the beginning of Chap. 5 and in Sec. 14 of Chap. 5 of [87] and also in [62]. In this connection cf. Sec. 6 of Chap. 3 below.

For constant coefficients (without assumptions of metrizability of the locally compact spaces, in the remark on Sec. 1 of [64]† in [62, 64], and in Sec. 3 of [64] for locally constant coefficients $\mathscr{G}$ chains $C_n(X;\mathscr{G})$, $n\geqslant 0$ were constructed which define sheaves $\mathscr{C}_{\cdot}(\mathscr{G})$ which are always flabby, and for any open sets $U \subset X$ one has $\mathscr{C}_{\cdot}(U)(\mathscr{G})=C_{\cdot}(X, X\setminus U;\mathscr{G})$. The most complete description of chains generating such sheaves independent of cochains and cohomology was given in Sec. 3 of [68].

Such constructions are also applied to Massey chains which appeared later. Although these chains are constructed in [126] only for constant coefficients G (as $\mathrm{Hom}_R(C_c^*, G)$ for special cochains with compact supports C_c^* with coefficients in the ground ring R), since their definition has local character, by natural identifications on intersections of open sets they can be defined for any locally constant coefficients $\mathscr{G}$ (it is essential that the cochains C_c^* are free R-modules and, being a version of Alexander–Spanier cochains, turn out to be sections with compact supports of soft sheaves).

Cf. Secs. 3 and 4 of Chap. 3 for a brief description of the chains discussed.

2.3. Locally Compact Spaces. It is natural to single out this case since the homology groups of the second kind are of independent interest (point 1.3 above).

The essence of the construction of the sheaves of chains discussed above is the following. Although the chains in [62, 64, 68] are constructed independently of cochains, on sufficiently small open sets $U \subset X$, they (like Massey chains) have the form $C_*(X, X\setminus U;\mathscr{G})=\mathrm{Hom}_R(C_c^*(U; R), G)$, where G is the stalk of $\mathscr{G}$ and C_c^* are free cochains with coefficients in the ground ring R having compact supports. For any closed sets $Y \subset X$ (for example, for $Y = X\setminus U$) there arise exact sequences $0 \to C_c^*(U; R)\to C_c^*(X; R)\to C_c^*(Y; R)\to 0$ which split [since $C_c^*(Y; R)$ is free] in each separate dimension. Along with the softness of the sheaves of cochains $\mathscr{C}^*(R)$ whose sections are $C_c^*(U; R)$, this property also assures the existence for any sets $U \subset X$ of the relations $\mathscr{C}_*(\mathscr{G})(U)=C_*(X, X\setminus U;\mathscr{G})$ for the sheaves $\mathscr{C}_*(\mathscr{G})$ on X defined by the chains considered [in particular, the flabbiness of $\mathscr{C}_*(\mathscr{G})$ is a consequence of the splitting of the cochains of the pair (X, Y) indicated above].

The basic reason for the complexity of the construction of the homology from [85, 86] discussed above is that the authors do not have available free cochains with compact supports. With the help of rather artificial methods, using tools of homotopy theory, free cochains were first constructed in [128] for compact spaces. However these cochains are not sections of the sheaves defined by them (the fact that they are nonzero in all dimensions including negative ones in particular testifies to the artificiality of their construction). Similarly to [62, 64], free cochains with compact supports were defined in the category of locally compact spaces later in [39]. Apparently having the same properties as the cochains of the author and Massey, probably they can also be used for constructing a version of flabby sheaves of chains. Such questions however are not considered in [39].

2.4. Supports. Homology of Subspaces and Pairs. In accord with the observations made for any closed set $B \subset X$ the kernel of the epimorphism serves as the chains of B. Thus, the ordinary homology groups $\mathscr{C}_{\cdot}(\mathscr{G})(X)\to\mathscr{C}_{\cdot}(\mathscr{G})(X\setminus B)$ of the space X are defined by the complex of

†Cf. also Sec. 2 of [65].

sections of the differential sheaf $\mathscr{C}_*(\mathscr{G})$ with compact supports and for any subset $Y \subset X$, the groups $H_*^c(Y;\mathscr{G})$ are defined by the subcomplex consisting of sections with supports in Y. The homology groups of the second kind of the subspace $Y \subset X$ which are of independent interest are defined by all those chains of X (generally infinite) whose supports are contained in Y constituting the family θ of all sets which are closed in X and situated in Y. These homology groups, denoted by $H^{\theta}_*(Y;\mathscr{G})$, are defined by the complex of sections of $\mathscr{C}_*(\mathscr{G})$ with supports in θ. The groups $H^{\theta}_*(Y;\mathscr{G})$ obviously depend not only on the topology of Y, but also on the way Y is imbedded in X, they coincide with $H_*(Y;\mathscr{G})$ when Y is closed and with $H^c_*(Y;\mathscr{G})$ when the closure of Y in X is compact. We note that in all cases the homology groups of Y are simultaneously the homology groups $H^{\varphi}_*(X;\mathscr{G})$ of the space X with a suitable family of supports φ (which coincides with the restriction of the family c to Y in the case of ordinary homology and with θ in the case of homology of the second kind). The family φ in addition may be completely nonparacompactifying also (in the cases considered above, for example, this is so for any Y which is not open).

Thus, for different reasons one comes to considering the homology of the space X with supports in various families φ. The picture which arises in describing the homology groups H^{φ}_* for subspaces and pairs turns out to be dual to that which one has in the case of cohomology (cf. point 2.1). The homology groups of an arbitrary subspace $Y \subset X$ are defined by the subcomplex $C_*^{\theta(\varphi)} \subset C_*^{\varphi}(X;\mathscr{G})$ of sections of sheaves $\mathscr{C}_*(\mathscr{G})$ with supports in $\theta(\varphi)=\varphi|_Y$, i.e., by the kernel of the operation of restriction of sections to $X \setminus Y$ (in comparison: in the case of cochains, this kernel defined the cohomology groups of the pair $(X, X \setminus Y)$ under some additional conditions, cf. point 5.3 of Chap. 1), while at the same time, the homology groups of the pair $H^{\varphi}_*(X, Y;\mathscr{G})$ are defined by the corresponding quotient-complex which coincides with the complex of restrictions to $X \setminus Y$ of sections from $\Gamma_{\varphi}(\mathscr{C}_*(\mathscr{G}))=C^{\varphi}_*(X;\mathscr{G})$. Since the sheaves $\mathscr{C}_*(\mathscr{G})$ are flabby, for any closed Y this quotient complex coincides with the complex of of sections of sheaves $\mathscr{C}_*(\mathscr{G})|_{X\setminus Y}$ with supports in the family $\varphi|\cap(X\setminus Y)$ (obviously consisting of those sets which are closed in $X \setminus Y$ whose closures in X belong to φ). Under the condition that the family φ is paracompactifying, analogous assertions are also valid for open Y, and if X in addition is hereditarily paracompact, then for any $Y \subset X$ too (the reasons for the requirement of being paracompactifying for φ being essential are the same as in the case of cohomology). In each case there arises the exact homology sequence of the pair

$$\ldots \to H^{\varphi}_n(Y;\mathscr{G}) \to H^{\varphi}_n(X;\mathscr{G}) \to H^{\varphi}_n(X, Y;\mathscr{G}) \to H^{\varphi}_{n-1}(Y;\mathscr{G}) \to \ldots .$$

Here $H^{\varphi}_n(Y;\mathscr{G})=H^{\theta(\varphi)}_n(Y;\mathscr{G})=H^{\theta(\varphi)}_n(X;\mathscr{G})$ and in the case of closed Y, $H^{\varphi}_n(X, Y;\mathscr{G})=H^{\varphi\cap(X\setminus Y)}_n(X\setminus Y;\mathscr{G})$ also [since for open U the sheaf $\mathscr{C}_*(\mathscr{G})|_U$ is the same as $\mathscr{C}_*(\mathscr{G})$ but constructed on the locally compact space U]. When φ is the family of all closed sets and Y is closed, the symbol φ can be omitted in the exact sequence. In addition $H_n(X, Y;\mathscr{G})=H_n(X\setminus Y;\mathscr{G})$ [in particular, the homology groups $H_n(X; G)$ of a noncompact space X with coefficients in the R-module G coincide with the reduced homology groups $H_n(\tilde{X}, *; G)$ of the compactification $\tilde{X}$ of the space X by one point $*$]. For nonclosed Y the symbol φ (in the case of all closed sets) can be omitted in the homology groups of X and (X, Y), replacing it by θ in the terms corresponding to Y.

2.5. General Spaces. Sheaves of chains exist (although with slightly less remarkable properties) not only for locally compact spaces.

For an open set U of a locally compact space B, the restriction $\mathscr{C}_*(\mathscr{G})|_Y$ of the sheaf $\mathscr{C}_*(\mathscr{G})$ to the closed set $Y = B \setminus U$ defines, by virtue of the preceding point, by its sections, homology groups of the pair (B, U), and clearly differs from the sheaf $\mathscr{C}_*(\mathscr{G}|_U)$ constructed similarly to $\mathscr{C}_*(\mathscr{G})$ for the locally compact space Y. The association to each open set $V \subset B$ of the sections of $\mathscr{C}_*(\mathscr{G}|_Y)$ on $V \cap Y$ [i.e., of chains of the pair $(Y, Y \setminus (V \cap Y))$] defines the direct image $\mathscr{C}_*(\mathscr{G})^Y$ of the sheaf $\mathscr{C}_*(\mathscr{G}|_Y)$ under the inclusion $Y \subset B$. This sheaf on B coincides with $\mathscr{C}_*(\mathscr{G}|_Y)$ on Y and is equal to zero on $B \setminus Y$. Since the restrictions of cochains $C^*_c(V; R) \to C^*_c(V \cap Y; R)$ are epimorphic, and the chains (at least on small pieces) are obtained from the cochains as $\mathrm{Hom}_R(C^*_c, G)$, for the sheaves of chains there arise inclusions $\mathscr{C}_*(\mathscr{G})^Y \subset \mathscr{C}_*(\mathscr{G})$ [in particular, $\mathscr{C}_*(\mathscr{G}|_Y) \subset \mathscr{C}_*(\mathscr{G})|_Y$ always so $\mathscr{C}_*(\mathscr{G})^Y \subset \mathscr{C}_*(\mathscr{G})_Y$) too]. Such inclusions are locally isomorphic at points $y \in \mathrm{Int}\, Y$. It follows from this among other things that $\mathscr{C}_*(\mathscr{G}) = \varinjlim_C \mathscr{C}_*(\mathscr{G})^C = \bigcup_C \mathscr{C}_*(\mathscr{G})^C$ is the direct limit over all compact subsets $C \subset B$.

Now let X be an arbitrary (paracompact and Hausdorff) space, $\mathscr{G}$ be a system of coefficients on X. For any closed locally compact subset $B \subset X$ we shall denote by $\mathscr{C}_*(\mathscr{G})^B$ the direct image of the sheaf $\mathscr{C}_*(\mathscr{G}|_B)$ under the inclusion $B \subset X$. This sheaf is flabby (as direct image of a flabby sheaf). To the inclusions B ⊂ F of closed locally compact subspaces in X obviously correspond inclusions $\mathscr{C}_*(\mathscr{G})^B \subset \mathscr{C}_*(\mathscr{G})^F$. Thus, there arises a inductive system directed by inclusion of flabby sheaves $\mathscr{C}_*(\mathscr{G})^B$. Let $\mathscr{C}_*(\mathscr{G}) = \varinjlim_B \mathscr{C}_*(\mathscr{G})^B = \bigcup_B \mathscr{C}_*(\mathscr{G})^B$, where the B are all closed, locally compact subspaces.

In accord with Proposition 3.6 of [22] (cf. also [34]) $\mathscr{C}_*(\mathscr{G})$ are soft sheaves. They are also c-soft: for any closed Y ⊂ X the map $\Gamma_c(\mathscr{C}_*(\mathscr{G})) \to \Gamma_c(\mathscr{C}_*(\mathscr{G})|_Y)$ is epimorphic (one defined φ-soft sheaves analogously for any family φ). Indeed if ξ' is a section on Y with compact support |ξ'| then [from the fact that $\mathscr{C}_*(\mathscr{G})$ is the union of $\mathscr{C}_*(\mathscr{G})^B$] one can find a compact set C ⊂ X containing |ξ'| such that the nontrivial part of ξ' turns out to be defined on C ∩ Y by a section of $\mathscr{C}_*(\mathscr{G})^C$ [one uses the openness of $\mathscr{C}_*(\mathscr{G})^C$ in $\mathscr{C}_*(\mathscr{G})$ and the compactness of |ξ'|] which extends to a section ξ of the flabby sheaf $C_*(\mathscr{G})^C \subset C_*(\mathscr{G})$ which coincides with ξ' on Y. We note however that the family c in X is not paracompactifying in general so although the sheaves $\mathscr{C}_*(\mathscr{G})$ are soft, it is not clear whether they are c-acyclic (φ-acyclicity of φ-soft sheaves is established for paracompactifying families φ cf. Secs. 3.6 and 4.13 of Chap. 2 of [107] of Sec. 9 of Chap. 2 of [87]).

The sheaves $\mathscr{C}_*(\mathscr{G})$ are sheaves of chains of X: all of them are sections, the chains of X with (closed) locally compact supports defining the as yet little studied homology groups of the second kind $H_n(X; \mathscr{G}) = \varinjlim_B H_n(B; \mathscr{G})$ (cf. the end of point 1.3 above). Indeed let ξ be any section of $\mathscr{C}_*(\mathscr{G})$, $\xi(x) \neq 0$ for some point $x \in X$. One can find a B such that $\xi(x) \in \mathscr{C}_*(\mathscr{G})^B \subset \mathscr{C}_*(\mathscr{G})$ so $\xi(U) \subset \mathscr{C}_*(\mathscr{G})^B$ for some neighborhood U of the point x. Consequently, the support |ξ| is a loally compact (closed) subspace of X. Suppose in addition B = |ξ|. For any closed, locally compact set F ⊂ X containing B, the sections of $\mathscr{C}_*(\mathscr{G})^F$ with supports in B, being chains of the space B, are sections in $\mathscr{C}_*(\mathscr{G})^B \subset \mathscr{C}_*(\mathscr{G})^F$. Thus, the section ξ [of the sheaf $\mathscr{C}_*(\mathscr{G})$] considered belongs to $\mathscr{C}_*(\mathscr{G})^B$ if it is a section of a $\mathscr{C}_*(\mathscr{G})^F$. But due to the softness of $\mathscr{C}_*(\mathscr{G})$ it can be represented as a locally-finite sum $\xi = \sum_\lambda \xi_\lambda$ with compact $|\xi_\lambda|$ hence locally, near each point $x \in B$ the section ξ belongs to a $\mathscr{C}_*(\mathscr{G})^F$, $B \subset F$ [the arguments are the same as in the proof of the c-softness of $\mathscr{C}_*(\mathscr{G})$ above]. Consequently, ξ is a section of $\mathscr{C}_*(\mathscr{G})^B$.

Thus, $\Gamma(\mathscr{C}_*(\mathscr{G})) = \varinjlim_B C_*(B; \mathscr{G}) = \varinjlim_B \Gamma(\mathscr{C}_*(\mathscr{G})^B) = C_*(X; \mathscr{G})$ are chains of X with any (closed) locally compact supports B ⊂ X (the symbol $\varinjlim$ can be replaced here by the summation sign Σ or that of the union ∪).

Since for locally compact B, as noted above, the sheaves $\mathscr{C}_*(\mathscr{G})$ are exhausted by the subsheaves $\mathscr{C}_*(\mathscr{G})^C$ with respect to compact C ⊂ B and in general $\mathscr{C}_*(\mathscr{G}) = \varinjlim_C \mathscr{C}_*(\mathscr{G})^C = \bigcup_C \mathscr{C}_*(\mathscr{G})^C$, where C ⊂ X are all compact subspaces. Hence, $\Gamma_c(\mathscr{C}_*(\mathscr{G})) = \varinjlim_{C \subset X} C_*(C; \mathscr{G}) = \varinjlim_{C \subset X} \Gamma(\mathscr{C}_*(\mathscr{G})^C) = C_*^c(X; \mathscr{G})$ are chains of X with compact supports (the symbol $\varinjlim$ can be replaced by Σ or ∪). They dedefine homology groups with compact supports $H_*^c(X; \mathscr{G})$. Since $C_*^c(X; \mathscr{G}) \subset C_*(X; \mathscr{G})$ there arise natural transformations $H_n^c(X; \mathscr{G}) \to H_n(X; \mathscr{G})$ which are contained in an exact homology sequence (cf. with the case of locally compact spaces, cf. point 1.3 above).

Similar inferences are also valid with respect to sections of $\mathscr{C}_*(\mathscr{G})$ on any open sets U ⊂ X; they are the chains of U, respectively, with locally compact or compact supports. Not all locally compact B' which are closed in U have the form U ∩ B for analogous B ⊂ X hence the sheaves $\mathscr{C}_*(\mathscr{G})$ are apparently not flabby. As usual, the homology groups (H_* or H_*^c) of the subspace Y ⊂ X are defined by chains [sections of the sheaves $\mathscr{C}_*(\mathscr{G})$] with supports (of the necessary type) in Y while at the same time the homology groups of the pair (X, Y) are defined by the corresponding quotient-complex. When Y = U is open, this quotient-complex (due to the softness and c-softness of the sheaves $\mathscr{C}_*(\mathscr{G})$] coincides with the complex of sections (of corresponding type) of the restriction $\mathscr{C}_*(\mathscr{G})|_{X \setminus U}$ [thus defining $H_*(X, U; \mathscr{G})$ and $H_*^c(X, U; \mathscr{G})$), respectively]. Hence for compact C, $H_n(X, X \setminus C; \mathscr{G}) = H_n^c(X, X \setminus C; \mathscr{G})$.

<u>2.6. Local Homology Sheaves.</u> As C above one can take any point $x \in X$ and we get $H_p(X, X \setminus x; \mathscr{G}) = H_p^c(X, X \setminus x; \mathscr{G}) = H_p^x(\mathscr{G})$ which are the local homology groups. But $H_p^x(\mathscr{G}) = H_p^c(X, X \setminus x; \mathscr{G})$ are obviously equal to the direct limits $\varinjlim_U H_p^c(X, X \setminus U; \mathscr{G})$, i.e., are the stalks of the local homology sheaves $\mathscr{H}_p(\mathscr{G})$ defined by the presheaves $U \to H_p^c(X, X \setminus U; \mathscr{G})$. Analogously,

since the chains of $X \setminus x$ are exhausted by chains with supports in the sets $X \setminus U$ for all neighborhoods U of the point x, we have $H_p^x(\mathcal{G}) = H_p(X, X \setminus x; \mathcal{G}) = \varinjlim_U H_p(X, X \setminus U; \mathcal{G})$. These limits also coincide with $\varinjlim_N H_p(X, X \setminus N; \mathcal{G})$, where N runs through all closed neighborhoods of points $x \in X$. In this way however the derived sheaves $\mathcal{H}_*(\mathcal{G})$ of the differential sheaf $\mathcal{C}_*(\mathcal{G})$ are defined. The transformations of functors $H_p^c \to H_p$ define sheaf homomorphism $\mathcal{H}_p(\mathcal{G}) \to \tilde{\mathcal{H}}_p(\mathcal{G})$ which are isomorphic on the fibers, i.e., isomorphisms of sheaves. Thus the sheaves of local homology groups $\mathcal{H}_p(\mathcal{G})$ are the derived sheaves of the differential sheaf of chains $\mathcal{C}_*(\mathcal{G})$. In terms of singular homology groups of the second kind for locally compact spaces the analogous construction was first considered by Cartan. It is essentially more complicated and carries less information (cf. point 2.1 above).

2.7. Cartan Spectral Sequences. Since the sheaves $\mathcal{C}_*(\mathcal{G})$ are soft, under the condition $\dim_G X < \infty$ (G being the stalk of $\mathcal{G}$) for any closed subset Y of an arbitrary paracompact (Hausdorff) space X and paracompactifying family of supports φ there arise spectral sequences converging to $H^{\varphi}_{-p-q}(X, X \setminus Y; \mathcal{G})$ and $H^{\varphi|X\setminus Y}_{-p-q}(X \setminus Y; \mathcal{G})$ with second terms $E_2^{pq} = H^p_{\varphi}(Y; \mathcal{H}_{-q}(\mathcal{G}))$ and $E_2^{pq} = H^p_{\varphi}(X, Y; \mathcal{H}_{-q}(\mathcal{G}))$. In particular, ordinary cohomology groups (φ being the family of all closed sets) converge to the homology groups of the second kind. If the space X is locally compact, then the sheaves $\mathcal{C}_*(\mathcal{G})$ are flabby so the analogous results is also valid for open $Y \subset X$ and if X is hereditarily paracompact in addition, then generally for any $Y \subset X$ (in the case of closed Y the result is also true for nonparacompactifying φ). As the example of a countable product of circles (whose local homology groups are equal to zero) shows, the assumption of the finiteness $\dim_G X$ is essential.

In terms of singular theories (for locally compact spaces) such a spectral sequence was first constructed by Cartan (cf. the description in point 8.6 of Chap. 5 of [87]) and used to prove Poincaré duality by sheaf methods. Besides Cartan, the spectral sequence was studied (in the category of polyhedra) by Farey and Zeeman. In terms of the Borel–Moore homology (for typical restrictions for it cf. Sec. 6 of Chap. 3 below, and only for locally compact spaces) the spectral sequence is constructed in Sec. 8 of Chap. 5 of [87].

CHAPTER 3

MOST TYPICAL CONCRETE APPROACHES

Clearly there are many approaches to the description of the homology and cohomology of general (or on the other hand concrete) spaces. In concrete cases this lets one, considering the specifics of the problems considered, apply the most effective of them for the specific needs (depending functorially on the spaces where the maps are considered, using a specific type of covering and passages to the limit of Cech type chains and cochains in problems where one is considering dimensional properties, the specifics of the structure of the space or using the functor $\varprojlim^1$ or even the functors $\varprojlim^p$ limiting relations, etc.). Whenever the space is endowed with some additional structure (is locally compact, has some local properties, is a manifold or a polyhedron, etc.), on it there usually arise chains and cochains and special resolutions considering this structure. Thus, on manifolds, for example, besides the usual singular chains (hence also cochains) one can consider chains generated by smooth singular simplices, simplices, diffeomorphic (or homeomorphic) to a standard one, etc. The cochains corresponding to them (usually after localization, cf. below) are combined into acyclic sheaves which form resolutions under natural restrictions. On locally compact spaces the cochains with compact supports natural for such spaces turn out to be sections of soft sheaves. There are many versions (including ones with elements of continuity, smoothness) of Alexander–Spanier type cochains, also gathered into acyclic sheaves.†

†The division sometimes made of the methods of construction of homology and cohomology groups of general spaces into spectral and projectional, depending on whether the groups needed are defined as limits of homology (or cohomology) groups in auxiliary constructions (most often defined by the systems of coverings considered) or as homology (cohomology) groups of limits of chains (cochains) figuring in these constructions is not entirely justified since there is a set of methods of construction not connected with passages to the limit or coverings. For example, the approaches of the ancestors of the exact homology theory of N. Steenrod and K. A. Sitnikov, and also the approaches of Milnor, Borel and Moore, Massey (and some others) are such.

Below we shall elucidate constructions which have endured the test of time and previous examination in numerous applications. In particular, we shall describe chains which are collected into the differential sheaves discussed in the preceding chapter. Of course the homology groups defined by them coincide with those whose existence was established in Sec. 1 of Chap. 2 independently of any concrete definitions. Incidentally, approapriate new properties of homology and cohomology groups will be considered.

1. Singular Theory

This theory is irreplaceable in homotopic topology, in the theory of bundles, in the study of mapping spaces, and in problems of classification of maps. With its help one gets the simplest description of the groups of cell complexes, their homotopy invariance, and other basic properties.

1.1. Limits of Applicability. It is well known (and follows easily from the contents of Sec. 2 of Chap. 1), that the sheaves $\mathcal{S}^*$ of singular cochains with coefficients in the R-module G are soft if X is paracompact, and flabby if X is hereditarily paracompact (point 5.2 of Chap. 3 of [69]). Their section are "localized" cochains: for paracompact X there is an epimorphism $C_s^*(X; G) \to \Gamma(\mathcal{S}^*)$ of the complex of singular cochains whose kernel is the cochains which vanish on singular simplices of the fineness of a certain open covering of X (depending on the cochain). Since the subcomplex of $C_s^*(X; G)$ consisting of these cochains is acyclic (cf. point 5.2 of Chap. 3 of [69]), the singular cohomology groups $H_s^n(X; G)$ coincide with $H^n(\Gamma(\mathcal{S}^*))$. This still does not mean that $H_s^n(X; G) = H^n(X; G)$, the sheaf cohomology, because in general $\mathcal{S}^*$ is not a resolution of G, a necessary and sufficient condition for this (point 2.3 of Chap. 1 above) being the vanishing for any open $U \subset X$ of the restrictions to sufficiently small neighborhoods of points $x \in U$ of any elements $h \in H_s^n(U; G)$ (the cohomology groups in dimension zero are reduced). The condition holds, if for any neighborhood U of an arbitrary point $x \in X$ one can find a smaller neighborhood V (depending on n), for which the map $H_s^n(U; G) \to H_s^n(V; G)$ is zero (cohomological local connectedness of X over G; the cohomology groups in dimension zero are reduced). This is so, in particular, if X is weakly locally contractible (the neighborhoods V above are contractible in U to x). Under the local conditions formulated $H_s^n(X, U; G) = H^n(X, U; G)$ also for any open paracompact U. The analogous equation for the pair (X, Y) with closed Y only holds when along with X, Y also satisfies the local conditions formulated.

In the category of pairs of weakly locally contractible (Hausdorff, paracompact) spaces, the singular homology groups H_*^s coincide with the homology groups H_*^c discussed in the preceding chapter: in the category indicated any homology groups with compact supports which have the standard properties (to the extent of the Eilenberg–Steenrod axioms) and vanish in negative dimensions coincide with the theory H_*^c (Theorem 8 of [63]).

As noted in the introduction, in applications of homology and cohomology theory it is usually impossible to avoid consideration of subspaces of the space studied. Familiar difficulties in applying singular theory are explained by this: the local properties cited above are lost in passing to subspaces and although the sections $\mathcal{S}^*$ on such subspaces define cohomology groups as before, they stop coinciding with the singular ones. In correspondence with the material recounted in Chaps. 1 and 2, all the "pathologies" which are present in singular theory outside the limits of the category of weakly locally contractible spaces (cf. the introduction), testify to nothing but the fact that outside specific limits the singular groups H_*^s and H_s^* simply stop being homology and cohomology groups.

1.2. Some Properties. Let $C_*^\alpha \subset C_*^s(X; G)$ be the subcomplex of the complex of singular chains generated by the singular simplices of the fineness of some open covering α. For any such α the inclusion of chain complexes induces an isomorphism of homology groups H_*^s. Correspondingly, the restriction of the cochains to simplices of the fineness of α induces an isomorphism of cohomology groups H_s^* (this is precisely what lets one effect the localization of cochains, cf. the preceding point).

Even when Y is closed in X, for a set U which is open in X of the multiple $U \subset Y$, the inclusion $(X \setminus U, Y \setminus U) \to (X, Y)$ induces an isomorphism of the groups H_*^s and H_s^* (excision property), generally only under the condition that the closure $\overline{U}$ is contained in Int Y (cf. point 5.4 of Chap. 1 and point 5.4 of Chap. 3 below). One can see this by passage to chains and cochains of fineness of the covering $\alpha = \{\mathrm{Int}\, Y, X \setminus \overline{U}\}$. Analogously, for $X_1, X_2 \subset X$ the (ordinary and additional) Mayer–Vietoris exact sequences of the triad generally hold only under the condition that† $X'=X_1\cup X_2=\operatorname{Int}X_1\cup\operatorname{Int}X_2$ and the existence of a homology isomorphism induced by the inclusion $C_*^{\alpha}\subset C_*^{s}(X';G)$, $\alpha=\{\operatorname{Int}X_1, \operatorname{Int}X_2\}$ is one of the numerous conditions (equivalent with one another) figuring in the description of "excised" triads (cf., e.g., [99]). For the theories H_*^c and H^* considered by us, similar conditions are exceptional (at least for closed and open X_i cf. below Sec. 2 of Chap. 4).

1.3. Theories of the Second Kind. For locally compact spaces, in addition to finite ones, one not infrequently considers arbitrary locally finite singular chains (the set of images of singular simplices occurring in the chain is locally finite), the groups they define are called the singular homology groups of the second kind (cf. also points 2.1, 2.6, and 2.7 of Chap. 2). Analogously, the subcomplex of the complex of singular cochains of a locally compact X consisting of cochains with compact supports (for each occhain ξ one can find a compact set of whose complement ξ vanishes), defines singular cohomology groups of the second kind (i.e., with compact supports). Such ξ are characterized by the fact that they are nonzero only on a finite number of singular simplices from any locally finite collection of such simplices. Singular theories of the second kind were defined in [91].

2. Alexander–Spanier Cohomology

2.1. Basic Definitions. The role of n-dimensional "singular" simplices is played by ordered collections of points $(x_0,\ldots,x_n)$ of the space X, by cochains one initially means functions on such collections with values in the R-module G, the coboundary operator is analogous to that for singular cochains. In contrast with the singular theory however any cocycle ξ in this approach turns out to be a coboundary: $\xi = d\eta$, where $\eta(x_0,\ldots,x_{n-1}) = \xi(y, x_0,\ldots,x_{n-1})$ and y is a fixed point of the space X. The situation is corrected by considering localized Alexander–Spanier cochains $\widetilde{A^n}(X)$ defined by factoring the groups $A^n(X)$ of functions indicated above by the subgroup consisting of functions ξ vanishing on the elements of an open covering of X (depending on ξ). The groups $H^n(\widetilde{A^*}(X))$ are called the Alexander–Spanier cohomology groups of the topological space X with coefficients in G. For any subspace $Y \subset X$ there is an obvious epimorphism $\widetilde{A^*}(X)\to\widetilde{A^*}(Y)$ whose kernel defines the cohomology of the pair (X, Y). In the standard way one gets the cohomology exact sequence of the pair (X, Y) and also the cohomology exact sequence corresponding to a short exact sequence of coefficient modules.

The zero-dimensional cochains are obviously functions of an argument $x\in X$. In terms of localized cochains the condition $d\xi = 0$ for such a function ξ means that its values are equal on elements of some sufficiently fine open covering of X, i.e., that ξ is locally constant and $H^0(X; G)$ is the group of locally constant functions on X with values in G (cf. point 2.4 of Chap. 1).

2.2. Connection with Sheaves. For the presheaf $U \to A^*(U)$ we have $H^q(A^*(U)) = 0$ for $q > 0$ at the same time that $H^0(A^*(U))$ is the group of locally constant functions on U, so for any topological space X the differential sheaf $\mathscr{A}^*$ which arises is a resolution of the constant sheaf G. Just as in Sec. 1 above, for paracompact X the sheaves $\mathscr{A}^*$ turn out to be be soft and their sections $\mathscr{A}^*(X)$ are the (localized) Alexander–Spanier cochains $\widetilde{A^*}(X)$ (points 3.7 and 3.9 of Chap. 2 of [107], Sec. 2 of Chap. 3 of [87], point 5.3 of Chap. 3 of [69]). Now if X is hereditarily paracompact, the sheaves $\mathscr{A}^*$ turn out to be flabby (point 5.3 of Chap. 3 of [69]), and the equality $\mathscr{A}^*(U)=\widetilde{A^*}(U)$ holds for any open $U \subset X$. Thus, for paracompact spaces the Alexander–Spanier cohomology groups coincide with the sheaf groups.

We turn out attention to the external divergence in the definitions of the cohomology groups of subspaces Y and pairs (X, Y) by Alexander–Spanier cochains and by sections of sheaves $\mathscr{A}^*$ (under conditions, of course, when they coincide). In the first case the cohomology groups of a pair are defined by the subcomplex $\widetilde{A^*}(X)'$ consisting of all cochains with restriction to Y zero, and in the second by the subcomplex $\widetilde{A^*}(X)''\subset\widetilde{A^*}(X)'$ each cochain of which vanishes in a sufficiently small neighborhood of Y (since the cohomology groups of Y are defined by sections of $\mathscr{A}^*|_Y$ which does not in general coincide with $\widetilde{A^*}(Y)$). Under paracompactness assumptions appropriate to the case however (cf. above) the sections of $\mathscr{A}^*$

†The operation Int here is in the space X'.

extended to sections of $\mathscr{A}^*|_Y$, map onto $\overbrace{A^*(Y)}$ and this map coincides with the factorization of $\mathscr{A}^*(X)=\overbrace{A^*(X)}$ with respect to the pair of subcomplexes indicated. Since $\overbrace{A^*(Y)}$ is the sections of a resolution of type $\mathscr{A}^*$ constructed on Y, by virtue of the apparatus discussed in Chap. 1 of the comparison homomorphism, the epimorphism of sections of $\mathscr{A}^*|_Y$ onto $\overbrace{A^*(Y)}$ (which can be realized by a map of acyclic resolutions) induces an isomorphism of the cohomology groups of Y. The analogous isomorphism for pairs is a consequence of the five lemma. For similar arguments applied to closed $Y \subset X$, cf. point 4.10 of Chap. 2 of [107].

2.3. Remarks. In applications various modifications of Alexander–Spanier cochains occur. Thus, in [126] the subcomplex of all locally finite-valued cochains in $\overbrace{A^*(X)}$ is applied successfully. Sometimes one uses continuous cochains (with coefficients in the field R of real numbers), among them those which can be represented as sums of a finite number of products of functions of one variable [30]. One of the versions is described in the next section. Under appropriate paracompactness assumptions all similar cochains are easily assembled into acyclic sheaves so the cohomology groups defined by them are always equivalent to the sheaf groups. In each case in consideration of the cohomology groups of subspaces and pairs one must consider the possible difference between the cochains of subsets (and pairs) and the corresponding sections of the differential sheaf-resolutions which arise, which however (in contrast with the case of singular theory) do not lead to deviations in the cohomology groups. Of course a similar picture can arise in using cochains of any other type also.

3. Free Massey Cochains and the Chains Associated with Them

3.1. Construction of Cochains. By the support $|\xi|$ of the Alexander–Spanier cochain ξ (analogously, of the element of the cohomology group represented by the cocycle ξ) we mean the (minimal) closed set on whose complement ξ vanishes. Let B be a locally compact (paracompact and Hausdorff) spaces. The subcomplex of the complex $\overbrace{A^*(B)}$ of Alexander–Spanier cochains composed of all cochains with compact supports, defines the cohomology of the second kind (cohomology with compact supports). These groups are obviously functorial in relation to proper maps, are invariant with respect to proper homotopies, etc. (cf. [141]). The group $H_c^0(B; G)$ consisting of locally constant functions is nonzero only if B contains compact open-closed subsets.

In [126] the cohomology of locally compact spaces is constructed with the help of finite-valued Alexander–Spanier cochains with compact supports. One can establish the fact that the same groups H_*^c are defined in this way (this question is not considered in [126]) by the tools of sheaf theory. Namely, in the complex of such cochains $C_c^*(B; G)$ let $C_M^*(B; G)$ be the subcomplex consisting of cochains with supports in the set $M \subset B$. Let $\mathscr{C}^*(G)$ be the graded differential sheaf defined by the presheaf $U \to D^*(U)=C_c^*(B; G)/C^*_{B\setminus U}(B; G)$. The sheaves $\mathscr{C}^*(G)$ are soft and $\Gamma_c(\mathscr{C}^*(G)|_U)=C_c^*(U; G)$ for any open set $U \subset B$. Indeed a cochain $\xi \in C_c^*(B; G)$ defined the zero element in $D^*(U)$ only under the condition that $|\xi| \subset B\setminus U$ so the presheaf satisfies condition (S1) of point 1.1 of Chap. 1 and the natural map $r: D^*(U) \to \mathscr{C}^*(G)(U)$ is a monomorphism, i.e., to a cochain ξ with support in U corresponds a section of $\mathscr{C}^*(G)$ with the same support. Conversely, if s is a section of $\mathscr{C}^*(G)$ with compact support $|s| \subset U$ then for any point $x \in |s|$ one can find a neighborhood $V_x \subset U$ and an element $c_x \in D^*(V_x)$ such that $r(c_x)$ coincides with s on V_x. Choosing a finite (subcovering) from the covering $\{V_x\}$ of the set $|s|$ and using the fact that r is a monomorphism, it is easy to find a cochain ξ with support $|s|$ which coincides (under the inclusion r) with the section s. Analogous arguments establish the c-softness of $\mathscr{C}^*(G)$ from which (since B is locally compact and paracompact) the softness follows easily. The fact that $\mathscr{C}^*(G)$ is a resolution of G follows from general considerations recounted in points 2.3 and 2.4 of Chap. 1 (since the reduced cohomology groups of a point are equal to zero, cf. Example 1 in Sec. 1.1 of [126]; in dimension 0 one can use the arguments of point 2.1 above). Similar arguments are given in more detail in Sec. 7 of [62] (although for a different type of cochain).

A remarkable property of the cochains considered is that for any open $U \subset B$ the complexes $C_c^*(U; R)$ are free† (R being any principal ideal ring). Moreover, the modulus $C_c^p(U; R)$ are

†A brief account of this and of the possibility of constructing a homology theory on this basis (cf. the next point) was published in [125].

direct summands of $C_c^p(B; R)$ (Sec. 4.4 of [126]). This means in particular (cf. Sec. 5 of Chap. 1 above), that the sections of the sheaves $\mathscr{C}^*(R)$ with compact supports over any closed $Y \subset B$ are also free cochain complexes.

3.2. Chains of Locally Compact Spaces. The naturality of the existence of free cochains and also the forms of their connection with chains is corroborated by the arguments cited in points 1.2 and 1.3 of Chap. 2. The chains in Chap. 4 of [126] (cf. also [125] are defined in precisely this way: $C_*(B; G) = \mathrm{Hom}_R(C_c^*(B; R), G)$. To the exact sequence $0 \to C_c^*(U; R) \to C_c^*(B; R) \to C_c^*(F; R) \to 0$ with closed $F = B \setminus U$ corresponds (by virtue of the splitting in each dimension) a short exact sequence of chains $0 \to C_*(F; G) \to C_*(B; G) \to C_*(U; G) \to 0$ and consequently, a homology exact sequence of the pair (B, F), in which $H_*(B, F; G)$ is the homology of the complex $C_*(B, F; G) = C_*(U; G)$. The subcomplex $C_*^c(B; G) \subset C_*(B; G)$ of chains with compact supports is defined in [126] as the union of all $C_*(F; G)$ with respect to compact $\mathscr{C} \subset B$. One can also give the definition of the support $|\xi|$ of a (not necessarily compact) chain $\xi \in C_*(B \cdot G)$ directly: it is the minimal (closed) set such that ξ vanishes on all cochains with (compact) supports in $B \setminus |\xi|$. Any cochain from $C_c^*(B, R)$ by virtue of the softness of these sheaves can be represented as a sum of cochains with supports of any fineness, so the support of a nonzero chain is always nonempty.

These facts are obviously sufficient for developing a theory along the lines of Sec. 2 of Chap. 2 (in particular, for the construction of flabby sheaves of chains). As noted in point 2.2 of Chap. 2, chains are also constructed analogously for locally constant systems of coefficients $\mathscr{G}$.

3.3. General Case. The inclusion of the closed set $F \subset B$ induces an inclusion of chains $C_*(F; \mathscr{G}) \subset C_*(B; \mathscr{G})$ so for an arbitrary (paracompact, Hausdorff) space X, complexes of chains $C_*^c(X; \mathscr{G}) = \bigcup_{C \subset X} C_*(C; \mathscr{G})$ and $C_*(X; \mathscr{G}) = \bigcup_{B \subset X} C_*(B; \mathscr{G})$ where C is always compact and B is closed locally compact subspace X, consequently, and homologous $H_*^c(X; \mathscr{G})$ and $H_*(X, \mathscr{G})$ of the first (i.e., with compact supports) and second kinds. Analogously one defines chains and homology groups of arbitrary pairs (X, Y) [with respect to the corresponding pairs (C, C') and (B, B'), and $C' \subset Y$ and $B' \subset Y$]. These chains are assembled into soft sheaves, cf. point 2.5 of Chap. 2. In [126] only H_*^c are considered, defined as $H_q^c(X, Y; G) = \varinjlim H_q(C, C'; G)$ (Chap. 9 of [126]).

3.4. Remarks. A useful advantage of Massey chains and cochains is their functorial dependence (for fixed coefficients) on the space X. There are also inadequacies. Thus, chains and cochains are different from zero in all positive dimensions, in connection with which, for example, one has to prove specially (cf. Theorem 3.21 of Chap. 3 of [126]) that the homology and cohomology groups of a manifold M are equal to zero in dimensions $q > \dim M$. The approach is in applicable for obtaining relations connecting H_* with Cech homology $\check{H}_*$. It becomes necessary to verify (cf. Sec. 4.11 of [126]) the invariance of homology under change of ring (there is no such invariance, for example, in Borel–Moore theory, cf. below Sec. 6). The question of invariance does not arise in direct definitions of chains (independently of cochains), cf. the next section.

4. Cech Type Chains and Cochains

4.1. Free cochains of Cech type with coefficients in a principal ideal ring R, defining the cohomology groups H_c^* with compact supports, were constructed in [62] (for spaces without the second axiom of countability, in the remark on Sec. 1 in [64]; cf. also Sec. 2 in [65] or the remarks after Theorem 3.1 of [68]). For this locally finite coverings α of a locally compact (paracompact Hausdorff) space B consisting of "canonical" compact sets F (coinciding with the closures of Int F) such that $\mathrm{Int}\, F \cap \mathrm{Int}\, F' = \varnothing$ for any $F \neq F'$ from α were considered. For such coverings α, β let $\alpha \wedge \beta$ be the covering composed of closures of sets $\mathrm{Int}\, F \cap \mathrm{Int}\, F'$, $F \in \alpha$, $F' \in \beta$. Let ω be any system of such coverings α satisfying the following requirements: 1) for $\alpha, \beta \in \omega$ there exists a $\gamma \in \omega$ inscribed in $\alpha \wedge \beta$ $(\alpha \wedge \beta < \gamma)$; 2) coverings from ω can be locally refined (for any neighborhood U of the point $x \in B$ one can find an $\alpha \in \omega$ such that $F \subset U$ if $x \in F$, $F \in \alpha$). The projections of the nerves of coverings from ω inscribed in one another are single-valued, are proper surjective maps, so for any R-module G there is defined a complex $C_{\omega c}^*(B; G) = \varinjlim_{\alpha \in \omega} C_c^*(\alpha; G)$ of cochains with compact supports (which is obviously the union of its subcomplexes $C_c^*(\alpha; G)$]. For any system ω of the type indicated, it also defines (cf. below) groups $H_c^*(B; G)$.

Instead of infinite coverings α to construct the theory it is more convenient to use pairs (α, α'), where $\alpha \in \omega$ and α' is any part of α containing all the elements of α except for a finite number of them. Considering that $(\alpha, \alpha') < (\beta, \beta')$ if $\alpha < \beta$ and the union of β' is contained in the union of all $F \in \alpha'$ and considering that $C_c^*(\alpha; G) = \varinjlim_{\alpha'} C^*(\alpha, \alpha'; G)$ we conclude that $C_{\omega c}^*(B; G) = \varinjlim_{(\alpha, \alpha')} C^*(\alpha, \alpha'; G)$. If G = R, the modules $C^q(\alpha, \alpha'; R)$ are free and finitely generated and the inductive system of such modules satisfies the conditions of [114], so $C_c^q(B; R)$ are free R-modules [containing $C^q(\alpha, \alpha'; R)$ as direct summands].

One can treat the pairs (α, α') as canonical coverings of the compactification $\tilde{B}$ (if B is noncompact) by one point $*$, so $C_{\omega c}^*(B; G)$ is simultaneously the cochain complex of the compact pair $(\tilde{B}, *)$. Due to the commutation of the operations $\varinjlim$ and H^q the cohomology groups of $C_{\omega c}^*(B; G)$ coincide with the limit $\varinjlim H^q(\alpha, \alpha'; G)$ which, as is familiar, is simply the Cech cohomology $H^q(\tilde{B}, *; G)$. From this, $H_c^*(B; G) = H^*(\tilde{B}, *; G)$ (and in particular the cohomology is independent of of the choice of ω).

The requirement that the coverings $\alpha \in \omega$ consist of canonical sets is rather burdensome (for example in considering the cohomology groups of subspaces of B), but it is essential. One can consider locally finite coverings consisting of any closed sets (possibly with repetitions) not all of which are even compact. However along with the requirements 1) and 2) above, to each pair $\alpha < \beta$ there should correspond a surjective projection of their nerves coinciding with the compositions of the intermediate ones for $\alpha < \gamma < \beta$. Such coverings arise, for example, when ω is the intersection with B of a system of locally finite coverings by canonical closed sets of any space X containing B. In this case the cochains $C_{\omega c}^*(B; G)$ should be defined in terms of limits of cochains of pairs (α, α') in which α' is a system of all sets from α except for a finite number of compact ones. All the basic assertions about the complex $C_{\omega c}^*(B; G)$ asserted above (including their freeness for G = R) remain valid. Moreover, the restriction of ω to any closed set $A \subset B$ (obtained from the intersections $F \cap A$, $F \in \alpha$, $\alpha \in \omega$) defines cochains $C_{\omega c}^*(A; G)$ and an epimorphism $C_{\omega c}^*(B; G) \to C_{\omega c}^*(A; G)$ (which splits for G = R since the cochains of A are also free in each separate dimension).†

One easily defines the support of a cochain from $C^*(\alpha, \alpha'; G)$ (the union of the intersections of these sets of α on which the cochain is different from zero), hence any cochain of the complex $C_{\omega c}^*(B; G)$, as the union of the complexes of pairs (α, α') (with respect to all $\alpha \in \omega$ and all α'). Hence, just as in Sec. 3, one defines presheaves of cochains leading to sheaves $\mathscr{C}_\omega^*(G)$ on B, whose sections with compact supports are $C_{\omega c}^*(B; G)$. From general considerations recounted in Sec. 2 of Chap. 1, it follows that they are c-soft and consequently (cf. with Sec. 3), also soft. Since for compact neighborhoods A of points $x \in B$ the cohomology groups of the complex $C_\omega^*(A; G)$, being direct limits of the cohomology groups $H^n(\alpha', G)$ over $\alpha' = \alpha \cap A$, coincide with $H^n(A; G)$ and the limits of these groups over such neighborhoods of points coincide with the cohomology groups of a point, $\mathscr{C}_\omega^*(G)$ is a soft resolution of the constant sheaf G (in dimension zero a cochain of a covering considered as a section of $\mathscr{C}_\omega^0(G)$ is a cocycle precisely when it assumes constant values on the connected components of the nerve of α and considered as a function on the elements of the covering α is locally constant on B, cf. point 2.4 of Chap. 1). From this it again follows in particular that the definition of the cohomology groups is independent of the system of coverings ω which participate in it.

4.2. Chains of Locally Compact Spaces. These were constructed in [62] (for spaces without the second axiom of countability, in the remark on Sec. 1 of [64]). They are defined as $C_*^\omega(B; G) = \varprojlim_{\alpha \in \omega} C_*(\alpha; C) = \varprojlim_{(\alpha, \alpha')} C_*(\alpha, \alpha'; G)$, where ω, α, and α' are the same as in the preceding point. Just as for the cohomology groups H_c^* there arise relations between $H_*(B; G)$ and the homology groups of the one-point compactification of B. In the same way, besides systems ω of canonical coverings one can consider a larger class [satisfying requirements 1) and 2), with single-valued surjective projections of nerves, consisting closed locally finite coverings].

The definition of chains considered is obviously independent of cochains (cf. point 3.4 above). However there is a connection (which is the basis of the definition in Sec. 3): $C_*^\omega(B; G) = \mathrm{Hom}_R(C_{\omega c}^*(B; R), G)$ [a consequence of the obvious relation $\mathrm{Hom}(\varinjlim \ldots, G) = \varprojlim \mathrm{Hom} \times (\ldots, G)$]. A consequence of it, in particular, is the independence of the homology groups

†Since $C_{\omega c}^*(B, G) = C_{\omega c}^*(B; R) \otimes_R G$ obviously, the splitting also holds for any R-module of coefficients G.

which arise from the system of coverings ω figuring in the definition. If σ is another such system, then for $\tau = \omega \wedge \sigma$ there are maps of complexes of free cochains (with coefficients in R) $C^*_{\omega c} \to C^*_{\tau c}$, $C^*_{\sigma c} \to C^*_{\tau c}$ inducing the identity isomorphism of cohomology groups and which are therefore homotopy equivalences (Sec. 4 of [62]). Consequently, the chain maps $C^\tau_* \to C^\omega_*$ and $C^\tau_* \to C^\sigma_*$ are homotopy equivalences also. Another consequence [considering the properties of the sheaves $\mathscr{C}^*_\omega(R)$] is the possibility of treating the chains (cf. with Sec. 3) as sections of flabby sheaves, making use of the theory in the outline of Sec. 2 of Chap. 2.

Things are more complicated for locally constant coefficients $\mathscr{G}$ (with stalk G). The chain complexes $C^\omega_*(B; G)$ are defined in the same way (here one imposes an additional requirement on ω: the elements of the coverings $\alpha \in \omega$ should be sufficiently fine that the restriction of $\mathscr{G}$ to each set $F \in \alpha$ is constant). As to the sheaves of chains $\mathscr{C}^\omega_*(\mathscr{G})$ since one can identify chains with functions on free generators of the complex $C^*_{\omega c}(B; R)$ with values in the stalks of $\mathscr{G}$ over the supports of the generators, these sheaves are constructed in the same way as for constant coefficients and have the same properties (cf. Sec. 3 of [64]; in particular, since the $\mathscr{C}^*_\omega(\mathscr{G})$ are locally isomorphic to the flabby sheaves $\mathscr{C}^\omega_*(G)$ they are also flabby according to point 3.1 of Chap. 2 in [107]).

It is not obvious however, even so, why $C^\omega_*(B; \mathscr{G}) \neq 0$ because there are examples of projective systems of free groups of finite rank with epimorphic maps but with zero (or countably generated) limits [38]. Nevertheless this is so; there are even epimorphisms $C^\omega_*(B; \mathscr{G}) \to C_*(\alpha, \alpha'; \mathscr{G})$ (Lemma 3.2 of [68]). One is able to show (Lemma 3.3 of [68]), that to an exact sequence of coefficient systems $0 \to \mathscr{G}' \to \mathscr{G} \to \mathscr{G}'' \to 0$ there corresponds a short exact sequence of the corresponding chain complexes (for constant coefficients this is an obvious consequence of the connection with free cochains). From this, using the existence of a functorial imbedding of any R-module G in the compactum S = Hom(Hom(G, S^1), S^1) (S^1 being the group of real modulo 1), we construct for $\mathscr{G}$ a short exact sequence of systems $0 \to \mathscr{G} \to \mathscr{S} \to \mathscr{K} \to 0$ in which $\mathscr{S}$ has compact stalks S, and considering that $\underset{(\alpha, \alpha')}{\overleftarrow{\lim}}{}^p C_*(\alpha, \alpha'; \mathscr{S}) = 0$ for $p \geqslant 1$ with the help of the corresponding short exact sequence of projective systems

$$0 \to C_*(\alpha, \alpha'; \mathscr{G}) \to C_*(\alpha, \alpha'; \mathscr{S}) \to C_*(\alpha, \alpha'; \mathscr{K}) \to 0$$

the corresponding exact sequences of the derived functors $\overleftarrow{\lim}{}^p$ we conclude that $\underset{(\alpha, \alpha')}{\overleftarrow{\lim}}{}^p C_*(\alpha, \alpha'; \mathscr{G}) = 0$ for $p \geqslant 1$ also for any coefficient systems $\mathscr{G}$.

Finer analysis of the constructions shows (Sec. 3 of [68]) that $\underset{(\alpha, \alpha')}{\overleftarrow{\lim}}{}^p Z_n(\alpha, \alpha'; \mathscr{G}) = 0$ for $p \geqslant 2$ ($Z_n(\alpha, \alpha'; \mathscr{G})$ are the cycles of the complexes $C_*(\alpha, \alpha'; \mathscr{G})$). It follows from this (Theorem 3.2 of [68]) that $\underset{(\alpha, \alpha')}{\overleftarrow{\lim}}{}^p H_k(\alpha, \alpha'; \mathscr{G}) = 0$ for $p \geqslant 2$ and there are exact sequences

$$0 \to \underset{(\alpha, \alpha')}{\overleftarrow{\lim}}{}^1 H_{k+1}(\alpha, \alpha'; \mathscr{G}) \to H_k(B; \mathscr{G}) \to \underset{(\alpha, \alpha')}{\overleftarrow{\lim}} H_k(\alpha, \alpha'; \mathscr{G}) \to 0 \tag{1}$$

(cf. in this connection points 1.1 and 1.2 of Chap. 8 in [69]). With the help of special systems of coverings and the tools of homological algebra one can establish (end of Sec. 3 in [68]) that the extreme terms of this sequence are independent of the choice of ω. Comparing such sequences for ω, σ, and $\tau = \omega \wedge \sigma$ as above, with the help of the five lemma we conclude that the definition of the groups $H_k(B; \mathscr{G})$ is independent of ω (Theorem 3.1 in [68]).

The homology groups $H_*(Y; \mathscr{G})$ of any closed subset $Y \subset B$ are independent of the choice of ω and are defined by the intersection of Y with the system ω. The independence of the homology groups of the pairs (B, Y) from ω is a consequence of the method of comparison indicated above of the homology sequencs of pairs and the five lemma. Finally, the independence of the homology groups of any subspaces $M \subset B$ from ω follows from their representations as direct limits with respect to suitable (closed in B) supports.

The complex $C^\omega_*(Y; \mathscr{G})$ defining the homology of a closed set $Y \subset B$ (obtained with the help of a system of coverings of Y which are the intersections with Y of the coverings of ω) coincides with the subcomplex of $C^\omega_*(B; \mathscr{G})$ consisting of all chains with supports in Y. Indeed the support of the chain $\xi \in C^\omega_*(B; \mathscr{G})$ is the (minimal) closed set $|\xi|$ such that for any point $x \in B \setminus |\xi|$ one can find a neighborhood U of this point and an $\alpha \in \omega$ such that the image of ξ in $C_*(\alpha, \alpha'; \mathscr{G})$ (let the sets from α' not intersect U) has zero coefficients for all intersections of sets from α (corresponding to simplices of the nerve of α) having nonempty intersection with U. For chains with sufficiently small supports, the assertion follows from the coincidence of this definition of support with the definition given in Sec. 3 for constant coefficients [applied to chains of $C_*^{\omega}(B;\mathcal{G})$] for any decomposition of them as sums of chains with sufficiently small supports.

4.3. General Case. In order to get chains for the homology groups of an arbitrary (paracompact Hausdorff) space X, we fix on X a system ω of locally finite coverings by closed canonical sets satisfying the requirements 1) and 2) above, for the elements of each covering of which it follows from F ≠ F' that $\operatorname{Int} F \cap \operatorname{Int} F' = \varnothing$. Intersecting ω with each compact set C ⊂ X or any closed locally compact B ⊂ X we get complexes $C_*^{\omega}(C;\mathcal{G})$, $C_*^{\omega}(B;\mathcal{G})$ for them, and as the direct limits of such complexes with respect to the C and B indicated, chain complexes $C_*^{\omega c}(X;\mathcal{G})$ and $C_*^{\omega}(X;\mathcal{G})$ which also define the homology groups H_*^C and H_* of the space X needed, and upon considering supports and so on, also of any subspaces of X and pairs. The independence of the homology groups which arise from ω follows, for example, from relations of the type $H_n^c(X;\mathcal{G}) = \varinjlim_{C \subset X} H_n(C;\mathcal{G})$, $H_n(X;\mathcal{G}) = \varinjlim_{B \subset X} H_n(B;\mathcal{G})$ etc.

4.4. Some Remarks. An inadequacy of the construction of chains (and also cochains) considered in the present section is the necessity of verifying the independence of the theories which arise from the choice of the system of coverings defining the chains (and cochains); actually the proofs of their topological invariance are similar to the proof of the independence of the classical groups of polyhedra from the choice of their concrete triangulations. Inadequacies appear basically in defining the homomorphisms induced by continuous maps of spaces. The chains and cochains considered however are convenient in that (where this is possible) they let one, by the choice of appropriate systems of coverings, consider some additional information (countability considerations — in the metric case one can use as ω sequences of refinements of coverings inscribed in one another; dimensional considerations — by the choice of coverings of the multiplicity needed on the whole space or on certain subspaces, etc.). The quivalenc eof the approaches to the description of the homology groups considered in Secs. 3 and 4 for constant coefficients is established in Sec. 6 of [65]. The equivalence is also known for locally constant coefficients (although the proof is unpublished).

The idea of the description of homology and cohomology groups with the help of chains and cochains defined by coverings with unique inscription arose in the thirties. Under the influence of L. S. Pontryagin's theory of characters however it became traditional to treat the homology groups as the character groups of the cohomology groups and they are usually considered with coefficients in compact groups G and the topology of the inverse limit came to be considered in their definition (and hence in the consideration of the groups of cycles and boundaries also). By tradition this came to be done also for noncompact G and hence to operate with nonclosed subgroups which, in contrast with the groups of cycles, upon passage to direct limits usually turned out to be subgroups of the boundaries, was considered slightly natural, such groups were as a rule closed, which usually led to Cech type homology groups. In the case of noncompact spaces the necessity of considering infinite coverings, and in their consideration, the compactness factor (in accord with which, cf. the Introduction and Sec. 1 of Chap. 2, it is natural to define chains of general spaces not as inverse limits of chains of coverings, but as direct limits of such inverse limits, first considered for compact subspaces of the space studied) was not always considered. The existence of a large number of logically possible versions of the original definition of homology groups also led to the fact that the homology groups of N. Steenrod (1940) and K. A. Sitnikov (1951) which lies at the base of the modern theory were not immediately noticed and their value only became conclusively understood at the beginning of the sixties. In connection with the history of the development of homology groups, the influences on this of the inexactness of the inverse limit functor and some other factors, cf. [62], and also point 2.1 of Chap. 2 and point 2.4 of Chap. 5 of [69].

5. Some Inferences from the Apparatus of Chains and Cochains

5.1. Universal Coefficient Formulas. Let Y be a closed subset of the paracompact space X, φ be a paracompactifying family of supports. For a finitely generated module G over a principal ideal ring R, or more generally, for any finitely representable module over a hereditary ring (which can even be non-Noetherian) one has the following exact sequences (universal coefficient formulas)

$$0 \to H_{\varphi}^n(X,Y;R)\otimes_R G \to H_{\varphi}^n(X,Y;G) \to \operatorname{Tor}_R\left(H_{\varphi}^{n+1}(X,Y;R),\ G\right) \to 0. \qquad (2)$$

The restriction on G is explained by the fact that the groups of cochains of a covering $C^n(\alpha; \mathcal{G})$ are isomorphic to infinite products ΠG for which $\Pi G = (\Pi R) \otimes_R G$ precisely for finitely representable modules G (Lemma 7 of [62]). Thus, (2) does not hold, for example, for the zero-dimensional cohomology group of a countable discrete space.

Considering that the groups H^* coincide with the Cech cohomology groups $\check{H}^*$ in the case when φ is the family of all closed sets, the relations (2) are a consequence of the existence of such formulas for polyhedra and the permutability of the functors $\otimes$ and Tor with direct limits [with respect to open coverings α of the space X with the help of which the groups $\check{H}^*$) are defined.

By the support of a cochain $\xi \in C^*(\alpha; G)$ is meant the closure of the union of all those intersections of sets from α (corresponding to simplices of the nerve of α) on which ξ is different from zero. The subcomplexes $C^*_\varphi(\alpha; G) \subset C^*(\alpha; G)$ composed of cochains with supports in φ are compatible with the maps corresponding to the inscribed coverings, and their cohomology groups, upon passage to the direct limit, define $\check{H}^*_\varphi(X; G) = H^*_\varphi(X; G)$ (cf. point 2.2 of Chap. 4 of [69]). Moreover, the complexes $C^*_\varphi(X; R)$ are torsion-free and $C^*_\varphi(X; G) = C^*_\varphi(X; R) \otimes_R G$ so the relations (2) are true in general also.

One can also make these inferences starting from the complex of Alexander–Spanier cochains. Here one gets an additional fact: the sequence (2) splits (the splitting is a consequence of the general construction of (2) in homological algebra, cf. Theorem 3.3 of Chap. 6 of [92], it is natural in the argument G but not in (X, Y)).

From the description of cochains given in point 4.1, it follows quickly that $C^*_{\omega c}(B; G) = C^*_{\omega c}(B; R) \otimes_R G$ for any G so in the case of the cohomology groups H^*_c of locally compact spaces, the formulas (2) (they are split) hold for any modules G over hereditary rings (cf. Theorem 1 of [62]). In using Massey cochains of Sec. 3 this inference is more complicated (cf. Sec. 4.11 in [126] for R = Z and the editor's appendix to Chap. 4 of [126] for the general case).

It is clear from the description of the chains of locally compact spaces (points 3.2 and 4.2 above) that in each separate dimension they are isomorphic to the infinite product ΠG so $C_*(B; G) = C_*(B; R) \otimes_R G$ only for finitely representable G. The chains of an arbitrary (paracompact Hausdorff) space X as well as any subspaces of X and pairs, can be described as direct limits over closed locally compact subsets of a family φ (not necessarily compactifying), and hence also satisfy the relation indicated. Thus, for any finitely representable module G over a hereditary ring, and any family of supports φ (in particular, for $\varphi = c$, for homology groups with compact supports) one has the splitting exact sequences

$$0 \to H^\varphi_n(X, Y; R) \otimes_R G \to H^\varphi_n(X, Y; G) \to \mathrm{Tor}_R\left(H^\varphi_{n-1}(X, Y; R), G\right) \to 0 \tag{3}$$

(universal coefficient formulas). The fact that the restriction on G is essential is demonstrated by the one-point compactification of a countable discrete set.

The relations (3) are obtained in Theorems 2 and 2' of [62] (cf. also formulas (2.2) and (2.3) of [65]). They are also obtained with the help of Massey chains (cf. the editor's appendix to Chap. 4 of [126]). The validity of (2) and (3) (for modules G of finite type and the family φ of all closed sets) in the category of polyhedra [locally compact ones for (3)] also follows from Theorem 10 of Sec. 5 of Chap. 10 in [141].

Since for the chains of a locally compact space B one has $C_*(B; G) = \mathrm{Hom}_R(C^*_c(B; R), G)$ for any module G over a hereditary ring R, the following universal coefficient formulas hold (F being closed in B)

$$0 \to \mathrm{Ext}_R\left(H^{n+1}_c(B, F; R), G\right) \to H_n(B, F; G) \to \mathrm{Hom}_R\left(H^n_c(B, F; R), G\right) \to 0 \tag{4}$$

These exact sequences split (cf. the general construction in Sec. 3 of Chap. 6 in [92]), and split naturally with respect to the argument G. The relations are obtained in Theorem 3 of [62] (cf. also (2.1) in [65]). For compact spaces for R = Z they were previously obtained in [128] (and in terms of the original definition of N. Steenrod, in 1942 by S. Eilenberg and S. MacLane, cf. Sec. 2 of [65]). The relations are also obtained with the help of Massey chains (Sec. 4.8 of [126] for Abelian groups and the editor's supplement to Chap. 4 of [126], for modules).

For topological spaces endowed with some additional structure (homologically locally connected, with finitely generated homology or cohomology groups with coefficients in R, etc.) some other relations also hold (for example, expressing the cohomology in terms of the homology), superficially not differing from universal coefficient formulas, but holding in situations when the standard connections between chains and cochains by means of the functors $\otimes$ and Hom are absent and proved with the help of different tools of homological algebra. Moreover, in specific situations, in the absence of concrete forms of connection between functors, the so-called functorial dependence introduced by G. Bredon holds when the existence of some functors under concrete continuous maps of spaces implies the isomorphism of others (thus depending on the first). Cf. points 4.2 and 4.3 of Chap. 8 in [69] for a survey of results of this kind.

5.2. Passage to the Limit. Connection with Cech Homology Groups. For compact X (for example, upon replacing a locally compact space by its one-point compactification, which has essentially the same homology groups) the relations (1) of Sec. 4 (which generally play an auxiliary role there in checking the definition) give exact sequences

$$0 \to \varprojlim_{\alpha}{}^1 H_{n+1}(\alpha; \mathcal{G}) \to H_n(X; \mathcal{G}) \to \check{H}_n(X; \mathcal{G}) \to 0, \tag{5}$$

relating the groups H_n with the Cech homology groups $\check{H}_*$.

In this form this result is given in the editor's supplement to Chap. 4 of [126] (one can use both closed and open coverings α here, cf., e.g., Sec. 3 of [68]). For constant coefficients G the result was obtained earlier in [36] (assertion 3 of Theorem 2 and assertion 1 of Theorem 3, the group Pext^1 figuring in them coincides with $\varprojlim{}^1$ by virtue of Theorem 6 of [32]) and was afterwards reproved in [40]. For metric compacta the relations were known from the beginning of the sixties (Milnor, Roos, Deheuvels, etc.).

The relations (5) were used for finding various conditions (on X and $\mathcal{G}$) under which $H_n = \check{H}_n$. This is so, for example, for $n = \dim X$ [62], for homologically locally connected X [62] (for metrizable locally compact X, $H_*^c(X; Z) = \check{H}_*^c(X; Z)$ [44, 76]) in the case of metric X for countable G and $H_n(X; G)$ (since for countable projective systems consisting of countable groups A_i either $\varprojlim_i{}^1 A_i = 0$ or the cardinality of $\varprojlim_i{}^1 A_i$ is at least that of the continuum, cf. Lemma 3 in [76] or the more general result, Lemma 1.6 of [66]), when $H^{n+1}(X; Z)$ is the direct sum of a free group and a torsion group [36] (and in other cases). Cf. also point 1.3 of Chap. 8 of [69]. The isomorphism $H_*(X; G) = \check{H}_*(X; G)$ holds for all compact X precisely under the condition that the group G is algebraically compact [36] (in particular, finite, simply compact, divisible, or is the group of a field). For a locally compact space X with the second axiom of countability (for example, for a locally finite polyhedron) it follows from (5) that for a finitely generated module G over a countable ring R, the modules $H_n(X; G)$ [analogously, $H^n(X; G)$ if X is also homologically locally connected] are either finitely generated or have cardinality of the continuum [68].

For algebraically compact G, since $H_* = \check{H}_*$ in the category of compact spaces the theory H_* is continuous [36]: $H_n\left(\varprojlim_{\lambda} X_\lambda; G\right) = \varprojlim_{\lambda} H_n(X_\lambda; G)$. In general there is only a transformation $\sigma\colon\ H_n(X; G) \to \varprojlim_{\lambda} H_n(X_\lambda; G)$, where $X = \varprojlim_{\lambda} X_\lambda$. Since the chain complexes $C_*(X)$ and $C_*^\lambda = C_*(X_\lambda)$ have the form $\mathrm{Hom}_R(C_\lambda^*, G)$, where C_λ^* are free cochain complexes, the transformation σ is contained in an exact sequence of the form

$$0 \to \varprojlim_{\lambda}{}^1 \mathrm{Hom}\,(H^{n+1}(C_\lambda^*), G) \to H_n\left(\varprojlim_{\lambda} C_*^\lambda\right) \xrightarrow{\sigma} \varprojlim_{\lambda} H_n(C_*^\lambda) \to \varprojlim_{\lambda}{}^2 \mathrm{Hom}\,(H^{n+1}(C_\lambda^*), G) \to 0. \tag{6}$$

The existence of the exact sequences (6) (not only for chain and cochain complexes of compact spaces X_λ but also formally in a somewhat more general situation) was established in [36] (Lemma 1 and the beginning of Sec. 2, and also assertion 1 of Theorem 5). The authors studied the structure of the transformation σ in detail, in particular, they showed the nontriviality in general of the fourth term in (6) (Example 2 in [36]). For finitely generated groups A_λ (in particular, for the cohomology groups of nerves of finite coverings), and also for countable projective systems $\varprojlim_{\lambda}{}^p \mathrm{Hom}\,(A_\lambda, G) = 0$ for $p \geqslant 2$ and from the universal coefficient formulas $\varprojlim_{\lambda}{}^1 \mathrm{Hom}\,(H^{n+1}(C_\lambda^*), G) = \varprojlim_{\lambda}{}^1 H_{n+1}(C_*^\lambda)$ so the relations (6) become the exact sequences (5) (or their version in more general situations), cf. assertion 2 of Theorem 5 in

[36].† Later, relations (6) (by application of the Roos spectral sequence to projective systems composed of universal coefficient formulas) were also obtained in [127]. For $p\geqslant 2$ it was shown in addition that

$$\varprojlim_{\lambda}{}^{p} H_n(C_*^{\lambda})=\varprojlim_{\lambda}{}^{p}\operatorname{Hom}(H^n(C_\lambda^*),G)\oplus\varprojlim_{\lambda}{}^{p+2}\operatorname{Hom}(H^{n+1}(C_\lambda^*),G).$$

This permitted the author to extend (6) to an exact sequence which is infinite in both directions, with the groups $\varprojlim_{\lambda}{}^{2k} H_{n+k}(C_*^{\lambda})$ on the right and $\varprojlim_{\lambda}{}^{2k-1} H_{n+k}(C_*^{\lambda})$ on the left, also in terms of the derived functors of an arbitrary left exact covariant additive functor F (cf. [41, 127]). (Such an extension however does not carry additional information.) Cf. also points 1.1 and 1.2 of Chap. 8 of [69].

An analogous technique, together with the interpretation of chains and cochains as sections of flabby differential sheaves which is essential for these purposes is applicable for obtaining other limit relations. Despite the apparent multitude, they can be divided into types: a) connection of the cohomology groups of Y with the cohomology groups of subspaces $Y_\lambda\subset Y=\bigcup_\lambda Y_\lambda$, $Y_\lambda\subset \operatorname{Int} Y_\mu$ for $\lambda<\mu$; b) connection of the cohomology groups of the pair (X, Y) for open $Y\subset X$ with the cohomology groups of the pairs (X, Y_λ) (for the Y_λ indicated); c) connection of the homology of the second kind of a locally compact space Y with the homology groups of open $Y_\lambda\subset Y$ (of the type indicated); d) for an open subspace Y of a locally compact space X, a connection of the homology groups of the second kind $B = X\setminus Y$ with the homology groups of the sets $B_\lambda = X\setminus Y_\lambda$ enclosing B. In each case there arise spectral sequences which by means of the functors $\varprojlim{}^{p}$ connect the groups indicated for Y with systems of gropus corresponding to Y_λ which degenerate under natural supplementary conditions (for example, for countable systems Y_λ) into exact sequences of the form (5) (with the functor $\varprojlim$ figuring in them instead of $\check{H}_n$). Information about the conditions for $\varprojlim{}^{1}$ vanishing, δ about the cardinality of the values assumed by this functor, let one get extensive information about the structure of the homology and cohomology groups studied. The information known in this way is rather fully reflected in points 1.4 and 1.5 of Chap. 8 of [69].

5.3. Additive Properties. Connections with Questions of Axiomatics. Besides the basic properties concentrated in the expressions in the Eilenberg–Steenrod axioms or following directly from them, outside the limits of the category of finite polyhedra, in homology and cohomology theories one can necessarily find some which do not follow from these axioms. Some of them turn out to be obvious, considered clear in themselves, and usually one pays little attention to them. For this reason however one sometimes forgets to consider them in the development (in the large) of some new concrete theories (cf. also the Introduction). Additivity properties are the first of these.

When $X=\bigcup_\lambda X_\lambda$ is the discrete union of its closed subspaces X_λ for the homology groups H_*^c with compact supports there is obviously a natural isomorphism $H_n^c(X;\mathcal{G})=\bigoplus_\lambda H_n^c(X_\lambda;\mathcal{G})$ (the property of Σ-additivity) while at the same time for the theory H_* of the second kind, $H_n(X;\mathcal{G})=\prod_\lambda H_n(X_\lambda;\mathcal{G})$ (property of Π-additivity). An analogous picture holds (if one compares with respect to supports) for cohomology groups (cf. point 2.7 of Chap. 1).

In the compact category, instead of discrete unions one considers compact bouquets, unions of closed subspaces X_λ whose pairwise intersections $X_\lambda\cap X_\mu$, $\lambda\neq\mu$ consist of a unique point $x_0\in X$ each neighborhood of which contains all but a finite number of the X_λ. For any such bouquet, one can form a system ω defining chains $C_*^\omega(X;\mathcal{G})$ and cochains $C_\omega^*(X;\mathcal{G})$ (cf. Sec. 4) from the coverings α each of which contains all X_λ except a finite number as one of the elements. In this case the chains (respectively, cochains) of the pair (X, x_0) split

†In the case of a countable projective system $\{X_i\}$ of metric compacta, relations of the form (5) for $X=\varprojlim_i X_i$ for any extraordinary homology groups h_* of Steenrod type, Sec. 8.5 of [103].

It is known [56] that any extraordinary homology theory defined on the category of finite polyhedra extend to a Steenrod type theory (on the category of arbitrary compacta), while such an extension is Π_c-additive (cf. the next point). Cf. [103] (and some other papers) for such an extension to the category of metric compacta.

naturally into the direct product (respectively, direct sum) of chains (cochains) of pairs (X_λ, x_0) from which analogous splitting into a direct product and direct sum, respectively, of the groups $H_n(X;\mathcal{G})$ and $H^n(X;\mathcal{G})$ follow quickly, the properties of Π_c-additivity of homology groups and Σ_c-additivity of cohomology groups.

Formulated as axioms, the additivity properties play a paramount role in problems of axiomatics of homology and cohomology groups in appropriate categories of topological spaces (including polyhedra), containing the category of finite polyhedra. The value of such axioms in the categories of infinite polyhedra and metric compacta were first noted in [128, 129].

In using the uniqueness of homology and cohomology theories properly one speaks of numerous purely logical observations specially made for this in Chaps. 1 and 2. In the case of the cohomology groups of a fixed space one of the confirmations of this is the axiomatic characterization of sheaf cohomology groups in terms of derived functors (point 4.6 of Chap. 1). Uniqueness theorems based on Eilenberg–Steenrod type axioms characterize the homology and cohomology groups in specific categories of topological spaces for fixed groups (modules) of coefficients G. We note that the Eilenberg–Steenrod axioms themselves which constitute the basis of a general list are often required with a refined excision property (depending on the category considered): the homology (and cohomology) groups of the pair (X, Y) coincide with the reduced homology (cohomology) groups of the quotient-space X/Y if Y is compact or with the groups of the complement $X \setminus Y$ for closed $Y \subset X$ in the case of theories of the second kind. Striking evidence of the regularity of the theories considered is their complete and unique reconstructability from the list of Eilenberg–Steenrod axioms modified in the indicated way supplemented by a single new requirement, the additivity requirement suitable to each case. This is the situation in categories of infinite polyhedra (including separately locally finite ones), in the category of metric compacta, in the category of metrizable locally compact spaces and their proper maps, in the category of metrizable locally compact spaces with any maps (and in some others).

In some cases the requirement of additivity can be replaced by a no less natural condition on the local behavior of homology or cohomology groups (cf. below point 3.1 of Chap. 4). In the case of homology groups the requirement of additivity is closely connected with the property of compact supports (and in some cases is equivlent to it). Questions of axiomatics are completely elucidated in Chap. 7 of [69].

We note that in contrast with the Eilenberg–Steenrod axioms which have received unconditional recognition and are widely known, the supplementary requirements, in particular the simple and explicit requirements of additivity have not had their proper influence and the corresponding papers on axiomatics have remained long unnoted. In this sense the Eilenberg–Steenrod axioms have even inflicted a certain "harm" on the development of homology theory: by a homology (or cohomology) theory one should mean any functors (including occasionally ones which diverge from the usual homology groups in their restriction to the subcategory of polyhdra, cf. in this connection [65]), satisfying these axioms. In presenting a "new theory," authors consider it their duty to verify precisely these axioms (cf., e.g., [122], the homology groups considered in this paper turned out not to be additive and not equal to zero in dimension −1 [124]; cf. also point 1.1 of Chap. 2).

Again we recall that the Eilenberg–Steenrod axioms are complete only in the category of finite polyhedra, that even, for example, in categories of infinite polyhedra, besides the two independent homology theories H_*^c and H_*, as noted in the Introduction, quite random functors may satisfy these axioms. Thus, beyond the limits of the category of finite polyhedra the Eilenberg–Steenrod axioms should necessarily be supplemented by new requirements. The additivity requirements are obviously the most transparent among them.

<u>5.4. Some Other Observations.</u> In correspondence with point 4.3 of Chap. 1, to each short exact sequence of sheaves of coefficients on the space X corresponds the cohomology exact sequence of this space (with coefficients in these sheaves). For constant coefficients this fact is a consequence of any concrete version of the description of cochains (Secs. 1-4). The existence of analogous exact sequences for homology groups with constant coefficients is a consequence of the description of chains by means of means of the functor $\mathrm{Hom}_R(\ldots,G)$ of free cochains and passage to direct limits over corresponding families of supports, cf. Secs. 3 and 4. An exact homology sequence also corresponds to a short exact sequence $0 \to \mathcal{G}' \to \mathcal{G} \to \mathcal{G}'' \to 0$ of locally constant coefficient systems although it is somewhat more complicated to establish this (cf. point 4.2 above).

The excision property holds to the full extent for the homology groups H_*^c and H_* (as well as for the cohomology groups, cf. point 5.4 of Chap. 1): for a closed $Y \subset X$ and any set $U \subset Y$ which is open in X, the inclusion $(X\setminus U, Y\setminus U)\subset(X, Y)$ induces an isomorphism of homology groups. For a locally compact X such an isomorphism in a consequence of the equality $H_n(X, Y;\mathcal{G})=H_n(X\setminus Y;\mathcal{G})$ (obtained as a result of the description of the chains as sections of th flabby sheaves $\mathcal{C}^*(\mathcal{G})$ point 2.4 of Chap. 2). The general case is a consequence of passage to the direct limit with respect to the family of supports appropriate to the situation. In the case of compact Y, analogous arguments confirm the equality of the homology groups of the pair (X, Y) with the reduced homology groups of the quotient-space X/Y.

6. Borel–Moore Homology

The Borel–Moore theory [86] which appeared at the beginning of the sixties played a fundamental role in the formation of the modern theory of homology groups of general spaces. At this time the basic value of the Steenrod–Sitnikov homology groups [128] was already clear but the means employed in their description (cf. point 2.3 of Chap. 2) had not revealed any prospective paths for applying the homology theory. For the fist time the foundations of an apparatus letting one subsequently apply the homology (along with the cohomology) groups widely to problems arising naturally, using effective tools of homological algebra and sheaf theory, was created in [86].

From the very beginning however along with the obvious advantages there appeared numerous inadequacies of the new theory, generally confirming its strainedness and artificiality for general coefficient systems. In the present section we analyze the most typical of them and reveal their common cause.

6.1. Basic Construction. Let B be a locally compact space and R be a principal idea domain, I^* be a canonical injective resolution $0 \to R \to I^0 \to I^1 \to 0$ of the ring R and $\mathcal{J}^*(B; R)$ be a canonical injective resolution (consisting of injective sheaves) of the constant sheaf R on B (cf. Chap. 5 of [87]). Its sections with compact supports $\Gamma_c(\mathcal{J}^*)$ form a (flabby) graded differential sheaf on B. By $D_*(\mathcal{J}^*(B; R))$ we denote the bigraded differential sheaf for which $D_n=\sum_{p-q=n}\mathcal{H}om(\Gamma_c(\mathcal{J}^p), I^q)$ [for a cosheaf Φ on B and any R-module I, we denote by $\mathcal{H}om(\Phi, I)$ the sheaf generated by the presheaf $U\to\mathrm{Hom}_R(\Phi(U), I)$ and the differential $d: D_n \to D_{n-1}$ is defined as $d = d'' + (-1)^n d''$, where d' is induced by the differential $I^q \to I^{q+1}$ from I^* and d'' by the map $\Gamma_c(\mathcal{J}^{p-1})\to\Gamma_c(\mathcal{J}^p)$ from the resolution $\mathcal{J}^*$. It turns out that $D_*(\mathcal{J}^*(B; R))$ is a flabby sheaf. By definition $H_n(\Gamma_\varphi(D_*(\mathcal{J}^*(B; R))))=H_n^\varphi(B; R)$. The module of n-dimensional chains (with any closed supports) has the form $C_n(B; R)=\mathrm{Hom}_R(\Gamma_c(\mathcal{J}^n), I^0)\oplus\mathrm{Hom}_R(\Gamma_c(\mathcal{J}^{n+1}), I^1)$.

The use of I^* in this definition instead of the ring R itself is a forced measure (cf. the end of the introduction to Chap. 2) connected with the fact that the basic cochain complex $\Gamma_c(\mathcal{J}^*)$ of the space B of the authors is not free (cf. Secs. 3 and 4). Actually they consider the hyperhomology (in the sense of [92]) of the complex $\Gamma_c(\mathcal{J}^*)$ for the functor $\mathrm{Hom}_R(\ldots,R)$. The hyperhomology of the complex C^* for this functor coincides with the homology of the ordinary complex of homomorphisms $\mathrm{Hom}_R(C^*, R)$ whenever C^* is composed of free R-modules so the Borel–Moore homology with coefficients in R coincides with that considered in Secs. 3 and 4. In particular, it is equal to zero in dimension $n = -1$ (since the chains in this dimension are different from zero, in the development of the theory one must watch the conditions which guarantee that H_{-1} is zero, cf., e.g., Theorems 5.10 and 5.12 of Chap. 5 of [87]).

Analogously, using a two-termed injective resolution of an arbitrary R-module G as I^*, one could also define the homology groups $H_*^\varphi(B; G)$ (and they would coincide with the groups constructed in Secs. 3 and 4). Instead of this, the authors, aiming at the inclusion of nonconstant coefficients, define $H_n^\varphi(B;\mathcal{G})$ with coefficients in any sheaf of R-modules $\mathcal{G}$ as the homology of the complex of sections with supports in φ on the sheaf $D_*(\mathcal{J}^*(B; R))\otimes_R\mathcal{G}$. Not having available any concrete constructions of chains for locally compact spaces (which could be assembled naturally into a sheaf), the authors define a sheaf formally, whose sections play the role of chains. The artificiality of this approach is note, for example, in the introduction to Chap. 5 of [87] where the theory which arises is called "a kind of co-cohomology theory associated with sheaf cohomology groups." Contradictions do not retard one saying: the construction in general is not compatible with the construction of chains discussed in point 5.1 above, and for compact spaces, for example, the universal coefficient formulas (3) turn out to be valid in the Borel–Moore theory not only for finitely generated, but also for arbitrary R-modules G (point 3.10 of Chap. 5 of [87]), at the same time that relations (4) from point 5.1 which are natural for any G are established only for G = R (Sec. 3 of Chap. 5 of [87]). This explains why in all the most typical applications of Borel–Moore $\mathscr{G}$ homology (Chap. 5 of [87]) the sheaves G = R are assumed to be locally constant with finitely generated stalks (it is precisely under this condition that the homology groups coincide with the Steenrod–Sitnikov groups considered in the present survey).

6.2. Some Observations. Due to what was said above about universal coefficient formulas (3) the Borel–Moore homology groups with coefficients in the field of rational numbers Q depend whether Q is considered as the ground ring or only as an Abelian group (cf. the examples in Sec. 14 of Chap. 5 of [87] or Sec. 1 of [62] for the noninvariance of the homology groups under change of ring). The Borel–Moore homology groups defined by complexes of sections of sheaves are Π-additive but by virtue of the universal coefficient formulas (3) are not Π_c-additive for infinitely generated modules G even in dimension 0 (cf. Sec. 5 of [63]) due to which the (reduced) homology of the one-point compactification of B may not coincide with the homology of B and the homology of pairs (B, A) for closed A with the homology of B \ A (cf. point 2.4 of Chap. 2), hence the conditions for such coincidences found in Corollary 5.9 of Chap. 5 of [87] for locally constant coefficients are not superfluous. The absence of the property of Π_c-additivity lets one easily find simple examples of metric compacta X with points $x_0 \in X$ for the homology groups of fundamental systems of compact neighborhoods A_i of which $\varprojlim_i^1 H_0(A_i, x_0; G) \neq 0$ cf. point 2.3 of Chap. 7 in [69] (this does not happen for the Steenrod–Sitnikov homology groups H_*^c of any spaces with the first axiom of countability, cf. point 3.1 of Chap. 4 below).

Moreover, besides the fact that it is not entirely proper, the basic construction also yields qualitatively to the constructions of Sec. 2 of Chap. 2. In particlar the flabbiness of the sheaves of chains of locally compact spaces (besides the confirmations there were in the present chapter, cf. the set of confirmations in Chap. 4). For $\mathscr{G} \neq R$ (and paracompactifying families φ) the sheaves $D_* \otimes_R \mathscr{G}$ above are only φ-fine, hence φ-soft (Sec. 3 of Chap. 5 in [87]). In connection with the artificiality the homology groups of subspaces A ⊂ B and pairs (B, A) are only defined for locally closed A while the fact that the chains of A can be realized as a subcomplex of the chains of the whole space B is far from obvious (cf. Sec. 5 of Chap. 5 in [87]). In contrast with those considered in the present paper (cf. points 2.4 and 2.5 of Chap. 2), the Borel–Moore homology groups $H_n^\varphi(B; \mathscr{G})$ do not always coincide with the direct limit over $K \in \varphi$ of the groups $H_n(K; \mathscr{G})$ (cf. the coincidence conditions in Theorems 5.2 and 5.5 of Chap. 5 in [87]) in any situation and $H_n^\varphi(A; \mathscr{G}) = H_n^{\varphi|A}(X; \mathscr{G})$ (cf. Corollary 5.6 of Chap. 5 in [87]).

6.3. Propriety Conditions. Despite many inadequacies, under certain natural restrictions the Borel–Moore homology groups coincide with the ordinary ones for any locally constant coefficients (not just with finitely generated stalks). Thus, under restrictions of the type of metrizability and local compactness, the Borel–Moore homology groups with compact supports of homologically locally connected spaces are isomorphic to the Steenrod–Sitnikov homology groups H_*^c (cf. [49, 51]). In the category of locally finite polyhedra and their proper maps, the addition to the usual Eilenberg–Steenrod axioms of the property of Π-additivity uniquely defines the homology groups H_* [50] so the Borel–Moore homology groups of such polyhedra coincide with the usual (i.e., simplicial) homology groups of the second kind. (Unfortunately, it is impossible to say this about certain proposed imitations suitable for more general spaces which are not locally compact; in categories of polyhedra they lead to functors which differ both from H_*^c and H_*, cf. [65] in this connection).

7. Cech Cohomology Groups

7.1. In Connection with Cech Homology Groups. The apparatus of nerves of sufficiently fine coverings of the space X reflects the structure of X itself so coverings came to be used in studying homology and cohomology groups long ago. The nonuniqueness of the projection of nerves of coverings inscribed in one another is the basic reason why it became the tradition to effect passages to the limit not in the chains and cochains of coverings but in their homology and cohomology groups, arriving at the Cech groups. The tradition in general was preserved even after systems of coverings with single-valued projections were found.

As to the Cech homology $\check{H}_*$ the characteristic points and basic obstructions to their development were actually already noted above, cf. point 4.4 (the use, under the influence of the theory of characters, of mainly compact coefficients, consideration of the topology of the inverse limit, the clarification of the role of infinite coverings, ignorance of the compactness factor, etc.). If to this one further adds the absence of exactness of the inverse limit functor, the inadequacies noted last, but which should hardly be considered the unique obstruction on the path of developing the homology theory, it becomes clear why the groups $\check{H}_*$ are considered at the present time only as auxiliary, not having independent interest (cf. also Sec. 1 of Chap. 5 in [69]).

Their role is revealed, for example, by the relations between H_* and $\check{H}_*$ of point 5.2 of Chap. 3. Also of independent interest in connection with this in the connection of the homology groups H_* of a space with the homology groups of an open covering $\alpha = \{U_i\}$. As shown recently in [75] in terms of singular homology groups, in the case of a compact space it is the same as for the cohomology groups: there is a spectral sequence of a covering which converges to $H^S_*(X; G)$ with second term $E^2_{pq} = H_p(\alpha; \mathcal{H}_q)$, where $\mathcal{H}_q$ are local coefficients on the nerve $|\alpha|$ under which to simplices $(i_0, \ldots, i_q) \in |\alpha|$ are associated the groups $H^s_q\left(\bigcap_j U_{i_j}; G\right)$. A consequence is an isomorphism $H^s_*(X; G) = H_*(\alpha; G)$ when α is made up of acyclic sets with acyclic (finite) intersections.

7.2. Role of Sheaves. As to the Cech cohomology groups $\check{H}^*$ up to now they are among the most commonly used ones. Despite the fact that the groups $\check{H}^n$ can be realized only for very special systems of coverings (or for a small class of compact spaces) as the cohomology groups of cochain complexes arising upon passage to the direct limit over coverings and turning out in addition to be sections of acyclic sheaf resolutions (cf. Sec. 5.10, Chap. 2 of [107]), the theory $\check{H}^*$ again confirms the naturality and effectiveness of the language and tools of sheaf theory. In more typical situations for sheaves arising in considering Cech type constructions (for example, for the construction of point 4.1 above), sections with arbitrary closed supports do not constitute such cochain complexes but define the cohomology groups H^* by virtue of the acyclicity of the sheaves which arise in the resolutions. The groups H^*, originally defined with coefficients in presheaves (in the classical case even in constant ones), under natural conditions turn out to depend only on the sheaves generated by these presheaves, so it is natural to consider them as cohomology groups with coefficients in shaves (using in their original definition the presheaves of sections of these sheaves). For any topological spaces the groups $\check{H}^0(X; \mathcal{G})$ (even the groups $H^0(\alpha; \mathcal{G})$ of any open covering) are obviously isomorphic to the groups of sections $\Gamma(\mathcal{G})$.

In all known situations where the groups $\check{H}^*$ are applied, they are isomorphic to the sheaf cohomology groups H^*. Analogous results are valid (at least) for cohomology groups with paracompactifying families of supports. In general there is a spectral sequence connecting both families (including one for cohomology groups with supports in families which along with each set contain sufficiently tight closed neighborhoods of this set).

The Cech theory $\check{H}^*$ is not related traditionally to a simple one. For this reason it is not dealt with in [87] (although the author of the book consiers this a striking omission). The account in [107] is actually rather complicated (although here and there indications of alternative simpler paths slip in). A largely simple and rather complete modern account of the theory $\check{H}^*$ is given in the editors appendix to the translation of [87] and also in Sec. 2 of Chap. 2 and Chap. 4 of [69] (where along with the basic ideas there is a brief exposition and history of the development of this theory).

7.3. Value of Infinite Coverings. The necessity of considering infinite coverings in the definition of $\check{H}^*$ is usually demonstrated by the example of a space homeomorphic to the line R: its map into the circle S^1 (in the complex plane) defined as $x \to e^{ix}$ extends to a noncontractible map $\beta R \to S^1$ of the Cech compactification and since $S^1 = K(Z; 1)$ is an Eilenberg–MacLane complex, $\check{H}^1_f(R; Z) = H^1(\beta R; Z) \neq 0 = H^1(R; Z)$. Here $\check{H}^*_f$ are the "finite" cohomology groups, defined by finite open coverings of the space. We consider that the n-dimensional cohomology groups $H^n(X; G)$ of a paracompact space coincide with the set of homotopy classes of maps $X \to K(G; n)$ [2].

For $n > 1$ an instructive example is $X = K(G, n)$. Since $K(G, n)$ has nontrivial cohomology groups in arbitrarily high dimensions, the space X is not contractible to a finite skeleton of itself and so maps homotopic to the identity do not extend to maps $\beta X \to K(G; n)$. This means that the natural transformation $\beta: \check{H}^n_f \to H^n$ (corresponding to the inclusion $X \subset \beta X$), is not epimorphic, in particular $\check{H}^n_f(X; G)$ and $H^n(X; G)$ differ.

Nevertheless, for paracompact X and closed $Y \subset X$ for $\dim X < \infty$ and $n > 1$ the equality $\check{H}^n_f(X, Y; G) = H^n(X, Y; G)$ holds for any finitely generated groups G [79] (an analogous result in the category of polyhedra was obtained in [90]), and the map $\check{H}^1_f(X, Y; G) \to H^1(X, Y; G)$ is epimorphic. In [90] it is shown that when it is reduced the group $\check{H}^1(X; Z)$ coincides (for $\dim X < \infty$) with the quotient-group of $\check{H}^1_f(X; Z)$ by a maximal divisible subgroup. For finite-dimensional X the transformations β are isomorphic when G is a finite group (Theorem 12 in [35]; in [79] an isomorphism is also established for pairs (X, Y)). Since the zero-dimensional cohomology groups are groups of locally constant functions (on X or βX), the transformation β in dimension 0 is always monographic but not epimorphic for infinite groups G (since in contrast with βX on X the functions indicated may also assume an infinite number of values).†

If $\varkappa$ is an infinite cardinal number greater than or equal to the cardinality of G, then the cohomology groups $H^p(X, Y; G)$ of a paracompact pair (X, Y) (with closed Y) are defined by open coverings whose cardinality does not exceed $\varkappa$ [3]. In particular, for finite or even countable G one can always consider countable coverings only.

CHAPTER 4

MOST TYPICAL APPLICATIONS

1. Homology and Cohomology of a Relation. Homology and Cohomology of a Surrounding of a Closed Set

In this section we single out as independent objects some new groups which have previously occurred in the literature only in special cases. In explaining their nature and interconnections the flabbiness of the sheaves of chains (and cochains) is essential. The construction promotes the formation of a new look at the problem about the apparatus of local homology and cohomology groups. The content of the section again confirms the influence of the compactness factor in homology theory (the problems which arise in the homological version disappear on passage to cohomology groups).

1.1. Homology Groups of a Join. Let A be an arbitrary subset of a paracompact space X and $B = X \setminus A$, $\mathcal{G}$ be locally constant coefficients. In using the chains of Sec. 3 or Sec. 4 of the preceding chapter there arises a short exact sequence of chain complexes

$$0 \to C^{\varphi}_*(A; \mathcal{G}) \oplus C^{\varphi}_*(B; \mathcal{G}) \to C^{\varphi}_*(X; \mathcal{G}) \to K^{\varphi}_* \to 0 \tag{1}$$

and the exact homology sequence corresponding to it

$$\ldots \to H^{\varphi}_n(A; \mathcal{G}) \oplus H^{\varphi}_n(B; \mathcal{G}) \to H^{\varphi}_n(X; \mathcal{G}) \to H^{\varphi}_n(A \div B; \mathcal{G}) \to \ldots \tag{2}$$

It is obviously natural to call the homology groups $H^{\varphi}_n(A \div B; \mathcal{G})$ of the complex of chains of X (with supports in φ) by the subcomplex of chains whose supports are in A and in B the homology groups of the relation between the complementary sets A and B. Clearly $H^{\varphi}_n(A \div B; \mathcal{G}) = H^{\varphi}_n(B \div A; \mathcal{G})$. Here φ is either the family of any closed sets (homology grous of the second kind) or the family of compact sets (homology groups H^c_* with compact supports), or arbitrary.

Since $C^{\varphi}_*(X, B; \mathcal{G}) = C^{\varphi}_*(X; \mathcal{G}) / C^{\varphi}_*(B; \mathcal{G})$ there also arises a short exact sequence of chains

$$0 \to C^{\varphi}_*(A; \mathcal{G}) \to C^{\varphi}_*(X, X \setminus A; \mathcal{G}) \to K^{\varphi}_* \to 0 \tag{3}$$

and an exact homology sequence

$$\ldots \to H^{\varphi}_n(A; \mathcal{G}) \to H^{\varphi}_n(X, X \setminus A; \mathcal{G}) \to H^{\varphi}_n(A \div X \setminus A; \mathcal{G}) \to \ldots . \tag{4}$$

If A is acyclic (for example, consists of one point), then $H^{\varphi}_n(A \div X \setminus A; \mathcal{G}) = H^{\varphi}_n(X, X \setminus A; \mathcal{G})$. It is also easy to note the existence of the short exact sequence of chain complexes

$$0 \to C^{\varphi}_*(X; \mathcal{G}) \to C^{\varphi}_*(X, A; \mathcal{G}) \oplus C^{\varphi}_*(X, B; \mathcal{G}) \to K^{\varphi}_* \to 0 \tag{5}$$

and consequently, the exact homology sequence

$$\ldots \to H^{\varphi}_n(X; \mathcal{G}) \to H^{\varphi}_n(X, A; \mathcal{G}) \oplus H^{\varphi}_n(X, B; \mathcal{G}) \to H^{\varphi}_n(A \div B; \mathcal{G}) \to \ldots . \tag{6}$$

In these observations A and B = X \ A are arbitrary subspaces of X. We consider all pairs P, Q of closed subspaces of X such that P ⊂ A, Q ⊂ B. Then the exact sequence (1) coincides with the direct limit (over such P and Q) of the short exact sequences

†Cf. also point 5.3 of Chap. 4 on connections of the cohomology groups of a space and its compactifications.

$$0 \to C_*^\varphi(P;\ \mathcal{G}) \oplus C_*^\varphi(Q;\ \mathcal{G}) \to C_*^\varphi(X;\ \mathcal{G}) \to C_*^\varphi(X,\ P \cup Q;\ \mathcal{G}) \to 0$$

at the same time that (2) coincides with the direct limit of the corresponding homology sequences. In particular, $H_n^\varphi(A \div B;\ \mathcal{G}) = \varinjlim H_n^\varphi(X,\ P \cup Q;\ \mathcal{G})$ is the direct limit over the P and Q indicated. This gives an intuitively clear interpretation of the homology groups of a relation.

Analogously, the homology sequences (4) and (6) can be represented as direct limits over sets $P \subset A$, $Q \subset B$ which are closed in X. In particular, if A is a subpolyhedron of the polyhedron X and A' is a regular neighborhood of it, then $H_n^\varphi(A \div X \setminus A;\ \mathcal{G}) = H_n^\varphi(X,\ A \cup (X \setminus \operatorname{Int} A');\ \mathcal{G})$.

For chains of the complex K_*^φ one also "designates" supports simultaneously intersecting A and B and defined up to supports of chains in A and B, so for any subspace $Y \subset X$ there is defined the subcomplex of K_*^φ generated by chains from Y. This subcomplex defines the homology of the relation between A and B on Y (which coincides with the homology of the relation in the space Y between $A \cap Y$ and $B \cap Y$). The corresponding quotient-complex defines the homology of the relation between A and B of the pair (X, Y) (etc.).

1.2. Homology of a Surrounding. Let $\{U_\lambda\}$ be a system of open subsets of X such that $X \setminus U_\lambda = P_\lambda \cup Q_\lambda$, P_λ and Q_λ being subsets of the sets A and B which are closed in X. There arises a projective system of chain complexes $\{C_*^\varphi(U_\lambda;\ \mathcal{G})\}$ with zero inverse limit (intersection). The hyperhomology (in the sense of Chap. 17 of [92]) of this system for the functor $\varprojlim$ which is naturally treated as a chain complex in the category of projective systems of Abelian groups (or modules) is of interest. Clearly, these hyperhomology groups should also shed light on the character of the relation between A and $B = X \setminus A$.

In general one is unable to establish a relations these hyperhomology groups and the groups of the relation $H_n^\varphi(A \div X \setminus A;\ \mathcal{G})$,† so we shall assume below that the set A is closed. Let W_λ be all neighborhoods of A in X. In correspondence with what was said above, in this case $H_n^\varphi(A \div X \setminus A;\ \mathcal{G}) = \varinjlim_{W_\lambda} H_n^\varphi(X,\ A \cup (X \setminus W_\lambda);\ \mathcal{G})$. For each $U_\lambda = W_\lambda \setminus A$ (a "surrounding" of A in X) there arises a short exact sequence of chain complexes

$$0 \to C_*^\varphi(U_\lambda; \mathcal{G}) \to C_*^\varphi(W_\lambda;\ \mathcal{G}) \to C_*^\varphi(X,\ X \setminus A;\ \mathcal{G}) \to 0 \tag{7}$$

[since $C_*^\varphi(W_\lambda,\ U_\lambda;\ \mathcal{G}) = C_*^\varphi(X,\ X \setminus A;\ \mathcal{G})$]. We shall denote the hyperhomology groups of the system $C_*^\varphi(U_\lambda;\ \mathcal{G})$ for the functor $\varprojlim$ in dimension n by $H_n^\varphi(A < X;\ \mathcal{G})$ and call them the homology groups of a surrounding of the closed set A in X.

The properties of projective sequences of the form (7) are closely connected with the properties of projetive systems of short exact sequences

$$0 \to C_*^\varphi(W_\lambda;\ \mathcal{G}) \to C_*^\varphi(X;\ \mathcal{G}) \to C_*^\varphi(X, W_\lambda;\ \mathcal{G}) \to 0. \tag{8}$$

Clearly $\varprojlim_\lambda C_*^\varphi(W_\lambda;\ \mathcal{G}) = \bigcap_\lambda C_*^\varphi(W_\lambda;\ \mathcal{G}) = C_*^\varphi(A;\ \mathcal{G})$. For $\varphi = c$ using all the properties of sheaves of chains from Sec. 2 of Chap. 2, one can show that $\varprojlim_\lambda C_*^c(X,\ W_\lambda;\ \mathcal{G}) = C_*^c(X, A;\ \mathcal{G})$. From this, and from the exact sequence of functors $\varprojlim^p$ corresponding to (8) it follows that $\varprojlim_\lambda^1 C_*^c(W_\lambda;\ \mathcal{G}) = 0$ so upon passing to the limits in (7) one gets the exact sequence

$$0 \to C_*^c(A;\ \mathcal{G}) \to C_*^c(X,\ X \setminus A;\ \mathcal{G}) \to \varprojlim_\lambda^1 C_*^c(U_\lambda; \mathcal{G}) \to 0. \tag{9}$$

Since it obviously coincides with (3), we have $K_*^c = \varprojlim_\lambda^1 C_*^c(U_\lambda;\ \mathcal{G})$.

It turns out that the transformation of the homology groups of the first two terms in (9) into the hyperhomology groups, respectively, of the second and third terms of (7) (which is an isomorphism of the pair $(X, X \setminus A)$] extends to a natural transformation $\tau: H_n^c(A \div X \setminus A;\ \mathcal{G}) \to H_{n-1}^c(A < X;\ \mathcal{G})$ of the homology groups of the relation between A and $X \setminus A$ into the $(n - 1)$-dimensional homology groups of a surrounding of A in X. There arises a transformation of the homology sequence (4) into the exact hyperhomology sequence corresponding to the short exact sequence of projective systems (7). In general, however, the transformation of $H_n^c(A;\ \mathcal{G})$ into

†Except for the natural transformation in the hyperhomology groups of a relation.

the hyperhomology groups for the functor $\varprojlim$ of the projective system $\{C^c_*(W_\lambda;\ \mathcal{G})\}$ is not an isomorphism (for example, for the set A of points of the plane $X = R^2$ whose abscissas are all integers and whose ordinates are equal to the rational numbers $1/n$, $n = 1, 2, \ldots,$ and zero; the set A coincides with the space $X^{(k)}$ from [124] for $k = 0$). Thus, the transformation τ is not an isomorphism in general.

The transformation τ is isomorphic for a compact subspace A in a locally compact X. In this case, the complexes $C_*^c(X,\ W_\lambda;\ \mathcal{G})$ being sections of flabby sheaves of chains over the sets $B_\lambda = X \setminus W_\lambda$ are weakly flabby in the sense of [113] and hence a $\varprojlim$-acyclic projective system. By (8), the system $\{C^c_*(W_\lambda;\ \mathcal{G})\}$ is also $\varprojlim$-acyclic, so the transformations of the homology groups of A into hyperhomology considered above are isomorphisms and it remains to apply the five lemma. The transformation τ is also isomorphic when A is compact and has a countable fundamental system of neighborhoods in X (for example, in a metric space): in this case the $\varprojlim$-acyclicity of the system $\{C^c_*(W_\lambda;\ \mathcal{G})\}$ is a consequence of the vanishing of $\varprojlim^p$ for $p\geqslant 2$ for countable projective systems.

The transformation τ is also isomorphic when the set A has a fundamental system of neighborhoods W_λ in X for each of which the map of homology groups $H^c_*(A;\ \mathcal{G})\to H_*^c(W_\lambda;\ \mathcal{G})$ is an isomorphism: in this case one can show that the homology groups of A coincide with the hyperhomology groups for the functor $\varprojlim$ of the projective system $\{C^c_*(W_\lambda;\ \mathcal{G})\}$. This condition holds, for example, for a subpolyhedron A of any cellular polyhedron X (W_λ being regular neighborhoods of A). In this case $H^c_p(A<X;\ \mathcal{G}) = H^c_p(U_\lambda;\ \mathcal{G}) = H^c_p(\partial W_\lambda;\ \mathcal{G})$ (∂W_λ being the boundary of W_λ).

The groups $H^\varphi_n(A<X;\ \mathcal{G})$ defined as hyperhomology groups are the homology groups of the double complex considered with respect to the total grading, including their nontriviality for $n < 0$. In the category of compacta, in particular, $H_{-1}(A<X;\mathcal{G}) = H_0(A\div X\setminus A;\ \mathcal{G})$ and this group is nonzero, for example, when X is a sequence of points converiging to a point $x_0 = A\subset X$ [it is easy to see this using the representation of $H_0(A\div X\setminus A;\ \mathcal{G})$ as a direct limit from point 1.1].

1.3. Cohomology Groups of a Surrounding and a Relation. Let $\mathscr{C}^*$ be a flabby Godement resolution of the sheaf of coefficients $\mathcal{G}$ on X (point 4.3 of Chap. 1). Without any restrictions on X and A, the cochains of the pair (X, A) are defined as the subcomplex of $\Gamma(\mathscr{C}^*)$ containing all sections with supports in $X \setminus A$ (point 5.3 of Chap. 1). It reduces to the sections indicated at least when A is open, when X is paracompact and A closed, and for any A in a hereditarily paracompact X (cf. also point 5.3 of Chap. 1). Under such conditions (hence we shall assume below that they hold) the cochains of (X, A) and (X, B), where $B = X \setminus A$ constitute a direct sum, there arises a short exact sequence of cochain complexes $\Gamma(\mathscr{C}^*)$

$$0\to C^*_\varphi(X,\ A;\ \mathcal{G})\oplus C^*_\varphi(X,\ B;\ \mathcal{G})\to C^*_\varphi(X;\ \mathcal{G})\to K^*_\varphi\to 0$$

and the exact cohomology sequence corresponding to it

$$\ldots\to H^n_\varphi(X,\ A;\ \mathcal{G})\oplus H^n_\varphi(X,\ B;\ \mathcal{G})\to H^n_\varphi(X;\ \mathcal{G})\to H^n_\varphi(A\div B;\ \mathcal{G})\to\ldots \tag{10}$$

It is natural to call the groups $H^n_\varphi(A\div B;\ \mathcal{G})$ the cohomology groups of the relation beween A and $B = X \setminus A$. Just as in point 1.1 there also arise exact cohomology sequences

$$\ldots\to H^n_\varphi(X,\ X\setminus A;\ \mathcal{G})\to H^n_\varphi(A;\ \mathcal{G})\to H^n_\varphi(A\div X\setminus A;\ \mathcal{G})\to\ldots \tag{11}$$

$$\ldots\to H^n_\varphi(X;\ \mathcal{G})\to H^n_\varphi(A;\ \mathcal{G})\oplus H^n_\varphi(B;\ \mathcal{G})\to H^n_\varphi(A\div B;\ \mathcal{G})\to\ldots \tag{12}$$

Again let P and Q be closed subspaces of X containing A and B. Then each of the cohomology sequences above can be represented as the direct limit of the corresponding exact sequences for P and Q (here, for example, $H^n_\varphi(A;\ \mathcal{G}) = \varinjlim_Q H^n_\varphi(X\setminus Q;\ \mathcal{G})$, $H^n_\varphi(X,\ A;\ \mathcal{G}) = \varinjlim_Q H^n_\varphi(X,\ X\setminus Q;\ \mathcal{G})$], etc.]. In particular, (10) is the limit of the sequences

$$\ldots\to H^n_\varphi(X,\ X\setminus Q;\ \mathcal{G})\oplus H^n_\varphi(X,\ X\setminus P;\ \mathcal{G})\to H^n_\varphi(X;\ \mathcal{G})\to H^n_\varphi(U;\ \mathcal{G})\to\ldots,$$

where $U = X\setminus(P\cup Q)$. Thus, $H^n_\varphi(A\div B;\ \mathcal{G}) = \varinjlim H^n_\varphi(X\setminus(P\cup Q);\ \mathcal{G})$.

The last gives an obvious interpretation of the cohomology of a relation. When A is closed, $U = W \setminus A$, where W runs through all neighborhoods of A in X and the groups $H_\varphi{}^n(A\div B;\ \mathcal{G})$ can be interpreted at the same time as the n-dimensional cohomology groups $H_\varphi{}^n(A<X;\ \mathcal{G})$ of a

surrounding of the closed set A in X. For this special case the sequence (11) was first obtained in [137] (the groups of a surrounding are defined in it as the direct limit of $H^n(U;\mathcal{G})$, $\dot{U}=W\setminus A$ and were called the local cohomology groups on the set A). For the special case when U are the complements of any compact subspaces of a locally compact spaces, an analogous construction was considered even earlier in [93] (and local cohomology using this construction figured in [85]).

Just as at the end of point 1.1, the groups $H^n(A\dot{-}B;\mathcal{G})$ are the cohomology groups of a relation of the entire space X. Analogous groups of the relation between A and B = X \ A are defined for subspaces Y ⊂ X and pairs (X, Y), there is an exact cohomology sequence of the pair (X, Y), etc.

Consideration of the cohomology groups of pairs ($\bar{U}$, Fr U), where $U=X\setminus(P\cup Q)$ (cf. above), $\bar{U}$ being the closure of U, Fr U being the boundary of U, gives an alternative approach to the cohomology groups of a relation. Since $\varprojlim_{U} C^*_\varphi(\bar{U}, \mathrm{Fr}\,U;\mathcal{G})=\cap C^*_\varphi(\bar{U}, \mathrm{Fr}\,U;\mathcal{G})=0$ it is natural to consider (just as in the case of homology groups) the hyperhomology groups for the functor $\varprojlim$ of a projective system of finite complexes. In general however it is hard to establish the connection of the new groups with $H^n_\varphi(A\dot{-}B;\ \mathcal{G})$.†

The hyperhomology groups indicated coincide with the cohomology groups of a relation (or surrounding) for any closed set A when φ is the family of all closed sets. Let W be a neighborhood of such an A in X. In the exact sequence of projective systems

$$0\to C^*(X,\ X\setminus W;\mathcal{G})\to C^*(X;\mathcal{G})\to C^*(X\setminus W;\mathcal{G})\to 0$$

the last (consisting of sections of flabby sheaves C* on X \ W) turns out to be weakly flabby in the sense of [113] and hence $\varprojlim$-acyclic. Consequently [since $\varprojlim C^*(X\setminus W;\ \mathcal{G})=\overleftarrow{C}^*(X\setminus A;\ \mathcal{G})$] the system $C^*(X,\ X\setminus W;\mathcal{G})=C^*(\bar{W}, \mathrm{Fr}\,W;\mathcal{G})$ is also $\varprojlim$-acyclic, and its limit (intersection) coincides with $C^*(X,\ X\setminus A;\mathcal{G})$. After passing to the limit in the projective system of exact sequences (here U = W \ A)

$$0\to C^*(\bar{U}, \mathrm{Fr}\,U;\mathcal{G})\to C^*(\bar{W}, \mathrm{Fr}\,W;\mathcal{G})\to C^*(A;\mathcal{G})\to 0$$

we get the exact sequence

$$0\to C^*(X,\ X\setminus A;\ \mathcal{G})\to C^*(A;\ \mathcal{G})\to \varprojlim_{W}{}^1 C^*(\bar{U},\ \mathrm{Fr}U;\mathcal{G})\to 0$$

[here $\varprojlim_{W}{}^p C^*(\bar{U},\ \mathrm{Fr}U;\ \mathcal{G})=0$ for p ≠ 1].

Thus, $K^*=\varprojlim_{W}{}^1 C^*(\bar{U},\ \mathrm{Fr}U;\ \mathcal{G})$. Hence, just as in point 1.2, the possibility arises of comparing the cohomology groups of the last short exact sequence with the hyperhomology groups of the preceding one and constructing a natural transformation of the groups $H^n(A\dot{-}X\setminus A;\ \mathcal{G})$ into the (n + 1)-dimensional hyperhomology groups for the functor $\varprojlim$ of the projective system of finite complexes $C^*(\bar{U},\ \mathrm{Fr}U;\ \mathcal{G})$. In this case the arguments of point 1.2 guarantee an isomorphism.

2. Pairs of Subspaces. Mayer–Vietoris Sequences

2.1. Homology and Cohomology of Pairs of Subspaces. Since the homology groups of general spaces (and pairs) can be represented as direct limits with respect to compact or closed locally compact support (point 2.5 of Chap. 2), they are considered below only for locally compact spaces. Let $A_1 \subset A_2$ be subsets of the locally compact space X. If X is not hereditarily paracompact, let us assume that A_1 is open or closed. The homology groups of the pair (A_2, A_1) with supports in the paracompactifying family φ are defined by chains wihich are sections of sheaves of chains $\mathcal{C}_*(\mathcal{G})$ (point 2.4 of Chap. 2) with supports in $\varphi|_{A_2}$ restricted to $X \setminus A_1$ or equivalently, to sections of $\mathcal{C}_*(\mathcal{G})$ on $X \setminus A_1$ with supports in $(\varphi\cap(X\setminus A_1))|_{A_2}$. In the obvious way one constructs an exact sequence of homology groups of pairs of subspaces participating in a triple $A_1 \subset A_2 \subset A_3$ (under the same restrictions on A_1 and A_2).

Dually one gets cohomology groups $H_\varphi^q(B_2, B_1;\mathcal{G})$ (with coefficients in the sheaf $\mathcal{G}$) of a pair of subspaces of a paracompact Hausdorff space X (point 5.3 of Chap. 1). In the case of

†Although there is a transformation into these groups.

an n-manifold X the oppositely numbered resolution of the orienting sheaf of the manifold (cf. point 6.1 below) turn out to be flabby sheaves of chains, so the p-dimensional homology groups of the pair (A_2, A_1) coincide with the $(n-p)$-dimensional cohomology groups (with the same supports but with coefficients in the sheaf indicated) of the pair $(X \setminus A_1, X \setminus A_2)$ (Poincaré–Lefschetz duality), and the exact homology sequence of a triple with the exact cohomology sequence of the complimentary triple $X\setminus A_3\subset X\setminus A_2\subset X\setminus A_1$.

2.2. Homology of a Triad. Just as above, we restrict ourselves to the category of locally compact spaces. As noted in point 1.2 of Chap. 3, the more complicated form of the excision property in singular theory is the paramount restriction figuring in the definition of proper triads (guaranteeing the existence of exact homology and cohomology sequences corresponding to them). The condition that the chains of $X_1\cup X_2$ coincide with the sum of chains of X_1 and X_2, which is equivalent (as is easily seen) to the coincidence of the chains of the pairs $(X_1, X_1\cap X_2)\subset(X_1\cup X_2, X_2)$, holds automatically (considering the flabbiness of the sheaves of chains and the description of the chains of pairs in point 2.4 of Chap. 2), for example, for sets X_1, X_2 closed in $X_1\cup X_2$, or for open ones when $X_1\cup X_2$ is paracompact.

Let $A_i \subset X_i$, $i = 1, 2$, where the X_i satisfy the conditions formulated and A_i the conditions imposed in point 2.1 on A_1. Using the interpretation of chains as sections of flabby sheaves $\mathscr{C}_*(\mathscr{G})$ for any paracompactifying family φ we get a short exact sequence of chain complexes

$$0\to C_*^{\varphi|(X_1\cap X_2)}(X_1\cap X_2,\ A_1\cap A_2;\ \mathscr{G})\xrightarrow{\mu} C_*^{\varphi|X_1}(X_1,\ A_1;\ \mathscr{G})\oplus C_*^{\varphi|X_2}(X_2,\ A_2;\ \mathscr{G})\xrightarrow{j}$$
$$\xrightarrow{j} C_*^{\varphi|(X_1\cup X_2)}(X_1\cup X_2,\ A_1\cup A_2;\ \mathscr{G})\to 0.$$

In it $\mu = \mu_1 - \mu_2$, where μ_i are induced by the inclusions $(X_1\cap X_2,\ A_1\cap A_2)\subset(X_i, A_i)$ and j by the inclusions $(X_i, A_i)\subset(X_1\cup X_2,\ A_1\cup A_2)$. The exactness in the first two terms is a sequence of the interpretation of chains as sections of sheaves (point 2.1). The chains of the third term are the sections with supports in $X_1\cup X_2$ restricted to $X\setminus(A_1\cup A_2)$ and the flabbiness of $\mathscr{C}_*(\mathscr{G})$ lets one decompose such sections into a sum of sections with supports in X_1 and X_2 which guarantees the surjectivity of j. Thee arises an exact homology sequence

$$\ldots\to H_p^{\varphi|(X_1\cap X_2)}(X_1\cap X_2,\ A_1\cap A_2;\ \mathscr{G})\to H_p^{\varphi|X_1}(X_1,\ A_1;\ \mathscr{G})\oplus H_p^{\varphi|X_2}(X_2,\ A_2;\ \mathscr{G})\to$$
$$\to H_p^{\varphi|(X_1\cup X_2)}(X_1\cup X_2,\ A_1\cup A_2;\ \mathscr{G})\to\ldots. \tag{13}$$

One separates two standard cases. In the first $A_1=A_2=\varnothing$ we get the exact homology sequence of a triad

$$\ldots\to H_p^{\varphi|(X_1\cap X_2)}(X_1\cap X_2;\ \mathscr{G})\to H_p^{\varphi|X_1}(X_1;\ \mathscr{G})\oplus H_p^{\varphi|X_2}(X_2;\ \mathscr{G})\to H_p^{\varphi|(X_1\cup X_2)}(X_1\cup X_2;\ \mathscr{G})\to\ldots. \tag{14}$$

In the second $X_1 = X_2 = X$ there arises an "additional" homology sequence of the triad $(X; A_1, A_2)$

$$\ldots\to H_p^{\varphi}(X,\ A_1\cap A_2;\ \mathscr{G})\to H_p^{\varphi}(X,\ A_1;\ \mathscr{G})\oplus H_p^{\varphi}(X,\ A_2;\ \mathscr{G})\to H_p^{\varphi}(X,\ A_1\cup A_2;\ \mathscr{G})\to\ldots. \tag{15}$$

2.3. Cohomology of a Triad. By consideration of the planar diagram containing the two short exact sequences of cochain complexes (with supports in φ cf. point 5.3 of Chap. 1) of pairs of subspaces $i_1: (Y_1,\ Y_1\cap Y_2)\subset(Y_1\cup Y_2,\ Y_2)$ (with the corresponding inclusion homomorphisms) and at the same time, the analogous sequences for the pair $i_2: (Y_2,\ Y_1\cap Y_2)\subset(Y_1\cup Y_2;\ Y_1)$, it is easy to conform that the map of cochains of pairs under the inclusion i_1 is an isomorphism if and only if the analogous map for i_2 is an isomorphism. Triads $(Y_1\cup Y_2;\ Y_1,\ Y_2)$ having this property will be called φ-excisive.† For such triads, as follows easily from the indicated diagram, there arise exact sequences of cochain complexes

$$0\to C^*_{\varphi\cap(Y_1\cup Y_2)}(Y_1\cup Y_2;\ \mathscr{G})\xrightarrow{\mu} C^*_{\varphi\cap Y_1}(Y_1;\ \mathscr{G})\oplus C^*_{\varphi\cap Y_2}(Y_2;\ \mathscr{G})\to C^*_{\varphi\cap(Y_1\cap Y_2)}(Y_1\cap Y_2;\ \mathscr{G})\to 0,$$

where $\mu = \mu_1 - \mu_2$, μ_i are induced by the restriction homomorphisms. The φ-excisiveness of triads with open Y_1, Y_2 in $Y_1\cup Y_2$ for closed Y_1, Y_2 for families φ which are compactifying (in $Y_1\cup Y_2$), for any Y_1, Y_2 in a hereditarily paracompact $Y_1\cup Y_2$ for a paracompactifying φ (generally for φ-rigid Y_1, Y_2).

†One of the broadest sufficient conditions for this is [98]: $Y_1\cup Y_2=\mathrm{Int}\,Y_1\cup\mathrm{Int}\,Y_2\cup(Y_1\cap Y_2)$ (the operation Int is in $Y_1\cup Y_2$).

In contrast with the above considerations, the definition of φ-excisiveness given in Sec. 13 of Chap. 2 in [87] involves the requirement that i_1 induces an isomorphism of cohomology groups (with supports in φ) for any sheaves of coefficients $\mathcal{G}$. In the formulation however of concrete conditions guaranteeing the presence of this property one actually uses the behavior of the cochains and their supports (in the interpretation of cochains as sections of flabby resolutions).

Considering the exactness of the short sequence of cochains for a φ-excisive triad $(B_1\cup B_2;\ B_1,\ B_2)$, where $B_i\subset Y_i$ we get a short exact sequence of cochain complexes

$$0\to C^*_{\varphi\cap(Y_1\cup Y_2)}(Y_1\cup Y_2,\ B_1\cup B_2;\ \mathcal{G})\to C^*_{\varphi\cap Y_1}(Y_1,\ B_1;\ \mathcal{G})\oplus C^*_{\varphi\cap Y_2}(Y_2,\ B_2;\ \mathcal{G})\to$$
$$\to C^*_{\varphi\cap(Y_1\cap Y_2)}(Y_1\cap Y_2,\ B_1\cap B_2;\ \mathcal{G})\to 0$$

and the cohomology exact sequence corresponding to it

$$\ldots\to H^q_{\varphi\cap(Y_1\cup Y_2)}(Y_1\cup Y_2,\ B_1\cup B_2;\ \mathcal{G})\to H^q_{\varphi\cap Y_1}(Y_1,\ B_1;\ \mathcal{G})\oplus H^q_{\varphi\cap Y_2}(Y_2,\ B_2;\ \mathcal{G})\to$$
$$\to H^q_{\varphi\cap(Y_1\cap Y_2)}(Y_1\cap Y_2,\ B_1\cap B_2;\ \mathcal{G})\to\ldots. \tag{16}$$

In particular, for $Y_1 = Y_2 = X$ there arises an additional cohomology sequence of a triad

$$\ldots\to H^q_\varphi(X,\ B_1\cup B_2;\ \mathcal{G})\to H^q_\varphi(X,\ B_1;\ \mathcal{G})\oplus H^q_\varphi(X,\ B_2;\ \mathcal{G})\to H^q_\varphi(X,\ B_1\cap B_2;\ \mathcal{G})\to\ldots, \tag{17}$$

and for $B_1 = B_2 = \emptyset$ the exact cohomology sequence of the triad

$$\ldots\to H^q_{\varphi\cap(Y_1\cup Y_2)}(Y_1\cup Y_2;\ \mathcal{G})\to H^q_{\varphi\cap Y_1}(Y_1;\ \mathcal{G})\oplus H^q_{\varphi\cap Y_2}(Y_2;\ \mathcal{G})\to H^q_{\varphi\cap(Y_1\cap Y_2)}(Y_1\cap Y_2;\ \mathcal{G})\to\ldots. \tag{18}$$

In the case when X is a manifold (cf. with the end of point 2.1), the sequences (13)-(15) coincide for $p + q = n$ with the sequences (16)-(18) in which $Y_i=X\setminus X_i$, $B_i=X\setminus A_i$ and the sheaf of coefficients depends on the coefficients for the homology groups (consequence of Poincaré duality, cf. point 6.1 below).

3. Local Behavior

3.1. Local Vanishing Homological Local Connectedness. As noted in point 3.1 of Chap. 1, the cohomology groups vanish locally: $\varinjlim_U H^p(U, x; \mathcal{G})=0$ (U being a neighborhood of the point x). The content of Sec. 2 of Chap. 1 demonstrates the fundamental value of this observation in constructing sheaf cohomology groups. This property, which can be interpreted as an improvement of the axiom of dimension (concerning the cohomology of a point), was used successfully in [63] for the axiomatics of cohomology groups in several natural categories of topological spaces (cf. with point 5.3 of Chap. 3). In the case of cohomology of the second kind (including that in the category of polyhedra) this requirement was also assumed to hold at infinity. The equivalence of the axiomatics obtained to the approach based on additivity properties confirms the equivalence to these properties of the condition that the cohomology vanishes locally.

For homology groups the picture is somewhat more complicated: even in the category of metrizable compacta for an element $h\in H_n(X;Z)$, $n>0$ there can in general exist points $x\in X$ such that h is contained in the image of $H^c_n(U;\ Z)\to H_n(X;\ Z)$ for arbitrarily small neighborhoods U of the point x (cf. the remark after Lemma 8 in [62]). This cannot be if X is homologically locally connected (the condition of this phenomenon with homological local connectivity has not yet been completely clarified). All the same, the homology groups H^c_* at least for spaces X with the first axiom of countability, vanish locally: $\varprojlim H^c_n(U,\ x;\ \mathcal{G})=0$, U being neighborhoods of an arbitrary point $x\in X$ (Lemma 8 in [62]). Moreover, in this case $\varprojlim_U{}^1 H^c_n(U;\mathcal{G})=0$ also, cf. Lemma 8 of [76] (we recall that for countable projective systems $\varprojlim{}^p=0$ for $p > 1$).

As an improvement of the axiom of dimension, the condition of local vanishing was used in the axiomatics of homology groups in some natural categories of topological spaces [63] (including at infinity for simplicial homology groups of the second kind). In accord with the results of [50] for finitely generated groups (or modules) of coefficients G, this condition is equivalent to the requirment of additivity used in the alternative approach (cf. point 5.3 of Chap. 3). For G not finitely generated this is not so; a counterexample is the

Borel—Moore theory which is not Π_c-additive (Sec. 6 of Chap. 3 above).

Natural refinements of the local vanishing property are the requirement of homological ($h\ell c_R$) and cohomological ($c\ell c_R$) local connectedness of the space X (over the ring of coefficients R): for any natural number n and any point $x \in X$ there should exist arbitrarily small neighborhoods $V \subset U$ of this point with trivial maps of homology groups H_*^c (respectively, cohomology groups H^*) of these neighborhoods up to dimension n. The requirements $h\ell c_R$ and $c\ell c_R$ are equivalent (for locally compact spaces). One defines homological and cohomological local connectedness over an arbitrary R-module of coefficients G analogously. There are many consequences of these requirements (in particular, the homology and cohomology groups of compacta satisfying these requirements are finitely generated in each dimension) and also a number of equivalent conditions. Rather complete information on such questions can be found in point 2.1 of Chap. 8 in [69] (cf. also [66]).

A metrizable locally compact space X which is homologically locally connected up to dimension n (for short $h\ell c^n$) at points of $X \setminus x_0$ also has this property at the point x_0 if one can find a neighborhood U of this point for which the images $H_p^c(U; Z) \to H_p^c(X; Z)$ [in particular, the groups $H_p^c(X; Z)$ themselves] are finitely generated for $p \leqslant n$ [101]. The one-point compactification of a non-point compactification of a noncompact $h\ell c^n$-space X is $h\ell c^n$ if and only if the groups $H_p(X; Z)$ are finitely generated for $p \leqslant n$ [101].

3.2. Local Groups. In point 2.6 of Chap. 2 we have already encountered the local homology groups $H_p^x(G) = H_p^c(X, X \setminus x; G) = H_p(X, X \setminus x; G)$. By local cohomology groups $I_x^p(G)$ we mean the direct limits of the groups $H^p(U \setminus x; G)$ over neighborhoods U of points $x \in X$. It follows easily from consideration of the cohomology exact sequences of pairs $(U, U \setminus x)$ that $I_x^p(G) = H^{p+1}(X, X \setminus x; G)$. The following local groups are also widely used in the literature: $I_p^x(G) = \varprojlim_U H_p^c(U \setminus x; G)$ and $H_x^p(G) = \varinjlim_U H^p(X, X \setminus U; G)$. In [5, 62] transformations $\delta_x : H_p^x(G) \to I_{p-1}^x(G)$ and $\delta^x : I_x^p(G) \to H_x^{p+1}(G)$ are defined which, for spaces with the first axiom of countability, are contained in exact sequences

$$0 \to \varprojlim_U{}^1 H_p^c(U \setminus x; G) \to H_p^x(G) \to I_{p-1}^x(G) \to 0 \tag{19}$$

(cf. [36, 76]) and

$$0 \to \varprojlim_U{}^1 H^p(X, X \setminus U; G) \to I_x^p(G) \to H_x^{p+1}(G) \to 0 \tag{20}$$

(cf. [76]). Here δ_x turn out to be isomorphisms for $h\ell c_G$-spaces [76], and δ^x for peripherally cohomologically locally connected spaces over G [54] (cf. the terms below). When G = R is a countable principal ideal ring and X is a metrizable locally compact $h\ell c_R$-space, (20) splits, its extreme terms admit description in terms of $\mathrm{Ext}_R(A, R)$ and in terms of inverse limits of certain concrete modules and the groups $H_p^x(R)$ are countable [132]. Conditions guaranteeing splitting of (19) are also found in [132].

These constructions, the history of their creation, and the development of the ideas are dealt with in point 2.2 of Chap. 8 in [69]. Here we describe a new point of view about the nature of the local groups and some applications of them. It follows from the exact sequence (4) of point 1.1 above (at least for p > 0) that $H_p^x(G)$ are the homology groups of the relation of the point x with its complement, and from the exact sequence (11) of point 1.3 that for any p the groups $I_x^p(G)$ are the cohomology groups of the relation (surrounding) of the point x so they should not be compared with $I_{p-1}^x(G)$ and $H_x^{p+1}(G)$ (as above), but, respectively, with the (p − 1)-dimensional hyperhomology groups for the functor $\varprojlim$ of the projective system of chain complexes $C_*^c(U \setminus x; G)$ and the (p + 1)-dimensional hypercohomology groups of the projective system of cochain complexes $C^*(X, X \setminus U; G)$. The comparison is always an isomorphism in cohomology and is in homology for locally compact spaces, for spaces with the first axiom of countability, and for locally contractible spaces (cf. point 1.2 above). The transformations δ_x, δ^x factor through the standard spectral sequences for hyperhomology groups [in the case of local cohomology groups for example, with second term of the form $E_2^{pq} = \varprojlim_U{}^p \times H^q(X, X \setminus U; G)$] the conditions for δ_x, δ^x being isomorphisms coincide with the conditions for their collpase.

Thus, in relation to the "true" local groups $H_p^x(G)$ and $I_x^p(G)$ the groups $I_{p-1}^x(G)$ and $H_x^{p+1}(G)$ occupy the same position as the Cech homology groups in relation to the Steenrod—Sitnikov homology groups. Nevertheless these groups play a specific independent role, for example, in the homological theory of dimension (cf. below). Groups of the type of $\varprojlim_U^p H^q(X, X\setminus U; G)$ are also considered in the literature sometimes but not as terms of the spectral sequences indicated above, but as alternative versions of local groups [of the type of $H_x^q(G)$], cf. [119, 132].

Local groups found many applications to the homological theory of dimension (cf. below). They also turned out to be useful tool in studying various local properties of spaces. For example, the point $x \in X$ is called peripheral (in relation to the coefficient module G) if for any neighborhood U of it one can find a small neighborhood V such that its images $H^p(X, X\setminus V; G) \to H^p(X, X\setminus U; G)$ are equal to zero; points which are not peripheral are called inner [120]. The point $x \in X$ is called homologically unstable [135] if $H_x^p(G) = 0$ for all p. If G = R is a countable principal ideal ring or any field and X is a finite-dimensional metrizable space, then the sets of peripheral and homologically unstable points coincide with simultaneously for arbitrary G they are independent [53]. For countable G or countable generated modules G over Artinian rings, the peripheral points are characterized by the fact that for them all $I_x^p(G)$ are equal to zero [52]. If C is a compact subspace consisting of peripheral points (with respect to the ring indicated above), then the inclusion $X \setminus C \subset X$ induces an isomorphism of cohomology groups with coefficients in any R-module G [53]. In addition, in special cases one finds a connection with the Vietoris theorem for imbeddings (point 5.2 below). Such a result is obtained in [29] for the homology groups of the second kind of locally compact metrizable spaces [C being a closed set at whose points $H_p^x(G) = 0$], and also for the homology groups H_*^c of arbitrary metrizable spaces. The connection between such points, defined in terms of homology and cohomology, is considered in [52]; for commutative rings R they are equivalent under additional restrictions of the type of ordinary or peripheral (cf. below) homological local connectedness. In the case of homology, by definition the inner points are those at which X, for some r forms an r-dimensional barrier in the sense of P. S. Aleksandrov [76]. If G = R is a field, they are characrerized by the condition $H_r^x(R) \neq 0$ (in particular, they are homologically stable). In a space X of homological dimension n (over G) the set of points at which $H_n^x(G) \neq 0$ has the same homological dimension and the dimension of the complementary set is smaller [76].

<u>3.3. Peripheral Local Conditions.</u> Any point x of a polyhedron X has neighborhoods U for which $X \setminus U$ are deformation retracts of $X \setminus x$ in particular, the homology and cohomology groups of the pair $(X \setminus x, X \setminus U)$ are trivial. For spaces X with the first axiom of countability $\varprojlim_U H^p(X\setminus x, X\setminus U; G)=0$ cf. [5]. It is easy to see that $\varinjlim_U H_p^c(X\setminus x, X\setminus U; G)=0$ also. This justifies the definitions given in Sec. 5 of [65]. The space X is called peripherally homologically locally connected over G (for short $ph\ell c_G$) if for any n and any neighborhood U of an arbitrary point x one can find a smaller neighborhood V with trivial homomorphisms $H_p^c(X\setminus x, X\setminus U; G) \to H_p^c(X\setminus x, X\setminus V; G)$ for $p \leqslant n$. One defines peripheral cohomological local connectedness (for short $pc\ell c_G$) analogously in terms of H^*.

As established in [54], these new conditions are independent of the presence of $h\ell c_R$. To some degree their role is revealed by the example of a contractible locally contractible two-dimensional compactum X which is the compactification by one point x of an infinite polyhedron for which $I_x^2(Z)=H^3(X, X\setminus x; Z)\neq 0$ while at the same time obviously $H_x^3(G) = 0$ (Example 4.6 of [66]); such a phenomenon is impossible in the presence of $ph\ell c_G$ [since $I_x^p(G)=H_x^{p+1}(G)$ cf. the preceding point). For countable G the requirement of $pc\ell c_G$ is equivalent to the countability of all $I_x^p(G)$, $x \in X$ [54]. In the class of locally compact $h\ell c_R$-spaces the requirements of $ph\ell c_R$ and $pc\ell c_R$ are equivalent and if X has finite homological dimension and R is countable, the presence of $h\ell c_R$ and $ph\ell c_R$ simultaneously is equivalent to the finite generation of all modules $H_p^x(R)$ [or $H_p^x(R)$ and $I_x^p(R)$ together], in addition for $H_p^x(G)=I_{p-1}^x(G)$ and $I_x^p(G)= H_x^{p+1}(G)$ all the usual universal coefficient formulas are valid (i.e., those with the functors ⊗ and Tor, and also those with Hom and Ext) [54]. In the presence of $h\ell c_R$ and $ph\ell c_R$ simultaneously one can calculate the local gropus of a product (Künneth relations) [54].

Less typical conditions guaranteeing the isomorphism $I_x^p(R)=H_x^{p+1}(R)$ and the finite generation of the modules $I_x^p(R)$ are discussed in [132]. Interest in conditions guaranteeing the finite generation of the local groups is confirmed, for example, by the fact that statements about the finite generation of $H_p^x(Z)$ play an essential role in the study of local properties of neighborhood retracts and, in particular, conditions under which such spaces turn out to be homological manifolds [89]. The requirements $ph\ell c$ and $pc\ell c$ should apparently be useful in the homological theory of dimension (cf. below), the class of spaces which are simultaneously $h\ell c_R$ and $ph\ell c_R$ should be particularly interesting.

4. Homological Dimension

4.1. Application of Homology. For clear reasons (discussed in Chaps. 1 and 2) in solving problems connected with dimension theory one usually uses cohomological language. However, even at an early stage of development of the homological theory of dimension of compact spaces, the equivalent Cech homology groups with coefficients in the group S^1 of real numbers reduced modulo 1, which coincide with the character groups in the sense of Pontryagin of the corresponding integral cohomology groups, were widely used. Nevertheless, for many recent years interest has constantly emerged in the applicaiton of other, not just compact, coefficients. There is particular interest, first of all, in the group Z of integers. The proofs of many of the results listed below depend essentially on the presence of flabby sheaves of chains.

Many standard facts of the homological theory of dimension (monotonicity properties, various sum theorems, etc.) are obtained in [76]. It is established there that in the class of metrizable locally compact spaces the homological dimension $h\dim_G X$ with respect to the R-module of coefficients G does not exceed the cohomological one $\dim_R X$ and if R is a field, then $h\dim_R X = \dim_R X$. The dimension $h\dim_Z X$ over the integers can differ from $\dim_Z X$ by no more than 1, equality implies the dimensional polyvalue of X and is equivalent to it in the class of $h\ell c_Z$-spaces [76]. Characterizations dual to those known in the cohomological version of the theory are valid. For example, from $H_k(X, A; G) = 0$ for all closed $A \subset X$ it follows that for $p > 0$ one also has $H_{k+p}(X, A; G) = 0$ [45] (cf. also [42]). The condition $h\dim_G X \leqslant n$ is equivalent to the maps $H_n(A; G) \to H_n(X; G)$ being monomorphic [45] (cf. also [42]) and analogously, to the maps $H_n(X, A \cap A'; G) \to H_n(X, A; G) \oplus H_n(X, A'; G)$ in the homology Mayer–Vietoris sequences being monomorphic [48] (A and A' above being arbitrary closed sets).

In some bases there are results which usefully differentiate the homological version of the theory from the cohomological one. As noted in Sec. 5 of [66] for example, $h\dim_G X$ is determined by the homology groups of pairs (X, A), not only for closed, but also for open sets $A \subset X$ for (for cohomology groups this is not so, cf. below). Analogously [64] if $h\dim_G X < \infty$ then this number coincides with the maximal n for which one can find nonzero local groups $H_n^x(G)$ (as noted in point 3.2 above, the set of such $x \in X$ has full dimension). As the example of a countable product of circles shows, the condition that $h\dim_G X$ is finite is essential. For $n = h\dim_G X$ the presheaf $U \to H_n(U; G)$ on a locally compact space is a sheaf (Lemma 9 of [76]), and this observation characterizes $h\dim_G X$ [45]. This means that the homology groups in the highest dimension coincide with the zero-dimensional cohomology groups of X with coefficients in the sheaf $\mathcal{H}_n(G)$ (cf. below the appearance of Poincaré duality).

4.2. Cohomology. General Information. For a paracompact space X the condition that its cohomological dimension $\dim_{\mathcal{G}} X$ (over the coefficient sheaf $\mathcal{G}$) does not exceed n is equivalent to the existence for the sheaf $\mathcal{G}$ of soft resolutions of length $\leqslant n$ (Proposition 2.15.2 in [87]). In this the agreement of the nature of dimension in topology and of the dimension of objects used in homological algebra appears, in particular, the global dimension of a ring, injective and projective dimensions of modules, etc. [92].

In contrast with $h\dim_G X$ the dimension $\dim_G X$ is determined only by pairs (X, A) with closed sets A, the dimensional invariant $\mathrm{Dim}_{\mathcal{G}} X$ defined in the same way as $\dim_{\mathcal{G}} X$ but with pairs (X, U) with open $U \subset X$ can differ (there exist metric compacta X of dimension n at some points of which $H^{n+1}(X, X \setminus x; G) \neq 0$ cf. point 3.3 above for n = 2, and also [44] for n = 0 and [76] for n = 1). It follows from Exercise 22 in Chap. 2 of [87, 37] that for a hereditarily paracompact space X with the first axiom of countability the condition $\mathrm{Dim}_{\mathcal{G}} X \leqslant n$ is equivalent to the existence for the sheaf $\mathcal{G}$ of flabby resolutions of length $\leqslant n$. For any hereditarily paracompact space $\mathrm{Dim}_{\mathcal{G}} X \leqslant \dim_{\mathcal{G}} X + 1$ (Lemma 5.2 of [66]). Both invariants coincide for any locally constant coefficients $\mathcal{G}$ if X is a cellular polyhedron, for a locally constant sheaf $\mathcal{G}$ with finitely generated stalks if X is a generalized manifold, and for locally constant $\mathcal{G}$ with stalk a field if X is homologically locally connected over this field (Sec. 5 in [66]).

For paracompact spaces the condition $\dim_{\mathcal{G}} X \leqslant n$ is equivalent to the maps $H^n(X; \mathcal{G}) \to H^n(A; \mathcal{G})$ being epimorphisms for al closed $A \subset X$ (cf. [60] or [33]) and for $n \geqslant 1$ also to the maps $H^n(X, X_1; \mathcal{G}) \oplus H^n(X, X_2; \mathcal{G}) \to H^n(X, X_1 \cap X_2; \mathcal{G})$ in the Mayer–Vietoris sequences being epimorphisms for any closed sets X_1, X_2 [48] (the condition $n \geqslant 1$ is essential). There are also other characterizations in terms of homomorphisms of the higher terms of the Mayer–Vietoris sequences [48]. In the case of a locally compact space, $\dim_{\mathcal{G}} X \leqslant n$, $n \geqslant 1$ precisely under the condition that the presheaf $U \to H_c^n(U; \mathcal{G})$ is a cosheaf (V. G. Miryuk, B. L. Okun', cf. [48]).

In contrast with $h\dim_G X$ the cohomological dimension $\dim_G X$ is not determined by the local groups $I_x^{p-1}(G)$ or $H_x^p(G)$ [76] even in the category of metric compacta (the unsuitability of $I_x^{p-1}(G)$ is demonstrated by the examples cited above). For a locally compact space X with the second axiom of countability, under the condition that $\dim_G X < \infty$ this invariant coincides with the maximal number n for which $H_x^n(G)\oplus I_x^n(G)\neq 0$ at some $x\in X$, where the set of such points has cohomological dimension n [76] (that the assumption of finite-dimensionality is essential is demonstrated by a countable product of circles). For pcℓc-spaces of finite dimension (over Z), $\dim_G X$ coincides with the maximal n for which $I_x^{n-1}(G)=H_x^n(G)\neq 0$ at some points [55].

Since the dimension is unchanged in passing to one-point compactifications, for locally compact spaces in the original definitions of both $h\dim_G X$ and $\dim_G X$ one can use both ordinary homology and cohomology groups and the theories H_* and H_c^*.

The interpretation of the ordinary Lebesgue dimension dim as the cohomological dimension $\dim_Z$ lets one apply the algebraic apparatus of cohomology theory to it. One gets generalizations to paracompact spaces here rather simply and naturally of some known results. In particular, with the help of the Leray spectral sequence of a continuous map it is easy to get a theorem about dimension-lowering maps [57]: for a closed surjective map $f:X\to Y$ of paracompact Hausdorff spaces, $\dim X\leqslant\max_k(d_k+k)$, where d_k is the relative dimension (which coincides with the ordinary one in the case of a space with countable base) in Y of the set M_k (i.e., the maximal dimension of sets contained in M_k which are closed in X) of all $y\in Y$ for which $\dim f^{-1}y\geqslant k$. Cf. [57, 59] for some corollaries, special cases, and generalizations to nonclosed maps. One of the corollaries is that closed zero-dimensional maps do not lower dimension, $\dim Y\geqslant\dim X$ for them always. If for such a map $\dim Y\geqslant\dim X+m$ then for any k = 0, 1, 2,...,m one gets an estimate $\dim Y\leqslant d_k+k$, where d_k is the relative dimension in Y of the set of all points y for which $f^{-1}y$ contains at least k + 1 preimages. In particular, the map f is at least (m + 1)-fold (one can find points $y\in Y$ having at least m + 1 preimages). Such results were obtained in [22, 71] using tools of algebraic topology and sheaf theory.

In [22] a connection was observed of a similar kind of result with the spectral sequence corresponding to a resolution which arises from the map of the constant sheaf R on Y (R being a commutative ring). First constructed only for finitely-multiple closed maps [22], such resolutions were defined along with simplifications afterwards for any closed maps [70, 72], and later also for any continuous ones [24] (along with corresponding spectral sequences). Such resolutions are of the greatest interest for zero-dimensional maps (when the Leray spectral sequence degenerates and actually does not carry any information), guaranteeing the characteristics of perfect zero-dimensional maps, in particular the ones that are open, the characteristics in the class of open perfect maps of the subclasses of zero-dimensional [151] and monotone [25] maps. A general approach to the description of this kind of resolution is given in [73], and in [74] there is an analysis of the spectral sequences which arises there. Special cases are the Cartan spectral sequence of a regular covering [23] and its generalization, the Borel spectral sequence of a semifree action of a finite group on a topological space [74], and the spectral sequences of coverings in Cech cohomology theory [24]. One also gets a natural description of the Snapper spectral sequence illuminating the connection between the cohomology groups of a finite group G and any subgroup H of it revealing its generality with the Serre–Hochschild spectral sequence defined when H is a normal subgroup of G [74].

In the situation when $f:X\to X/G$ is the canonical map onto the quotient-space under a free action on X by a finite group G, there arise spectral sequences of a new type, not included in the preceding framework [152]. The resolutions figuring here are also of interest from the point of view of the formal homotopy theory of locally homotopically contractible differential graded sheaves.

4.3. Bohshtein Inequalities. Realization of Dimension Functions. As is known (M. K. Bokshtein), there exists a countable system σ of Abelian groups such that from the cohomological dimensions $\dim_H X$ for $H\in\sigma$ one can define the cohomological dimension of a metric compactum X for any other group G. The system σ is made up of the following groups: the group of rational numbers Q, cyclic groups of prime orders Z_p, p-primary quasicyclic groups Q_p, and the groups of rational numbers R_p whose denominators are relatively prime with the prime number p. As M. F. Bokshtein showed, $\dim_G X=\sup_{H\in\sigma(G)}\dim_H X$, where $\sigma(G)$ is a supply of groups from σ depending on G (Sec. 4 of [33]). The following inequalities of M. F. Bokhstein are known (Sec. 4 in [33]):

TABLE 1

	R_p	Z_p	Q_p	R_q	q	Q_q	
$\dim_H FQ_n$	n	1	1	n	1	1	n
$\dim_H FR_{pn}$	n	n	n	n	1	1	n
$\dim_H FZ_{pn}$	n	n	$n-1$	1	1	1	1
$\dim_H FQ_{pn}$	n	$n-1$	$n-1$	1	1	1	1

$$\dim_{Q_p} X \leqslant \dim_{Z_p} X; \quad \dim_{Z_p} X \leqslant \dim_{Q_p} X + 1,$$
$$\dim_Q X \leqslant \dim_{R_p} X; \quad \dim_{Z_p} X \leqslant \dim_{R_p} X,$$
$$\dim_{Q_p} X \leqslant \sup\{\dim_Q X, \dim_{R_p} X - 1\},$$
$$\dim_{R_p} X \leqslant \sup\{\dim_Q X, \dim_{Q_p} X + 1\}.$$

By a dimension function D is meant an arbitrary map of the set σ into the set of all natural numbers satisfying these inequalities and such that if D(H) = 0 for some $H \in \sigma$ then D = 0. Examples are D(H) = $\dim_H X$ (for any fixed X). The question of whether each dimension function can be realized in this way (for some metrizable compactum), i.e., are the inequalities given above the only restrictions on the collection of cohomological dimensions of a compactum remained open for a very long time.

A positive answer to this question was recently obtained by Dranishnikov [17-19]. As established in [33], to solve the problem it suffices to construct for each prime p and any integer $n \geqslant 2$ so called basic compacta FQ_n, FR_{pn}, FQ_{pn}, and FZ_{pn} whose cohomological dimensions would be given by Table 1, cf. Sec. 6 in [33] ($q \neq n$). The compactum X corresponding to a given dimension function is obtained from these basic ones with the help of the operation of infinite compact bouquet. For n = 2, the basic compacta were constructed by Pontryagin, V. G. Boltyanskii, and Kodama, for n = 3 by Kuz'minov [31]. Dranishnikov constructed basic compacta for any n.

The following assertions [17] are corollaries of the basic result: for any natural numbers m < n and $1 \leqslant k \leqslant m$ there exist compacta X and Y such that dim X = m, dim Y = n, and dim × (X × Y) = n + k; for finitely generated groups G the condition $\dim_G X \leqslant n$ is equivalent to the existence of a G-acyclic map f onto X of some n-dimensional compactum [$\tilde{H}^*(f^{-1}(x); G) = 0$ for all $x \in X$). For G = Z, this is a theorem of Edwards [145].

4.4. Infinite-Dimensional Spaces. When the Lebesgue dimension dim is finite, it coincides with the cohomological one $\dim_Z$ (one of the basic theorems of the homological theory of dimension, due to S. P. Aleksandrov). How things are for dim X = ∞ is one of the basic problems of the homological theory of dimension (Aleksandrov's problem), which has resisted solution for a very long time: are there infinite-dimensional compacta X for which for some coefficient groups $\dim_G X < \infty$? As is known, for G = Z this problem is equivalent to the following no less complicated one: can cell-like maps raise dimension? [104, 145]? The history of this problem and also problems connected with the behavior of dimension under cell-like maps, including those relating to the theory of manifolds, were considered in many papers, cf. [138, 15]. It follows from the Vietoris–Begle theorem (cf. below point 5.3) that maps which are acyclic (over Z) do not raise dimension ($\dim_Z$) so cell-like maps can raise the dimension of finite-dimensional spaces only to infinity and this is how the two problems are related.

Until recently classes of infinite-dimensional compacta whose cohomology dimension also turns out to be infinite, were singled out. In [144] for example, similar subspaces of the Hilbert cube are described which are intersections of subsets separating its opposite faces Compacta of this type are contained as subspaces in familiar examples of compacta all of whose subspaces which are not zero-dimensional are infinite-dimensional so the cohomological dimension of the compacta in these examples is also infinite.

The existence of infinite-dimensional compacta of finite cohomological dimension established by A. N. Dranishnikov follows from the content of the preceding point: it suffices to take a countable compact bouquet of basic compacta of any of the four types considered, corresponding to all values of n (for fixed p). Here for $G = Z_q$, Q_q, R_q, and Q there arise infinite-dimensional compacta with $\dim_G X = 1$ (the answer to Aleksandrov's question in connection with the basic problem), even including some for which $\mathrm{dom}_{Z_p} X = 1$ for all p. In the original examples (including strongly infinite-dimensional compacta X) only the inequality $\dim_{Z_p} X \leqslant 2$ was achieved [16]. It is also shown that for any $m \geqslant 3$ there exists a compactum X_m of dimension m for which $\dim_{Z_n} X_m = 2$ for all n [16]. An analogous compactum is also constructed for $m = \infty$; it is strongly infinite-dimensional and $\dim_Z X = \infty$ [16]. In particular, the difference $\dim_Z X - \dim_{Z_n} X$ can assume any value including infinity (also answering one of the questions posed by Aleksandrov). The proof is based on the characterization of k-dimensional compacta with respect to $G = Z_p$ obtained by the author, in terms of their approximation by finite polyhedra, analogous to the famous result of R. Edwards [145] for $G = Z$.

The basic problem is of particular interest for $G = Z$. The characterizations of the dimensions $\dim X$ and $\dim_Z X$ in terms of extension of maps from subsets of X into the sphere S^n and into Eilenberg–MacLane polyhedra $K(Z, n)$, guaranteeing the equivalence of the conditions $\dim X \leqslant 1$ and $\dim_Z X \leqslant 1$ show that the solution of the problem for $n \geqslant 2$ depends on rather fine distinctions between the concepts of homotopy and weak homotopy equivalences. Dranishnikov was able to solve the problem for $n \geqslant 3$ (the case $n = 2$ remains obscure): there exists an infinite-dimensional compactum X for which $\dim_Z X = 3$. He also established the existence of cell-like maps of a) three-dimensional compacta and b) a standard cube of dimension $n = 7$ onto infinite-dimensional compacta. For $n = 3$ there are no such maps of a cube [117], for $n = 4$, 5, and 6 the question remains open.†

In [20] even the existence of infinite-dimensional compacta which are three-dimensional cohomological manifold over Z_p (simultaneously for all primes p) is established. Cohomological n-manifolds whose Lebesgue dimension exceeds n arise in studies connected with the famous Hilbert–Schmidt problem. The proof is based on the following result [20] which is of independent interest (cf. the end of point 4.3): for any set $\mathscr{P}$ of primes an arbitrary compactum X for which $\dim_{Z_p} X \leqslant n$ for $p \in \mathscr{P}$ is the image of an n-dimensional compacta with preimages of points Z_p-acyclic for all $p \in \mathscr{P}$.

4.5. Problems of Full-Valuedness of Dimension. Examples of dimensionally non-full-valued compacta (i.e., ones with noncoincident cohomological dimensions with respect to some coefficient groups) were known long ago (L. S. Pontryagin, V. G. Boltyasnkii, and V. I. Kuz'minov). Interest in them increased in the last decades in connection with the fact that a similar kind of space started to occur in some concrete problems of the theory of G-spaces, in particular in studies on the Hilbert–Schmidt problem.

The known examples of dimensionally non-full-valued compacta had complicated local homological structure. After [100] it was assumed for a long time that any $h\ell c$-spaces should be dimensionally full-valued (Dyer's problem). Afterwards attention was concentrated on the smaller class of finite-dimensional locally contractible compacta which coincides with the class of finite-dimensional compact ANR. It was shown that in this class for some p (depending on X) $\dim X = \dim_{Z_p} X$. It follows from this for example that $\dim X^n = n \dim X$ (for arbitrary compacta as proved by Boltyanskii this is not so). In accord with [148] any compact ANR has the homotopy type of a polyhedron. Some other distinctive properties of compact ANR corroborated K. Borsuk's opinion that they should be the homological analog of polyhedra. In 1978 he posed the problem of dimensional full-valuedness of compact ANR.

Dranishnikov [20] recently obtained the decisive solution of this problem: there exists a family of four-dimensional AR-compacta M_p indexed by all primes, such that $\dim(M_p \times M_q) = 7$ for any pair of indices $p \neq q$ (as is known, the dimensional full-valuedness of X is equivalent to the relations $\dim(X \times Y) = \dim X + \dim Y$ for any compacta Y). Thus, the WR-compacta M_p are not only not dimensionally full-valued, but the addition formula for dimensions can even fail when both factors are AR-compacta.

Problems of dimensional full-valuedness and the structure of homological n-manifolds over different fields of coefficients having infinite ordinary dimension are closely connected with the familiar reductions of the Hilbert–Schmidt conjecture (that any compact group which acts freely on a manifold is a Lie group). In [20] much attention is paid to the discussion of a number of profound new ideas relating to this (cf. also [21]).

†The characterization of spaces of cohomological dimension n (over Z) as cell-like images of n-dimensional ones is justified [104, 145, 153, and 154].

The idea that compact ANRs can serve as the homological analog of finite polyhedra is refuted in general by the example discussed in point 3.3 of a two-dimensional compactum: not only is the group $H^3(X, X \setminus x; Z)$ nonzero for it but it is a continuum. In the presence of the conditions $h\ell c_Z$ and $ph\ell c_Z$ as noted all local groups over Z are finitely generated. In [66] such locally compact spaces are called generalized polyhedra. The question of their dimensional full-valuedness is of interest.

5. Continuous Maps

5.1. Leray Spectral Sequence. The maps of homology and cohomology in equal degrees induced by continuous maps $f:X \to Y$ of topological spaces depend both on the disposition of the image $f(X)$ in Y and on the local structure of f characterized by the structure of the complete preimages of points $f^{-1}(y)$, $y\in Y$. For cohomology they are described by the Leray spectral sequence generated by f and converging to $H^*(X; \mathcal{F})$ whose second term has the form $E_2^{pq}=H^p(Y; \mathcal{H}^q(f))$, where the sheaves $\mathcal{H}^q(f)$ are generated by the presheaves $U\to H^q(f^{-1}(U); \mathcal{F})$. In particular, $\mathcal{H}^0(f)=f\mathcal{F}$ is the direct image of the sheaf $\mathcal{F}$ under the map f. The stalks of the sheaves $\mathcal{H}^q(f)$ at points $y\in Y$ are the direct limits of the groups $H^q(f^{-1}(U); \mathcal{F})$ over neighborhoods U of these points. They coincide with the cohomology groups $H^q(f^{-1}(y); \mathcal{F})$ in the following cases: a) for locally trivial fibrations $f:X \to Y$ over locally contractible Y and constant $\mathcal{F}(H^q(f^{-1}(U); \mathcal{F}) =H^q(f^{-1}(y)$ ($\mathcal{F}$ for sufficiently small U); b) for any closed maps f [property of rigidity of the cohomology groups of the subspaces $f^{-1}(y)$] in particular for any maps of compact spaces, for proper maps of locally compact spaces. In general there exist only natural transformations $\mathcal{H}^q(f)_y\to H^q(f^{-1}(y); \mathcal{F})$.

The spectral sequence guarantees the existence of homomorphisms ν: $E_2^{p0}=H^p(Y; f\mathcal{F})\to H^p(X; \mathcal{F})$ induced by f. In general the sheaf $f\mathcal{F}$ is rather specific. In order to "pass" through the spectral sequence the maps of cohomology groups with previously given coefficients induced by f, as $\mathcal{F}$ one takes the sheaf $f^*\mathcal{G}$, the inverse image under the map f of the sheaf $\mathcal{G}$ on Y. The homomorphisms $f^*: H^p(Y; \mathcal{G})\to H^p(X; f^*\mathcal{G})$ induced by f can be represented as compositions $H^p(Y; \mathcal{G})\to H^p(Y; f f^*\mathcal{G})\xrightarrow{\nu} H^p(X; f^*\mathcal{G})$ in which the first maps correspond to the natural transformation $\mathcal{G}\to ff^*\mathcal{G}$. The transformation is an isomorphism only for monotonic surjective f [in this case E_2^{p0} can be identified with $H^p(Y; \mathcal{G})$]. Analogous inferences are valid for cohomology groups with supports in a paracompactifying family φ for Y and in the family $f^{-1}(\varphi)$ for X.

For constant coefficients $f^*G = G$; hence in the case of bundles, the Leray spectral sequence becomes the ordinary spectral sequence of the bundle.

5.2. Vietoris-Type Theorems. An obvious corollary of the Leray spectral sequence is the following assertion (cohomology version of the Vietoris–Begle theorem, cf. point 5.4 below): For a closed map f of the paracompact space X onto the paracompact space Y which is acyclic up to dimension $N - 1$ (i.e., with trivial reduced cohomology groups of the complete preimages of points), the induced maps $f^*: H^p(Y; G)\to H^p(X; G)$ are isomorphic for $p\leqslant N-1$ and in dimension N the map is monomorphic. (This and the results below can also be obtained without the spectral sequence but using the ultimately equivalent language of exact pairs of Massey.) There are many generalizations and partial converses.

Let M_0 be the set of all those $y\in Y$ for which $H^0(f^{-1}(y); f^*\mathcal{G})\neq\mathcal{G}_y$, M_k, $k\geqslant 1$, be the set of all $y\in Y$ for which $H^k(f^{-1}(y); f^*\mathcal{G})\neq 0$, d_k in the relative dimension of M_k in Y (point 4.2). For a closed surjective map $f"X \to Y$ and a natural number N, let $n=1+\max_{0<k<N}(d_k+k)$ (as N, ∞ can also occur).† It turns our [59] that if $n < N$ (some K_k may be empty), the homonorphisms $f^*: H^i(Y; \mathcal{G})\to H^i(X; f^*\mathcal{G})$ are epimorphisms for $n\leqslant i<N$ $n<i\leqslant N$. As a consequence one gets the analogous result (Sec. 2 in [59]) for maps of pairs $f"(X, f^{-1}A) \to (Y, A)$ too (the set A being closed).

The converse of this result holds [59] in the following form. Let m be a number which is equal to zero if $d_k\leqslant 0$ for $k\leqslant N-1$ or to $1+\max_{\substack{d_k>0\\ k<N}}(d_k+k)$ if there is an M_k with positive d_k. Under the condition that $m\leqslant n<N$ it follows from the fact that f^* is epimorphic in dimensions $n\leqslant i<N$ and monomorphic for $n<i\leqslant N$ for all $y\in Y$ that $H^p(f^{-1}y; f^*\mathcal{G})=0$ for $n\leqslant p<N$, $p > 0$, and if $n = 0$, then also that $H^0(f^{-1}y; f^*\mathcal{G})=\mathcal{G}_y$.

†In order not to consider k for $M_k = \emptyset$ we set $d_k = -\infty$.

A different way of generalizing is given in [82]. The map $f:X \to Y$ being studied is considered over a space T in the sense that the preimages of points under the maps $g:Y \to T$ and $h = gf:X \to T$ are equal. One shows that the analog of the Vietoris–Begle theorem remains valid if the condition of acyclicity of preimages of points for the map f is replaced by the requirement that for $p \leqslant N-1$ all the maps $H^p(g^{-1}t) \to H^p(h^{-1}t)$, $t \in T$ are isomorphisms and in dimension N it is a monomorphism. In the same paper there is given the converse of the result. It is shown that if the sets M_k of points for which the maps $H^k(g^{-1}t) \to H^k(h^{-1}t)$ are not isomorphic are contained in an $M \subset T$ of relative dimension 0 (the case n = 0) and $H^p(g^{-1}t) = H^p(h^{-1}t) = 0$ for points $t \in T \setminus M$ for $0 < p < N$ then it follows from the fact that $H^p(Y) \to H^p(X)$ are isomorphisms for $p < N$ and a monomorphism in dimension N, that the indicated sets M_k are generally empty.

Both of these results are valid in the more general form indicted above [59]. Instead of the conditions of loss of acyclicity of f, figuring in the description of the sets $M_k \subset Y$ one uses the absence of an isomorphism $H^k(g^{-1}(t); \mathcal{G}) \to H^k(h^{-1}(t), f^*\mathcal{G})$ (M_k being subspaces of T). For the map f^* to be a monomorphism in the extreme dimension N one requires in additional that all $H^N(g^{-1}(t); \mathcal{G}) \to H^N(h^{-1}(t); f^*\mathcal{G})$ be monomorphisms. In the converse of the theorem, in contrast with [82] it is not required that the union of the M_k have relative dimension 0, and the requirement of acyclicity of the sets $g^{-1}(t)$ and $h^{-1}(t)$ at points $t \in T \setminus M_k$ is also discarded (the numbers m, n, and N preserve their former meanings). If $H^p(Y; \mathcal{G}) \to H^p(X; f^*\mathcal{G})$ are epimorphisms for $n \leqslant p < N$ and monomorphisms for $n < p \leqslant N$ then for all $t \in T$ the maps $H^p(g^{-1}(t); \mathcal{G}) \to H^p(h^{-1}(t); f^*\mathcal{G})$ are epimorphisms for $n \leqslant p < N$ and monomorphisms for $n < p \leqslant N$. If under the conditions indicated the relative dimension of the set of all $t \in T$ at which the map $H^N(g^{-1}t; \mathcal{G}) \to H^N(h^{-1}t; f^*\mathcal{G})$ is not a monomorphism is not greater than zero, then this set is generally empty.

Analogous results are valid for inclusions $i:X \subset Y$ as dense subsets, X and Y being paracompact spaces [59]. We consider the sets

$$M_0 = \{y \in Y \mid \varinjlim_{y \in U} H^0(U \cap X; \mathcal{G}) \neq \mathcal{G}_y\}, \quad M_k = \{y \in Y \mid \varinjlim_{y \in U} H^k(U \cap X; \mathcal{G}) \neq 0\}, \ k \geqslant 1,$$

where $\mathcal{G}$ is a sheaf on Y, U is a neighborhood of y in Y. These sets obviously lie in $Y \setminus X$. As above, let $1 + \max\limits_{0 \leqslant k < N} (d_k + k) = n$, where N is a natural number of ∞. For $n < N$ the maps i^*: $H^p(Y; \mathcal{G}) \to H^p(X; \mathcal{G})$ are epimorphisms for $n \leqslant p < N$ and monomorphisms for $n < p \leqslant N$ (Vietoris-type theorem for inclusions).

Analogously, let m = 0 if $d_k \leqslant 0$ for k < N or $m = 1 + \max\limits_{\substack{d_k > 0 \\ 0 \leqslant k < N}} (d_k + k)$ if there are positive d_k. Let $m \leqslant n < N$. If the maps i^* are epimorphic for $n \leqslant k < N$ and monomorphic for $n < k \leqslant N$ then $\varinjlim_{y \in U} H^p(U \cap X; \mathcal{G}) = 0$ for all $y \in Y$ for $0 < n \leqslant p < N$ and if n = 0, then $\varinjlim_{y \in U} H^0(U \cap X; \mathcal{G}) = \mathcal{G}_y$ also.

5.3. Applications. The behavior of the cohomology groups under continuous maps both from the positions of the Leray spectral sequence and generally, plays an important role in the study of maps lowering of or raising dimension (point 4.2 above). As simple consequences of the constructions of the beginning of point 5.2 one has the following [59]: the Hurewicz theorem on zero-dimensional maps (for $\dim Y > \dim X$ the relative dimension of the set of all $y \in Y$ having disconnected $f^{-1}(y)$ is at least $\dim Y - 1$; Dyer's theorem on acyclic maps $X \to Y$ (for $\dim Y < \infty$ one always has $\dim Y < \dim X$). In each case the classical result is extended to the larger category of closed maps of paracompact spaces and generalizations are obtained (to maps which are not zero-dimensional in the small or not acyclic in the small).

The Leray spectral sequence and the results of point 5.2 can be applied to many other problems. Below let $f:X \to Y$ be a closed surjective map (in the category of paracompact spaces). If for some $A \subset Y$ the spaces $X \setminus f^{-1}(A)$, $Y \setminus A$ are paracompact and dense in X and Y and for any point $y \in A$ we have $\varinjlim_{y \in U} H^p(U \cap (Y \setminus A); G) = H^p(y; G)$ then if the analogous conditions hold at points $x \in f^{-1}(A)$ the acyclicity of the map f on the set $Y \setminus A$ implies the acyclicity of f on all of Y (Theorem 2 of [61]). Such conditions arise when the sets A and $f^{-1}(A)$ are boundaries of generalized manifolds or have neighborhoods homeomorphic to their products by a half-interval, from which Theorems 1 and 2 of [118] follow quickly. If $d_k \leqslant 0$ for all $k \geqslant 0$

for $f:X \to Y$ (cf. the beginning of point 5.2) and for one of X and Y the cohomology modules with coefficients in an R-module of finite type G (R being a principal ideal ring) are finitely generated in each dimension, then the cohomology of the other space has precisely this same property provided that all the sets M_k are finite and the modules $H^p(f^{-1}(y); G)$ are finitely generated; if in addition all the homomorphisms $H^p(X; G) \to H^p(f^{-1}(y); G)$ are trivial [for example, the preimages $f^{-1}(y)$ are sufficiently small], then the cohomology groups of x and Y are isomorphic if and only if f is acyclic on all of Y (Theorem 1 of [61]). In particular, if f is not acyclic, X and Y are not homeomorphic. The analog of the first part of the formulated result also holds for suitable dense inclusions $X \subset Y$ (Lemma 4.4 of [136] and more completely, Theorem 1' of [61]).

The language of one-dimensional cohomology groups turns out to be irreplaceable in questions of classification of certain forms of compactifications of topological spaces [58, 59]. As is known, peripherally compact sets X (in particular, any locally compact ones) have compactifications (compact extensions) Y with zero-dimensional outgrowth $Y \setminus X$. Such compactifications, a special case of extensions with punctiform outgrowths (having relative dimension 0 in Y). If the space X has at least one compactification with punctiform outgrowth, then in the set of such extensions there is a unique maximal one denoted by μX (which in the case of peripherally compact X coincides with the familiar Freudenthal compactification with zero-dimensional outgrowth) and characterized by the fact that among them it is the unique perfect one (the outgrowth $Y \setminus X$ locally does not separate X at any point).

If Y_1, Y_2 are compactifications of X with punctiform outgrowths and $Y_1 < Y_2$ then the natural map $Y_2 \to Y_1$ (the identity on X) is zero-dimensional so $d_k = -\infty$ for $k > 0$, $d_0 = 0$, $n = 1$, and $N = \infty$. Thus, the map $f^*: H^p(Y_1; G) \to H^p(Y_2; G)$ is an isomorphism for $k \geqslant 2$ and an epimorphism for $k = 1$. For locally compact X the fact that one has an epimorphism in dimension 1 was established in [105]. If the space of quasicomponents of X is compact (in particular, if X is connected), then from the converse of the Vietoris theorem of point 5.2 it follows that if f^* is an isomorphism in dimension 1, then $Y_1 = Y_2$. For a connected locally compact X the result was obtained in [105].

For a fixed compactification Y_0 with punctiform outgrowth, let H be the kernel of the epimorphism $H^1(Y_0; Z) \to H^1(\mu X; Z)$ and H_Y be the kernel of the analogous homomorphism for a compactification of Y following Y_0. It turns out that H_Y is a serving subgroup of the torsion-free group H, and for connected X the assignment $Y \to H_Y$ is an isomorphism of the partially ordered set of all compactifications following Y_0 with punctiform outgrowths onto a set of such subgroups of H ordered by inclusion. The result is also of interest for locally compact spaces. In this case one can take as Y_0 the one-point compactification of X, and the set of all compactifications of X with zero-dimensional outgrowths is isomorphic to a set of serving subgroups of H ordered by inclusion. If Y_1, Y_2 are compactifications of X following Y_0 with punctiform outgrowths and $\pi_i: Y_i \to Y_0$ are the natural projections, then $Y_1 \leqslant Y_2$ precisely when one can find a homomorphism $\alpha: H^1(Y_1; Z) \to H^1(Y_2; Z)$ for which $\alpha\pi_1^* = \pi_2^*$. Analogous constructions are applicable for finding conditions under which a perfect surjective map f: $X_2 \to X_1$ of connected spaces extends to a map $Y_2 \to Y_1$ of certain compactifications of these spaces under the condition that Y_1 is a compactification of X_1 with punctiform outgrowth following an analogous compactification Y_0 for which such an extension does exist (in the category of locally compact spaces the maps discussed coincide with proper ones, as Y_0 one can always use the one-point compactification of X_1 so one can pose the question of extending f for any compactification Y_1 with zero-dimensional outgrowth).

Since the natural map of the Stone–Cech compactification βX onto μX is monotonic, the map $H^1(\mu X; Z) \to H^1(\beta X; Z)$ is monomorphic. As Y_2 above one can take extensions for which $H(Y_2; Z) \to H^1(\beta X_2; Z)$ is a monomorphism (in particular, acyclic in dimension 1). In this case it follows from the existence of an extension of the map f to a map $Y_2 \to Y_0$ onto an extension Y_0 (of the space X_1) with punctiform outgrowth (and this condition always holds in the category of locally compact spaces) that there exists an extension of f to a map $Y_2 \to Y_1$ with any extension Y_1 of the space X_1 have punctiform outgrowth. For $X_1 = X_2 = X$ and f the identity, in particular, if the compactification Y of the connected space X for which $H^1(Y; Z) \to H^1(\beta X; Z)$ is monomorphic follows at least one extension of X with punctiform outgrowth, then Y follows μX. When X is locally compact and Y is acyclic, this assertion is Theorem 4 in [105]. Thus the extension μX of the connected locally compact space X can be characterized as the minimal one among all compactifications Y for which $H^1(Y; Z) \to H^1(\beta X; Z)$ is a monomorphism. The connectedness requirement of the spaces considered in all these constructions can be replaced by weaker ones [58].

In some situations there arise transparent connections between the cohomology groups of X and the cohomology groups of compactifications of X. If the coefficient group G is finite or if the space of quasicomponents of X is pseudocompact, for any perfect extension Y the map $H^0(Y; G) \to H^0(X; G)$ is an isomorphism (Theorem 5 in [59]). It is clear that for arbitrary compactifications Y of a disconnected space X this is not so (on X one can find locally constant functions with values in G which cannot be extended to Y). Let Y be a compactification with the first axiom of countability at points of $Y \setminus X$. In order that the extension Y be perfect it is necessary and sufficient that for the constant sheaf G on X its direct image iG (under the inclusion $i: X \subset Y$) coincide with the constant sheaf G on Y (Theorem 6 in [59]). If Y is a perfect extension of X with the first axiom of countability at points of the outgrowth (for example, the Freudenthal extension μX of a peripherally compact space X of countable weight with compact space of quasicomponents), then the map $i^*: H^1(Y; Z) \to H^1(X; Z)$ is monomorphic [cf. with the cohomology groups $H^1(\beta X; Z)$ in point 7.3 of Chap. 3], and if there are compactifications of the space X with punctiform outgrowths, then for any of them different from μX (and if X is locally compact, then also for any preceding μX) the i^* indicated above has nonzero kernel (Theorem 7 in [59]). For any finite-dimensional paracompact space X there exist compactifications Y of the same dimension for which the maps $H^p(Y; G) \to H^p(X; G)$ $p>0$ (Theorem 11 in [59]).

5.4. Homology. Let $f: X \to Y$ be a surjective map of metric compacta and G be a countable group. If the homology of $f^{-1}(y)$, $y \in Y$ with coefficients in G is trivial up to dimension $N - 1$, then the induced map $H_p(X; G) \to H_p(Y; G)$ is isomorphic for $p \leqslant N-1$ and epimorphic for $p = N$. This result was recently proved in [47]. The analogous assertions, respectively, for $p \leqslant N - 2$ and $p = N - 1$ were previously obtained in [28]. For $G = Z$ and $N = \infty$ (without the metrizability condition) the assertion is proved in [87]. For compact groups and fields of coefficients the results were known long ago, cf. [80, 81] (Vietoris–Begle theorem).

There are no results of this kind yet for uncountable groups G and also beyond the limits of the compact category (for homology groups with compact supports). In relation to applications connected with continuous maps the content of the first three points of the present section demonstrate the advantage of cohomology as compared with homology.

5.5. Generalized Homotopy Property. Let $f, g: X \to Y$ be continuous maps of paracompact Hausdorff spaces for which there exists a connected compactum T with fixed points s and t and a map $F: X \times T \to Y$ such that the restrictions F_s, F_t of the map F to $X \times s$ and $X \times t$ coincide with f and g, and let i_s, i_t be the imbeddings of X in $X \times T$ corresponding to s and t. Under what conditions do the cohomology maps f^* and g^* induced by f and g (equivalently, i_s^*, i_t^*) and the homology maps f^*, g^* (equivalently i_s^*, i_t^*) coincide?

If X is compact, then $f^* = g^*$ for any connected (possibly almost noncompact) T and any coefficient group G [140, 97]. If G is a compact group or field, $f^* = g^*$ for compact T and any X [140]. The conditions for the equalities $i_s^* = i_t^*$ were studied in [46]. Let S be the compact space obtained by attaching to T at the points s and t the ends of the segment [0, 1]. The exact sequence [of cohomology groups of the pair (S, T)]

$$0 \to G \to H^1(S; G) \to H^1(T; G) \to 0 \tag{21}$$

plays a fundamental role.

It turns out that the equality $i_s^* = i_t^*$ holds for any paracompact X if and only if this sequence splits. For $G = Z$ this is so if $H^1(T; Z)$ has a finite number of generators (since being torsion-free it turns out to be free). If the sequence splits for $G = Z$, then $i_s^* = i_t^*$ for any other group G. If T is a solenoid, $i_s^* \neq i_t^*$ for some points s and t and the group $G = Z$.

For homology groups with compact supports the picture is the following [46]. For $G = Z$ th following assertions are equivalent: a) $i_{s*} = i_{t*}$; b) the sequence (21) splits; c) $i_{s*} = i_{t*}$ when X consists of one point. If the sequence (21) splits for $G = Z$, then $i_{s*} = i_{t*}$ for any group G.

The problem for the case $G = Z$ was considered from other points of view in [4]. In particular, $i_s^* = i_t^*$ in dimension p if and only if the connecting homomorphism $H^{p-1}(X; H^1(T; Z)) \to H^p(X; Z)$ corresponding to (21) is equal to zero (cf. with the material at the end of the point). When X is locally compact and homologically locally connected in the sense of singular theory H_*^S an analogous criterion is also obtained in terms of the connecting homomorphism $\mathrm{Hom}(H^s_{p-1}(X; Z), H^1(T; Z)) \to \mathrm{Ext}(H^s_{p-1}(X; Z), Z)$. Similar conditions (with the groups H^*

figuring in them instead of H^S_*) are found for the coincidence of the maps i_{s*} and i_{t*} in dimension p. As corollaries one gets new proofs (for G = Z) of the basic results of [46].

Conditions for the equality $i_s^* = i_t^*$ were later considered by Nguen Le An' for an arbitrary homotopy functor of X of the form [X, A], where A is an H-space having the homotopy type of a cell complex (results to be published). The following assertions are consequences of the general criteria obtained: a) $i_s^* = i_t^*$ always for one-dimensional cohomology groups (with any coefficients); b) for fixed dimension n > 1 the equality $i_s^* = i_t^*$ for a given group G is equivalent to (21) splitting; c) for k-theory (considered as the homotopy functor [X, A] on the category of paracompact spaces, A = BU × Z being the classifying space) the equality $i_s^* = i_t^*$ for all paracompact X is equivalent to the splitting of the exact sequence $0 \to Z \to k^1(S) \to k^1(T) \to 0$.

6. Poincaré Duality. Generalized Manifolds

6.1. (G − n)-Spaces. Duality. Let $\mathcal{G}$ be a locally constant system of coefficients with stalk G on the locally compact space X, $\mathscr{C}_*(\mathcal{G})$ be the graded differential sheaf of chains of the space X with coefficients in $\mathcal{G}$ (point 2.3 of Chap. 2), $n = \mathrm{h\,dim}_G X$ be the homological dimension. Let us assume that $n < \infty$. That this restriction is essential in the considerations below (as well as in point 2.7 of Chap. 2 or point 4.1 above) is demonstrated by the example of a countable product of circles. As usual, let $\mathscr{Z}_p \subset \mathscr{C}_p(\mathcal{G})$ be the kernel of d and $\mathscr{B}_p \subset \mathscr{Z}_p$ be the image of d so that $\mathscr{Z}_p/\mathscr{B}_p = \mathscr{H}_p(\mathcal{G})$ (cf. point 2.6 of Chap. 2). By virtue of point 4.1 above, in dimensions p > n there is no homology and all $\mathscr{Z}_p = \mathscr{B}_p$ hence they are flabby [n being the maximal number for which $\mathscr{H}_p(\mathcal{G}) \neq 0$]. Consequently, any homology groups in X (with any supports φ) are defined by the sheaf $\tilde{\mathscr{C}}_*(\mathcal{G})$ obtained from $\mathscr{C}_*(\mathcal{G})$ by by rejecting all $\mathscr{C}_p(\mathcal{G})$ for p > n and factoring $\mathscr{C}_n(\mathcal{G})$ by the flabby subsheaf $\mathscr{B}_n = \mathscr{C}_{n+1}(\mathcal{G})/\mathscr{Z}_{n+1}$.

When $\mathscr{H}_p(\mathcal{G}) = 0$ also for p < n we shall call X a (G − n)-space (not only manifolds, but also, for example, spaces of the type of a sphere with diametral plane turn out to be such). But for a (G − n)-space $\tilde{\mathscr{C}}_*(\mathcal{G})$ is a flabby resolution of the "orienting" sheaf $\mathscr{H}_n(\mathcal{G})$ numbered in the opposite order, so for any families φ we have $H_k^\varphi(X;\mathcal{G}) = H_\varphi^{n-k}(X;\mathscr{H}_n(\mathcal{G}))$. This is Poincaré duality.

Any other duality relations in manifolds are consequences of this basic construction and the description by means of the theory of sheaves of chains and cochains of subspaces, pairs, pairs of subspaces, triads, etc. (points 5.1 and 5.3 of Chap. 1, point 2.4 of Chap. 2, Sec. 2 above). Some such relations in particular are noted in points 2.1 and 2.3 above. We also note that for example the homology groups of the relation $H_n^\varphi(A \div B;\mathcal{G})$ between the complementary sets A and B = X \ A in the (G − n)-space X coincide with the cohomology groups of the relation $H_\varphi^{n-p}(A \div B;\mathscr{H}_n(\mathcal{G}))$. Analogous relations are valid for homology and cohomology groups of a surrounding of a closed set (cf. Sec. 1). From the basic construction there automatically follow the duality relations for complementary subsets of an acyclic manifold, the duality "laws" of Alexander, Pontryagin, Steenrod, and finally, for "nonclosed" sets, of Sitnikov. A rather complete description of all manifestations of duality (including those in terms of the ∩-multiplication) together with the history of their appearance and development, is given in Sec. 5 of Chap. 8 of [69]. A simple consequence of the constructions given there is the following result (recently proved in a rather complicated way in [146] for the case $\mathcal{G} = G$): for a pair (X, ∂X) in which ∂X is the boundary of the n-manifold X, the exact sequene of homology with coefficients in $\mathcal{G}$ and with supports in the paracompactifying family φ [the groups $H_p^\varphi(\partial X;\mathcal{G})$, $H_p^\varphi(X;\mathcal{G})$, $H_p^\varphi(X,\partial X,\mathcal{G})$] coincides with the cohomology sequence

$$\ldots \to H_\varphi^{n-p-1}(\partial X;\mathscr{H}_{n-1}(\mathcal{G})) \to H_\varphi^{n-p}(X,\partial X;\tilde{\mathscr{H}}_n(\mathcal{G})) \to H_\varphi^{n-p}(X;\tilde{\mathscr{H}}_n(\mathcal{G})) \to \ldots,$$

in which $\tilde{\mathscr{H}}_n(\mathcal{G})$ is the natural extension of $\mathscr{H}_n(\mathcal{G})$ from Int X to all of X and $\mathscr{H}_{n-1}(\mathcal{G})$ is the restriction of $\tilde{\mathscr{H}}_n(\mathcal{G})$ to the boundary ∂X.

So simple an approach to the proof of duality as that described in [64] became possible thanks to the existence of flabby sheaves of chains. From the point of view of the spectral sequence of a differential sheaf of chains the duality is described in [87]. There too the concept of (R − n)-space (R being the ground ring) is introduced. For homology and cohomology groups of pairs the results are somewhat less general since the sheaves of chains are flabby only when $\mathcal{G} = R$. The idea of and the first attempts at proving Poincaré duality by the tools of sheaf theory are due to Cartan (cf. point 2.7 of Chap. 2). However it is unfortunate that this extremely transparent and fruitful idea came to be forgotten! Thus, the new approach presented in the recent book on the theory of sheaves [110] is unjustifiably complicated, the proof which occupies two chapters (the 5th and 6th) is very involved and the results achieved here are far from complete.

6.2. Categorical Nature of Duality. As is clear from the contents of the preceding point, discovering Poincaré duality "visually," geometrically, by studying the (apparently) explicity global structure of combinatorial manifolds depends only on local conditions and has deeper character depending on the constructions of homological algebra. Upon failure of the local structure, i.e., without the requirement $\mathcal{H}_p(\mathcal{G}) = 0$ for $p < n$ as already noted there arises a spectral sequence (of Cartan, point 2.7 of Chapter 2), from which there follows the existence of transformations

$$H^p_\varphi(X, A; \mathcal{H}_n(\mathcal{G})) \to H^\varphi_{n-p}(X \setminus A; \mathcal{G}),\ H^p_\varphi(A; \mathcal{H}_n(\mathcal{G})) \to H^\varphi_{n-p}(X, X \setminus A; \mathcal{G}),$$

each of which is isomorphic for $p = 0$ and monomorphic for $p = 1$. Since one can choose as A any points $x \in X$ the transformations indicated are isomorphisms (one can see this by considering the second of them for the family φ of all closed sets) precisely for $(G - n)$-spaces.

Analogous relations occur in homological algebra. For example, if G is a left R-module of projective dimension n having a projective resolution of length n of modules of finite type (in particular, when the ring R is coherent and G is finitely representable), for any right R-module A there arise natural transformations $\gamma : \mathrm{Ext}^p_R(\mathrm{Ext}^n_R(G, R), A) \to \mathrm{Tor}^R_{n-p}(A, G)$ (cf. [67]), factoring through a spectral sequence (analogous to the Cartan spectral sequence) converging to $\mathrm{Tor}^R_{n-p-q}(A, G)$ with second term $E_2^{pq} = \mathrm{Ext}^p_R(\mathrm{Ext}^{n-q}_R(G, R), A)$. In particular, $\mathrm{Tor}^R_n(A, G) = \mathrm{Hom}_R(\mathrm{Ext}^n_R(G, R), A)$ and the transformation γ is monomorphic for $p = 1$. For fixed G the functors $\mathrm{Ext}^p_R(\mathrm{Ext}^n_R(G, R), A)$ are the right derived functors of the functor $\mathrm{Tor}^R_n(A, G)$ which is left exact with respect to the argument A. The transformations γ are isomorphisms precisely when $\mathrm{Ext}^q_R(G, R) = 0$ for $q \neq n$. In the relations considered such G obviously play the role of $(G - n)$-spaces. Modules of this type were studied in [111, 142] where similar results are considered from different points of view (it is nevertheless shown, cf. [142], that Poincaré duality in the case of groups [83] reduces to the isomorphism γ).

The vansihing of $\mathrm{Ext}^q_R(G, R)$ for $q \neq n$ in the case considered is equivalent to the requirement $\mathrm{Tor}^R_q(J, G) = 0$ for $q \neq n$ for any injective right modules J, so the results discussed are contained in the following scheme [67]. Let F be a right exact covariant additive functor (for example, the tensor product) for which the left derived functor F_{n+1} is equal to zero. Then the functor F_n is left exact and one can consider its right derived functors $(F_n)^p$. Let J^* be an injective resolution of the object G. Then there is a spectral sequence converging to $F_{n-p-q}(G)$ with second term $E_2^{pq} = H^p(F_{n-q}(J^*))$ in which $E_2^{p0} = (F_n)^p(G)$. Consequently, there are also transformations $(F_n)^p \to F_{n-p}$ which turn out to be an isomorphism in dimension 0 and a monomorphism for $p = 1$. They are isomorphic for all p if and only if the functors F_k vanish for $k < n$ on injective objects.

In the cited papers other versions of Poincaré type duality (corresponding to other known versions of derived functors) are also considered.

6.3. Homological Manifolds. The original definition of homological manifold, due to Wilder, was rather complicated and contained many requirements. It turned out that many of them are dependent and hence some can be derived from others. The constructions contained, for example, in [87] are also not conclusive. At the present time, by a homological n-manifold over an R-module G is meant a $(G - n)$-space for which the stalks of the sheaf $\mathcal{H}_n(G)$ are isomorphic to G. One usually considers homological manifolds over a ground ring R. The term "generalized manifold" is no less common (cf., e.g., [85, 86, 89], etc.). By a homological manifold X with boundary K we mean an $(R - n)$-space for which the stalks of the sheaf $\mathcal{H}_n(R)$ are isomorphic to R at points of $X \setminus K$ and equal to zero at points of K.

A homological manifold X with the second axiom of countability over a principal ideal ring is locally connected, $\dim_R X = \mathrm{h}\dim_R X$ and if R is countable or is a field, then X is always $\mathrm{h}\ell c_R$ [76] (the last assertion is also in [131]). If R is a principal ideal domain, then $X \setminus K$ is also always locally orientable [88] [i.e., the sheaf $\mathcal{H}_n(R)$ is locally constant). If R is a countable principal ideal ring and $\dim_R X < \infty$ then each of the following conditions is equivalent to the fact that X is a homological manifold over R with boundary K [76]: 1) the local Betti numbers $p^i(x; R)$ defined by the groups $H^i(X, X \setminus U; R)$ (U being neighborhoods of the point x) are equal to zero for $i \neq n$, $p^n(x; R) = 0$ for $x \in K$ but $p^n(x; R) = 1$ for all $x \in X \setminus K$; 2) the local Betti numbers $p_i(x; R)$ defined by the groups $H^c_i(U \setminus x; R)$ (reduced in dimension zero) are equal to zero for $i \neq n - 1$, $p_{n-1}(x; R) = 0$ for $x \in K$ but $p_{n-1}(x; R) = 1$ for $x \in X \setminus K$; 3) the local cohomology groups $I^i_x(R)$ (point 3.2 above) vanish for

$i \neq n-1$, $I_x^{n-1}(R) = 0$ for $x \in K$ but $I_x^{n-1}(R) = R$ for $x \in X \setminus K$. Thus, there are equivalent definitions of homological manifold in terms of any of four versions of local groups, while the homological and cohomological approaches are equivalent. It is established in [76] that the set K is always closed and nowhere dense, for countable rings $\dim_R K \leqslant n-1$ $(\dim_Z K = n-1)$ and $I_x^{p-1}(R) = H_x^p(R)$ for R = Z the boundary K is a homological (n − 1)-manifold over Z_2. Any (loally compact) hlc_R-space for which $\dim_R X = n < \infty$ and all $\mathscr{H}_p(R)$ are locally constant sheaves with finitely generated stalks, is a homological n-manifold [87]. In terms of singular homology groups the analogous result is valid without the requirement of local constancy of the sheaves $\mathscr{H}_p(R)$ under the assumption that for each $p \leqslant n$ the stalks of $\mathscr{H}_p(R)$ are isomorphic to one another [102].

Sometimes, by generalized manifolds one means homological manifolds satisfying some additional requirements (usually formulated in terms of singular theory, for example, the requirement of separation of images of two-dimensional discs). In some cases homological manifolds which are ANRs are called generalized. They coincide with the images under cellular maps of ordinary topological n-manifolds. In [96] a technique of partition of manifolds into Z-acyclic sets (not necessarily cellular) is developed with the goal of getting, in terms of such partitions, an analogous characterization of homological manifolds. Examples of such partitions are constructed, whose quotient-spaces either are or are not ANRs. The influence of the structure of a homological manifold on the property of being an ANR (or an ANR-divisor) is also studied in [26]. It is shown there that the suspension of a compact connected homological manifold over Z has the pointed shape of a polyhedron. If $f:M \to X$ is a surjective map of a compact homological n-manifold over an algebraically compact group G, then for dim × X < ∞ the compactum X is a homological n-manifold over G if any of the following conditions hold [26]: a) f is a shape fibration which is a strong shape equivalence; b) f is a hereditary shape equivalence; c) f is cell-like. (Cf. also point 7.2 below.)

The tools of homology theory can also be applied to study infinite-dimensional manifolds, in particular, Q-manifolds (modeled on the Hilbert cube Q). Thus, for a locally compact ANR-space X having the property of separating images of two-dimensional discs, the requirement of being a Q-manifold is equivalent to any of the following conditions [95]: a) for any two subspaces of X and any two elements of the homology groups of these subspaces (possibly of different dimensions) one can find disjoint compact supports of these elements; b) for any homology element one can find a compact support of infinite codimension; c) the points in X have infinite codimension and elements of the homology groups of any subspaces of X have finite-dimensional (compact) supports (without the property of separating discs each of these conditions is equivalent to the fact that a Q-manifold is the product of X by a square). Similar results find many applications [95] (attaching Q-manifolds, proper maps of Q-manifolds including their cellular maps onto spaces which turn out not to be Q-manifolds, the property of the product of two ANRs being a Q-manifold).

7. Other Examples

7.1. Theorems on Coincidences for Maps of Manifolds. Let M and N be connected topological n-dimensional manifolds (with boundaries ∂M, ∂N), $f, g: M \to N$ be continuous maps the first of which is compact [the closure of f(M) is compact] and the second proper and $g(\partial M) \subset \partial N$. By the Lefschetz number $\Lambda_{f,g}$ of this pair we mean the Lefschetz number of the compositions (with respect to all q) of the homomorphisms

$$H^q(N) \xrightarrow[f^*]{} H^q(M) = H_{n-q}(M, \partial M) \xrightarrow[g_*]{} H_{n-q}(N, \partial N) = H^q(N).$$

The coefficients for the homology and cohomology groups can be any field which however must be the field of residues modulo 2 if at least one of the manifolds is nonorientable. In [13, 14] it is established that for $\Lambda_{f,g} \neq 0$ in M one can find a point x for which f(x) = g(x).

Special cases of this result (basically for closed manifolds, frequently endowed with a combinatorial or smooth structure) have been proved by many authors. The compact case is considered in all completeness in [134]. For the case m = N and g the identity, the results formulated is the basic result of [133].

The tools of homology turned out to be irreplaceable in solving problems connected with far-reaching generalizations of theorems of Borsuk–Ulam, Bourgin–Yang, Conner–Floyd type and many others on identifying antipodes and more widely orbits in free (connected paracompact) Z_p-spaces (p being a prime number) under maps of such spaces into Euclidean spaces, spheres, and simply manifolds, on estimating the dimension of sets of identified orbits, on the dimension of sets of finite collections of points going into orbits under maps of the ambient space into Z_p-spaces, on the dimension of special sets of coincidences for pairs of maps of a Z_p-space, etc. Very advanced results on this subject are given in the recent papers [9-12, 115]. Poincaré–Lefschetz duality is used essentially (and in complete generality).

7.2. Linking of Cycles. Acyclic Coverings of Manifolds. Let $A_1,\ldots,A_k$ and $B_1,\ldots,B_\ell$ be two collections each of which consists of pairwise disjoint discs in three-dimensional Euclidean space R^3. If the points $x \neq y$ in R^3 are separated by their union, then for some pair of indices p, q, these points separate the union $A_p \cup B_q$ [84]. Analogous collections in the sphere S^n, $n > 2$ consisting of $(n - r - 2)$-acyclic closed sets $0 \leqslant r < n-2$ have the following property: if the set $A = (\bigcup_i A_i) \cup (\bigcup_j B_j)$ is linked by the element $h \in H_r^c(S^n \setminus A)$ (i.e., simply $h \neq 0$) then for some p, q, the element $i_*(h)$ links $A_p \cup B_q$, where i is the inclusion $S^n \setminus A \subset S^n \setminus (A_p \cup B_q)$ [149]. From this it is easy to get the following assertion: if the two indicated systems consisting of $(n - 1)$-acyclic sets cover the orientable topological manifold M^n then M^n is already covered by some two sets A_p and B_q. Results of this kind (and problems connected with them) also occur in other papers (cf., e.g., [150]).

At the same time that the cyclicity conditions figuring in similar assertions are essential, the presence in them of two collections of pairwise disjoint sets is a random phenomenon which is only important in that the multiplicity of their union is at most 2. Far-reaching generalizations and refinements of such results are obtained in [6]. The finiteness of the system of sets is not always essential.

For this let $m \geqslant l \geqslant 1$ be integers, $\{A_\lambda\}$ be a locally finite system of closed subspaces of the locally compact space X of multiplicity $\leqslant l$ and such that for any s different sets of it, $s < \ell$ we have $H_c^{m-s}(\bigcap_{j=1}^s A_{\lambda_j}; \mathcal{G}) = 0$. Then for any nonzero $h \in H_c^m(A; \mathcal{G})$ one can find indices $\lambda_1, \ldots, \lambda_q$, $q \leqslant l$ such that the restriction of h to $\bigcap_{i=1}^q A_{\lambda_i}$ is nonzero. Here $A = \bigcup_\lambda A_\lambda$, $\mathcal{G}$ is an arbitrary sheaf on X. In the case of finite systems $\{A_\lambda\}$ the analogous result is valid for ordinary cohomology groups H^* and any paracompact (Hausdorff) space X.

In particular, if A is a connected orientable generalized m-manifold over the principal ideal ring R, then in the system $\{A_\lambda\}$ one can find ℓ (or fewer) sets covering A. If $\alpha = \{A_\lambda\}$ is a locally finite covering of a connected orientable generalized m-manifold consisting of at least three closed sets where none of the sets of α can be thrown out, then for $H_c^{m-1}(A_\lambda; \mathcal{G}) = 0$ the multiplicity of the covering is at lest three (refinement of results proved in [150] for finite coverings of compact manifolds). If X above is a connected orientable generalized $(m + 1)$-manifold and $H_c^m(X; \mathcal{G}) = 0$ and $\{A_\lambda\}$ is the system of subsets of X indicated above and is such that $A = \bigcup A_\lambda$ separates X, then the union of some ℓ sets of the system $\{A_\lambda\}$ already separates X.

The general result formulated above in the case of a $(G - n)$-space X under the condition that $H_{r+1}^c(X; G) = 0$ gives the following assertion about linking of cycles. Let $n > l \geqslant 1$ be integers, $n - l > r \geqslant 0$ and let $\{A_\lambda\}$ be a locally finite system of closed subspaces of X of multiplicity $\leqslant l$ such that for any s different sets from it, $s < \ell$ we have

$$H_c^{n-r-s-1}(\bigcap_{i=1}^s A_{\lambda_i}; \mathscr{H}_n(G)) = 0.$$

If the set $A = \bigcup_\lambda A_\lambda$ is linked by a (nonzero) element $h \in H_r^c(X \setminus A; G)$ (this means that the image of h in X is equal to zero), then for some $\lambda_1, \ldots, \lambda_q$, $q \leqslant l$ the image of H in $X \setminus \bigcup_{i=1}^q A_{\lambda_i}$ links $\bigcup_{i=1}^q A_{\lambda_i}$. In the proofs of some results concerning (generalized) manifolds and $(G - n)$-spaces, duality is essential (cf. [6]).

7.3. Homology and Cohomology Spheres. Let $\{p_k\}$ be primes numbered in increasing order, Z_{p_k} be the corresponding fields of residues, and Q be the field of rational numbers. The following assertions are equivalent for a metricable compactum X [7]: a) X is a cohomology sphere over all the Z_{p_k} and Q; b) X is a cohomology sphere over all fields of coefficients; c) X is a homology sphere over all Z_{p_k} and Q; d) X is a homology sphere over all fields of coefficients. One can find an equivalent description of the possible valued of integral cohomology groups of X in dimensions n and n + 1 (the rest are trivial): $H^n(X; Z)$ is any torsion-free group of rank 1, $H^{n+1}(X; Z)$ is the direct sum of quasicyclic groups $Z(p_k^\infty)$ over all k for which $\alpha_k = \infty$, where $(\alpha_1, \ldots, \alpha_k, \ldots)$ is the characteristic of a nonzero element of $H^n(X; Z)$. In particular, the cohomology groups over Z of such compacta are in natural one-to-one correspondence with torsion-free Abelian groups of rank 1 (which are obviously continua).

Analogous (but essentially more complicated) descriptions are also given for cohomology spheres over fields of coefficients, the dimension of each of which depends on the field defining it [8]. These dimensions coincide if the compactum X is homologically locally connected over Z; in this case X is a cohomology sphere (of the same dimension) over Z too. In [8] there is also obtained a description of homology spheres over all Z_{p^k} and Q. By the homology groups corresponding to them all such "n-spheres" which are not homology n-spheres over Z are in one-to-one correspondence with infinite regular binary fractions. Incidentally one also gets the following result (which is used essentially in the classification): for a torsion-free group of rank 1 the group Ext(H; Z) is the direct sum of all $Z(p_k^\infty)$ for which $\alpha_k \neq \infty$ [$(\alpha_1, \ldots, \alpha_k, \ldots)$ being the characteristic of a nontrivial $h \in H$] and a continuum of copies of the group Q. It is shown that in comparison with the homological version of the classification (of homology spheres) the cohomological one is finer and that both versions coincide in the class of homologically locally connected compacta.

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